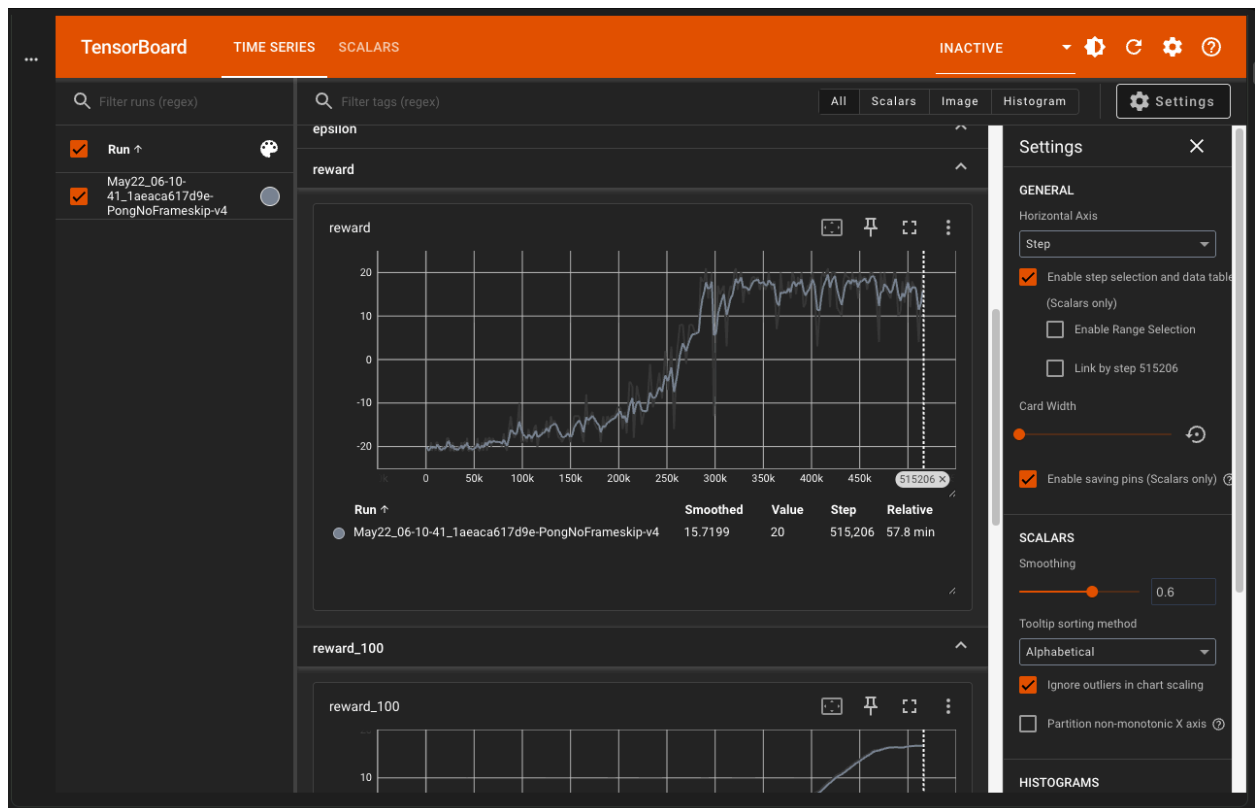
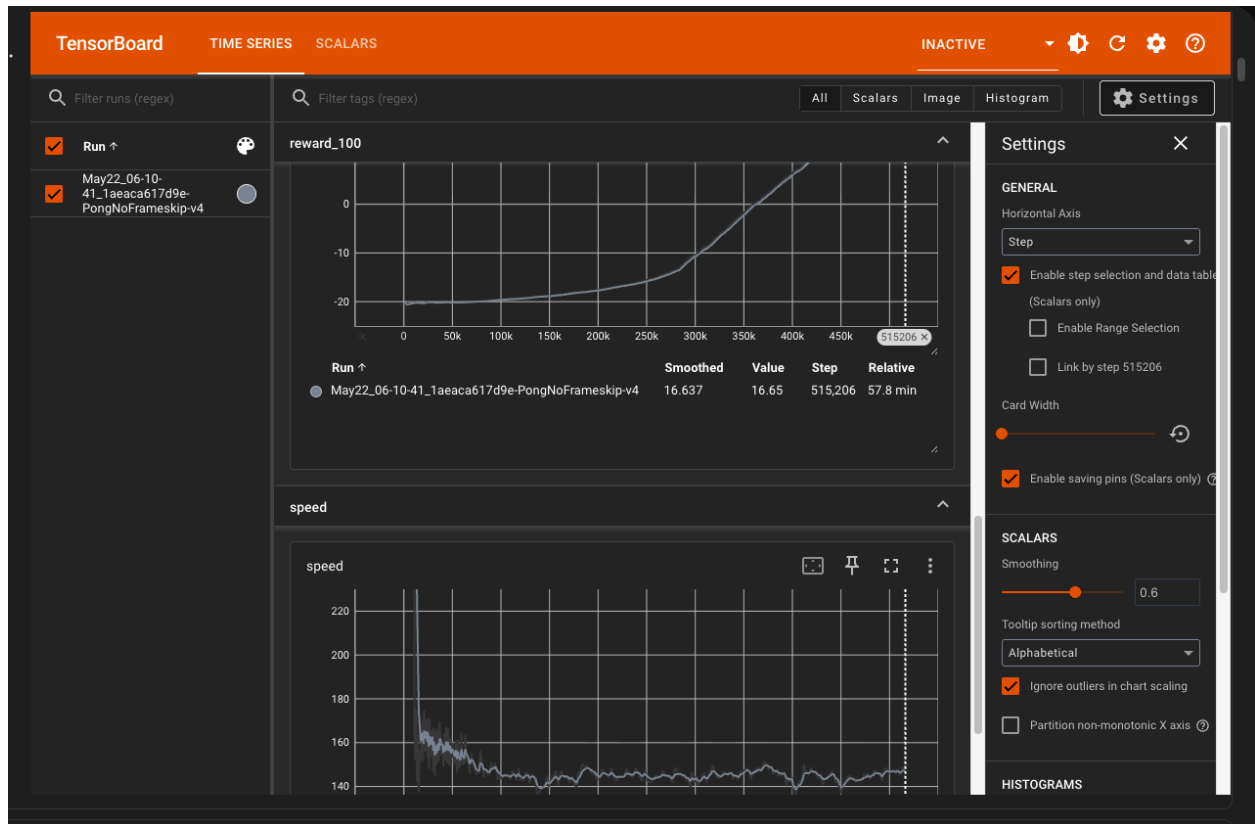


Apologies if this file is a bit incomprehensible. It just contains copy-paste from the outputs of the textboxes in the .ipynb file as I went along and ran everything. This was because if I was disconnected from the runtime in Colab, it would delete those text outputs. I just tried to ensure I could solve this by screenshotting and copy-pasting cell outputs.

Fri May 22 06:08:39 2026

NVIDIA-SMI 580.82.07			Driver Version: 580.82.07			CUDA Version: 13.0		
GPU	Name		Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr. ECC	
Fan	Temp	Perf	Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute M.	MIG M.
0	Tesla T4		Off	00000000:00:04.0	Off		0	
N/A	42C	P8	9W / 70W	0MiB / 15360MiB		0%	Default	N/A
Processes:								
GPU	GI	CI	PID	Type	Process name	GPU Memory		
	ID	ID				Usage		
No running processes found								

```
Cloning into 'Deep-Reinforcement-Learning-Hands-On'...
remote: Enumerating objects: 6277, done.
remote: Counting objects: 100% (39/39), done.
remote: Compressing objects: 100% (31/31), done.
remote: Total 6277 (delta 23), reused 13 (delta 8), pack-reused 6238 (from 1)
Receiving objects: 100% (6277/6277), 16.01 MiB | 13.19 MiB/s, done.
Resolving deltas: 100% (4161/4161), done.
/content/Deep-Reinforcement-Learning-Hands-On/Chapter06
```



A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)

[Powered by Stella]

DQN(

(conv): Sequential(

(0): Conv2d(1, 32, kernel_size=(8, 8), stride=(4, 4))

(1): ReLU()

(2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))

(3): ReLU()

(4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))

(5): ReLU()

)

(fc): Sequential(

(0): Linear(in_features=3136, out_features=512, bias=True)

(1): ReLU()

(2): Linear(in_features=512, out_features=6, bias=True)

)

)

962: done 1 games, mean reward -20.000, eps 0.99, speed 477.76 f/s

1726: done 2 games, mean reward -20.500, eps 0.98, speed 807.07 f/s

2634: done 3 games, mean reward -20.667, eps 0.97, speed 803.54 f/s

3533: done 4 games, mean reward -20.750, eps 0.96, speed 791.14 f/s

4486: done 5 games, mean reward -20.600, eps 0.96, speed 793.16 f/s

5406: done 6 games, mean reward -20.333, eps 0.95, speed 813.06 f/s

6198: done 7 games, mean reward -20.429, eps 0.94, speed 655.07 f/s

7103: done 8 games, mean reward -20.375, eps 0.93, speed 482.34 f/s

8005: done 9 games, mean reward -20.333, eps 0.92, speed 629.37 f/s

8969: done 10 games, mean reward -20.300, eps 0.91, speed 738.78 f/s

9959: done 11 games, mean reward -20.364, eps 0.90, speed 726.86 f/s

10924: done 12 games, mean reward -20.250, eps 0.89, speed 164.73 f/s

11839: done 13 games, mean reward -20.154, eps 0.88, speed 152.19 f/s

12824: done 14 games, mean reward -20.143, eps 0.87, speed 175.35 f/s

13588: done 15 games, mean reward -20.200, eps 0.86, speed 141.78 f/s

14502: done 16 games, mean reward -20.250, eps 0.85, speed 163.39 f/s

15341: done 17 games, mean reward -20.294, eps 0.85, speed 137.58 f/s

16332: done 18 games, mean reward -20.333, eps 0.84, speed 156.81 f/s

17281: done 19 games, mean reward -20.316, eps 0.83, speed 151.83 f/s

18170: done 20 games, mean reward -20.300, eps 0.82, speed 159.35 f/s

19111: done 21 games, mean reward -20.333, eps 0.81, speed 170.40 f/s

20041: done 22 games, mean reward -20.318, eps 0.80, speed 151.76 f/s

20911: done 23 games, mean reward -20.304, eps 0.79, speed 174.88 f/s

21994: done 24 games, mean reward -20.208, eps 0.78, speed 156.10 f/s

23158: done 25 games, mean reward -20.120, eps 0.77, speed 170.40 f/s

24213: done 26 games, mean reward -20.077, eps 0.76, speed 152.85 f/s

24996: done 27 games, mean reward -20.111, eps 0.75, speed 171.39 f/s

25839: done 28 games, mean reward -20.143, eps 0.74, speed 147.90 f/s
26729: done 29 games, mean reward -20.138, eps 0.73, speed 170.88 f/s
27726: done 30 games, mean reward -20.100, eps 0.72, speed 153.02 f/s
28638: done 31 games, mean reward -20.129, eps 0.71, speed 162.32 f/s
29420: done 32 games, mean reward -20.156, eps 0.71, speed 166.33 f/s
30213: done 33 games, mean reward -20.182, eps 0.70, speed 146.43 f/s
31167: done 34 games, mean reward -20.176, eps 0.69, speed 166.95 f/s
31950: done 35 games, mean reward -20.200, eps 0.68, speed 144.58 f/s
33004: done 36 games, mean reward -20.222, eps 0.67, speed 165.65 f/s
34004: done 37 games, mean reward -20.216, eps 0.66, speed 149.76 f/s
35054: done 38 games, mean reward -20.211, eps 0.65, speed 166.73 f/s
36014: done 39 games, mean reward -20.205, eps 0.64, speed 151.07 f/s
36998: done 40 games, mean reward -20.175, eps 0.63, speed 169.67 f/s
37841: done 41 games, mean reward -20.195, eps 0.62, speed 162.19 f/s
38689: done 42 games, mean reward -20.214, eps 0.61, speed 154.04 f/s
39662: done 43 games, mean reward -20.186, eps 0.60, speed 160.86 f/s
40772: done 44 games, mean reward -20.159, eps 0.59, speed 148.95 f/s
41691: done 45 games, mean reward -20.156, eps 0.58, speed 163.87 f/s
42733: done 46 games, mean reward -20.109, eps 0.57, speed 151.86 f/s
43636: done 47 games, mean reward -20.128, eps 0.56, speed 167.12 f/s
44567: done 48 games, mean reward -20.146, eps 0.55, speed 150.56 f/s
45410: done 49 games, mean reward -20.163, eps 0.55, speed 166.72 f/s
46610: done 50 games, mean reward -20.160, eps 0.53, speed 149.55 f/s
47794: done 51 games, mean reward -20.176, eps 0.52, speed 162.20 f/s
48665: done 52 games, mean reward -20.192, eps 0.51, speed 149.80 f/s
49860: done 53 games, mean reward -20.189, eps 0.50, speed 160.06 f/s
50979: done 54 games, mean reward -20.185, eps 0.49, speed 154.93 f/s
52182: done 55 games, mean reward -20.182, eps 0.48, speed 149.97 f/s
53651: done 56 games, mean reward -20.143, eps 0.46, speed 161.09 f/s
54631: done 57 games, mean reward -20.158, eps 0.45, speed 146.31 f/s
55734: done 58 games, mean reward -20.172, eps 0.44, speed 161.43 f/s
56856: done 59 games, mean reward -20.169, eps 0.43, speed 148.41 f/s
58253: done 60 games, mean reward -20.150, eps 0.42, speed 150.31 f/s
59304: done 61 games, mean reward -20.164, eps 0.41, speed 164.69 f/s
60615: done 62 games, mean reward -20.145, eps 0.39, speed 149.72 f/s
62112: done 63 games, mean reward -20.111, eps 0.38, speed 150.13 f/s
63268: done 64 games, mean reward -20.125, eps 0.37, speed 163.73 f/s
64757: done 65 games, mean reward -20.092, eps 0.35, speed 151.22 f/s
66462: done 66 games, mean reward -20.061, eps 0.34, speed 150.85 f/s
67827: done 67 games, mean reward -20.060, eps 0.32, speed 145.85 f/s
69513: done 68 games, mean reward -20.029, eps 0.30, speed 155.61 f/s
71135: done 69 games, mean reward -20.014, eps 0.29, speed 150.63 f/s
72680: done 70 games, mean reward -20.000, eps 0.27, speed 148.23 f/s
74030: done 71 games, mean reward -19.986, eps 0.26, speed 145.43 f/s

Best mean reward updated -20.000 -> -19.986, model saved
75741: done 72 games, mean reward -19.972, eps 0.24, speed 145.97 f/s
Best mean reward updated -19.986 -> -19.972, model saved
77458: done 73 games, mean reward -19.959, eps 0.23, speed 146.23 f/s
Best mean reward updated -19.972 -> -19.959, model saved
78721: done 74 games, mean reward -19.959, eps 0.21, speed 156.44 f/s
80823: done 75 games, mean reward -19.920, eps 0.19, speed 149.74 f/s
Best mean reward updated -19.959 -> -19.920, model saved
82133: done 76 games, mean reward -19.921, eps 0.18, speed 143.95 f/s
84069: done 77 games, mean reward -19.935, eps 0.16, speed 148.12 f/s
86057: done 78 games, mean reward -19.885, eps 0.14, speed 145.47 f/s
Best mean reward updated -19.920 -> -19.885, model saved
87998: done 79 games, mean reward -19.797, eps 0.12, speed 145.30 f/s
Best mean reward updated -19.885 -> -19.797, model saved
90021: done 80 games, mean reward -19.762, eps 0.10, speed 151.08 f/s
Best mean reward updated -19.797 -> -19.762, model saved
91769: done 81 games, mean reward -19.741, eps 0.08, speed 147.75 f/s
Best mean reward updated -19.762 -> -19.741, model saved
94063: done 82 games, mean reward -19.707, eps 0.06, speed 150.67 f/s
Best mean reward updated -19.741 -> -19.707, model saved
96895: done 83 games, mean reward -19.590, eps 0.03, speed 142.90 f/s
Best mean reward updated -19.707 -> -19.590, model saved
99072: done 84 games, mean reward -19.583, eps 0.02, speed 144.92 f/s
Best mean reward updated -19.590 -> -19.583, model saved
100896: done 85 games, mean reward -19.576, eps 0.02, speed 143.18 f/s
Best mean reward updated -19.583 -> -19.576, model saved
102848: done 86 games, mean reward -19.558, eps 0.02, speed 145.00 f/s
Best mean reward updated -19.576 -> -19.558, model saved
104691: done 87 games, mean reward -19.552, eps 0.02, speed 145.62 f/s
Best mean reward updated -19.558 -> -19.552, model saved
106423: done 88 games, mean reward -19.511, eps 0.02, speed 144.23 f/s
Best mean reward updated -19.552 -> -19.511, model saved
108358: done 89 games, mean reward -19.494, eps 0.02, speed 145.60 f/s
Best mean reward updated -19.511 -> -19.494, model saved
109822: done 90 games, mean reward -19.489, eps 0.02, speed 143.00 f/s
Best mean reward updated -19.494 -> -19.489, model saved
111995: done 91 games, mean reward -19.451, eps 0.02, speed 147.22 f/s
Best mean reward updated -19.489 -> -19.451, model saved
114128: done 92 games, mean reward -19.402, eps 0.02, speed 145.56 f/s
Best mean reward updated -19.451 -> -19.402, model saved
115731: done 93 games, mean reward -19.409, eps 0.02, speed 142.59 f/s
118127: done 94 games, mean reward -19.383, eps 0.02, speed 148.22 f/s
Best mean reward updated -19.402 -> -19.383, model saved
119695: done 95 games, mean reward -19.358, eps 0.02, speed 142.67 f/s

Best mean reward updated -19.383 -> -19.358, model saved
121513: done 96 games, mean reward -19.323, eps 0.02, speed 144.47 f/s
Best mean reward updated -19.358 -> -19.323, model saved
122909: done 97 games, mean reward -19.299, eps 0.02, speed 142.35 f/s
Best mean reward updated -19.323 -> -19.299, model saved
125265: done 98 games, mean reward -19.245, eps 0.02, speed 146.56 f/s
Best mean reward updated -19.299 -> -19.245, model saved
127270: done 99 games, mean reward -19.232, eps 0.02, speed 146.26 f/s
Best mean reward updated -19.245 -> -19.232, model saved
129689: done 100 games, mean reward -19.170, eps 0.02, speed 141.25 f/s
Best mean reward updated -19.232 -> -19.170, model saved
131583: done 101 games, mean reward -19.120, eps 0.02, speed 145.66 f/s
Best mean reward updated -19.170 -> -19.120, model saved
133556: done 102 games, mean reward -19.050, eps 0.02, speed 145.03 f/s
Best mean reward updated -19.120 -> -19.050, model saved
135650: done 103 games, mean reward -19.000, eps 0.02, speed 141.52 f/s
Best mean reward updated -19.050 -> -19.000, model saved
137561: done 104 games, mean reward -18.970, eps 0.02, speed 136.42 f/s
Best mean reward updated -19.000 -> -18.970, model saved
139313: done 105 games, mean reward -18.950, eps 0.02, speed 137.69 f/s
Best mean reward updated -18.970 -> -18.950, model saved
141320: done 106 games, mean reward -18.950, eps 0.02, speed 138.83 f/s
143087: done 107 games, mean reward -18.920, eps 0.02, speed 139.86 f/s
Best mean reward updated -18.950 -> -18.920, model saved
145167: done 108 games, mean reward -18.890, eps 0.02, speed 140.88 f/s
Best mean reward updated -18.920 -> -18.890, model saved
147442: done 109 games, mean reward -18.850, eps 0.02, speed 145.14 f/s
Best mean reward updated -18.890 -> -18.850, model saved
149595: done 110 games, mean reward -18.840, eps 0.02, speed 139.80 f/s
Best mean reward updated -18.850 -> -18.840, model saved
151716: done 111 games, mean reward -18.800, eps 0.02, speed 143.72 f/s
Best mean reward updated -18.840 -> -18.800, model saved
153703: done 112 games, mean reward -18.740, eps 0.02, speed 144.61 f/s
Best mean reward updated -18.800 -> -18.740, model saved
156423: done 113 games, mean reward -18.680, eps 0.02, speed 147.20 f/s
Best mean reward updated -18.740 -> -18.680, model saved
159013: done 114 games, mean reward -18.610, eps 0.02, speed 140.32 f/s
Best mean reward updated -18.680 -> -18.610, model saved
161636: done 115 games, mean reward -18.570, eps 0.02, speed 147.32 f/s
Best mean reward updated -18.610 -> -18.570, model saved
163523: done 116 games, mean reward -18.550, eps 0.02, speed 145.30 f/s
Best mean reward updated -18.570 -> -18.550, model saved
166307: done 117 games, mean reward -18.420, eps 0.02, speed 143.61 f/s
Best mean reward updated -18.550 -> -18.420, model saved

168178: done 118 games, mean reward -18.360, eps 0.02, speed 145.16 f/s
Best mean reward updated -18.420 -> -18.360, model saved

170372: done 119 games, mean reward -18.290, eps 0.02, speed 141.41 f/s
Best mean reward updated -18.360 -> -18.290, model saved

172774: done 120 games, mean reward -18.210, eps 0.02, speed 144.83 f/s
Best mean reward updated -18.290 -> -18.210, model saved

175343: done 121 games, mean reward -18.150, eps 0.02, speed 141.00 f/s
Best mean reward updated -18.210 -> -18.150, model saved

177699: done 122 games, mean reward -18.110, eps 0.02, speed 139.71 f/s
Best mean reward updated -18.150 -> -18.110, model saved

179761: done 123 games, mean reward -18.070, eps 0.02, speed 146.88 f/s
Best mean reward updated -18.110 -> -18.070, model saved

181403: done 124 games, mean reward -18.080, eps 0.02, speed 146.18 f/s

183805: done 125 games, mean reward -18.030, eps 0.02, speed 150.47 f/s
Best mean reward updated -18.070 -> -18.030, model saved

185700: done 126 games, mean reward -18.000, eps 0.02, speed 148.74 f/s
Best mean reward updated -18.030 -> -18.000, model saved

188173: done 127 games, mean reward -17.920, eps 0.02, speed 149.48 f/s
Best mean reward updated -18.000 -> -17.920, model saved

190364: done 128 games, mean reward -17.850, eps 0.02, speed 148.26 f/s
Best mean reward updated -17.920 -> -17.850, model saved

192743: done 129 games, mean reward -17.780, eps 0.02, speed 146.09 f/s
Best mean reward updated -17.850 -> -17.780, model saved

195918: done 130 games, mean reward -17.720, eps 0.02, speed 145.31 f/s
Best mean reward updated -17.780 -> -17.720, model saved

199026: done 131 games, mean reward -17.620, eps 0.02, speed 143.15 f/s
Best mean reward updated -17.720 -> -17.620, model saved

201727: done 132 games, mean reward -17.520, eps 0.02, speed 143.10 f/s
Best mean reward updated -17.620 -> -17.520, model saved

204081: done 133 games, mean reward -17.470, eps 0.02, speed 148.02 f/s
Best mean reward updated -17.520 -> -17.470, model saved

206339: done 134 games, mean reward -17.390, eps 0.02, speed 146.54 f/s
Best mean reward updated -17.470 -> -17.390, model saved

208214: done 135 games, mean reward -17.350, eps 0.02, speed 145.28 f/s
Best mean reward updated -17.390 -> -17.350, model saved

211227: done 136 games, mean reward -17.190, eps 0.02, speed 144.00 f/s
Best mean reward updated -17.350 -> -17.190, model saved

214094: done 137 games, mean reward -17.050, eps 0.02, speed 148.42 f/s
Best mean reward updated -17.190 -> -17.050, model saved

215605: done 138 games, mean reward -17.030, eps 0.02, speed 144.09 f/s
Best mean reward updated -17.050 -> -17.030, model saved

218785: done 139 games, mean reward -16.890, eps 0.02, speed 142.43 f/s
Best mean reward updated -17.030 -> -16.890, model saved

221390: done 140 games, mean reward -16.790, eps 0.02, speed 144.64 f/s

Best mean reward updated -16.890 -> -16.790, model saved
223929: done 141 games, mean reward -16.730, eps 0.02, speed 144.46 f/s
Best mean reward updated -16.790 -> -16.730, model saved
226469: done 142 games, mean reward -16.640, eps 0.02, speed 144.33 f/s
Best mean reward updated -16.730 -> -16.640, model saved
229166: done 143 games, mean reward -16.570, eps 0.02, speed 148.15 f/s
Best mean reward updated -16.640 -> -16.570, model saved
232553: done 144 games, mean reward -16.390, eps 0.02, speed 145.52 f/s
Best mean reward updated -16.570 -> -16.390, model saved
235465: done 145 games, mean reward -16.300, eps 0.02, speed 144.26 f/s
Best mean reward updated -16.390 -> -16.300, model saved
238623: done 146 games, mean reward -16.200, eps 0.02, speed 148.51 f/s
Best mean reward updated -16.300 -> -16.200, model saved
242164: done 147 games, mean reward -16.020, eps 0.02, speed 145.63 f/s
Best mean reward updated -16.200 -> -16.020, model saved
244951: done 148 games, mean reward -15.900, eps 0.02, speed 142.29 f/s
Best mean reward updated -16.020 -> -15.900, model saved
248169: done 149 games, mean reward -15.670, eps 0.02, speed 147.70 f/s
Best mean reward updated -15.900 -> -15.670, model saved
251139: done 150 games, mean reward -15.540, eps 0.02, speed 142.17 f/s
Best mean reward updated -15.670 -> -15.540, model saved
254588: done 151 games, mean reward -15.300, eps 0.02, speed 141.88 f/s
Best mean reward updated -15.540 -> -15.300, model saved
257351: done 152 games, mean reward -15.250, eps 0.02, speed 143.70 f/s
Best mean reward updated -15.300 -> -15.250, model saved
260576: done 153 games, mean reward -15.030, eps 0.02, speed 145.19 f/s
Best mean reward updated -15.250 -> -15.030, model saved
263483: done 154 games, mean reward -14.750, eps 0.02, speed 144.02 f/s
Best mean reward updated -15.030 -> -14.750, model saved
266493: done 155 games, mean reward -14.470, eps 0.02, speed 147.40 f/s
Best mean reward updated -14.750 -> -14.470, model saved
269911: done 156 games, mean reward -14.300, eps 0.02, speed 144.27 f/s
Best mean reward updated -14.470 -> -14.300, model saved
273353: done 157 games, mean reward -14.010, eps 0.02, speed 144.50 f/s
Best mean reward updated -14.300 -> -14.010, model saved
276365: done 158 games, mean reward -13.720, eps 0.02, speed 141.87 f/s
Best mean reward updated -14.010 -> -13.720, model saved
279479: done 159 games, mean reward -13.460, eps 0.02, speed 148.77 f/s
Best mean reward updated -13.720 -> -13.460, model saved
283038: done 160 games, mean reward -13.200, eps 0.02, speed 145.66 f/s
Best mean reward updated -13.460 -> -13.200, model saved
285018: done 161 games, mean reward -12.800, eps 0.02, speed 145.87 f/s
Best mean reward updated -13.200 -> -12.800, model saved
286880: done 162 games, mean reward -12.430, eps 0.02, speed 144.12 f/s

Best mean reward updated -12.800 -> -12.430, model saved
288769: done 163 games, mean reward -12.070, eps 0.02, speed 145.30 f/s
Best mean reward updated -12.430 -> -12.070, model saved
290435: done 164 games, mean reward -11.650, eps 0.02, speed 142.47 f/s
Best mean reward updated -12.070 -> -11.650, model saved
292718: done 165 games, mean reward -11.330, eps 0.02, speed 143.60 f/s
Best mean reward updated -11.650 -> -11.330, model saved
294855: done 166 games, mean reward -10.980, eps 0.02, speed 139.89 f/s
Best mean reward updated -11.330 -> -10.980, model saved
296543: done 167 games, mean reward -10.580, eps 0.02, speed 146.79 f/s
Best mean reward updated -10.980 -> -10.580, model saved
298646: done 168 games, mean reward -10.530, eps 0.02, speed 141.39 f/s
Best mean reward updated -10.580 -> -10.530, model saved
301128: done 169 games, mean reward -10.270, eps 0.02, speed 145.02 f/s
Best mean reward updated -10.530 -> -10.270, model saved
302986: done 170 games, mean reward -9.910, eps 0.02, speed 140.46 f/s
Best mean reward updated -10.270 -> -9.910, model saved
305165: done 171 games, mean reward -9.570, eps 0.02, speed 142.25 f/s
Best mean reward updated -9.910 -> -9.570, model saved
307048: done 172 games, mean reward -9.220, eps 0.02, speed 144.92 f/s
Best mean reward updated -9.570 -> -9.220, model saved
309031: done 173 games, mean reward -8.860, eps 0.02, speed 146.70 f/s
Best mean reward updated -9.220 -> -8.860, model saved
311930: done 174 games, mean reward -8.630, eps 0.02, speed 141.59 f/s
Best mean reward updated -8.860 -> -8.630, model saved
314133: done 175 games, mean reward -8.310, eps 0.02, speed 146.51 f/s
Best mean reward updated -8.630 -> -8.310, model saved
316078: done 176 games, mean reward -7.940, eps 0.02, speed 147.12 f/s
Best mean reward updated -8.310 -> -7.940, model saved
318136: done 177 games, mean reward -7.570, eps 0.02, speed 149.81 f/s
Best mean reward updated -7.940 -> -7.570, model saved
319852: done 178 games, mean reward -7.210, eps 0.02, speed 146.05 f/s
Best mean reward updated -7.570 -> -7.210, model saved
321505: done 179 games, mean reward -6.870, eps 0.02, speed 144.88 f/s
Best mean reward updated -7.210 -> -6.870, model saved
323392: done 180 games, mean reward -6.540, eps 0.02, speed 143.75 f/s
Best mean reward updated -6.870 -> -6.540, model saved
325267: done 181 games, mean reward -6.170, eps 0.02, speed 144.04 f/s
Best mean reward updated -6.540 -> -6.170, model saved
327028: done 182 games, mean reward -5.800, eps 0.02, speed 144.30 f/s
Best mean reward updated -6.170 -> -5.800, model saved
329081: done 183 games, mean reward -5.570, eps 0.02, speed 146.96 f/s
Best mean reward updated -5.800 -> -5.570, model saved
331219: done 184 games, mean reward -5.210, eps 0.02, speed 146.29 f/s

Best mean reward updated -5.570 -> -5.210, model saved
333169: done 185 games, mean reward -4.860, eps 0.02, speed 145.06 f/s
Best mean reward updated -5.210 -> -4.860, model saved
335121: done 186 games, mean reward -4.530, eps 0.02, speed 146.53 f/s
Best mean reward updated -4.860 -> -4.530, model saved
336986: done 187 games, mean reward -4.160, eps 0.02, speed 146.69 f/s
Best mean reward updated -4.530 -> -4.160, model saved
338800: done 188 games, mean reward -3.810, eps 0.02, speed 143.31 f/s
Best mean reward updated -4.160 -> -3.810, model saved
340582: done 189 games, mean reward -3.440, eps 0.02, speed 144.91 f/s
Best mean reward updated -3.810 -> -3.440, model saved
342425: done 190 games, mean reward -3.070, eps 0.02, speed 144.81 f/s
Best mean reward updated -3.440 -> -3.070, model saved
344225: done 191 games, mean reward -2.720, eps 0.02, speed 142.86 f/s
Best mean reward updated -3.070 -> -2.720, model saved
346414: done 192 games, mean reward -2.410, eps 0.02, speed 147.05 f/s
Best mean reward updated -2.720 -> -2.410, model saved
348361: done 193 games, mean reward -2.030, eps 0.02, speed 146.83 f/s
Best mean reward updated -2.410 -> -2.030, model saved
350153: done 194 games, mean reward -1.680, eps 0.02, speed 146.83 f/s
Best mean reward updated -2.030 -> -1.680, model saved
352071: done 195 games, mean reward -1.350, eps 0.02, speed 148.56 f/s
Best mean reward updated -1.680 -> -1.350, model saved
353895: done 196 games, mean reward -1.020, eps 0.02, speed 147.72 f/s
Best mean reward updated -1.350 -> -1.020, model saved
355566: done 197 games, mean reward -0.650, eps 0.02, speed 146.85 f/s
Best mean reward updated -1.020 -> -0.650, model saved
357420: done 198 games, mean reward -0.340, eps 0.02, speed 143.19 f/s
Best mean reward updated -0.650 -> -0.340, model saved
359355: done 199 games, mean reward 0.000, eps 0.02, speed 147.20 f/s
Best mean reward updated -0.340 -> 0.000, model saved
361161: done 200 games, mean reward 0.310, eps 0.02, speed 147.14 f/s
Best mean reward updated 0.000 -> 0.310, model saved
363882: done 201 games, mean reward 0.530, eps 0.02, speed 148.28 f/s
Best mean reward updated 0.310 -> 0.530, model saved
366259: done 202 games, mean reward 0.780, eps 0.02, speed 148.57 f/s
Best mean reward updated 0.530 -> 0.780, model saved
367945: done 203 games, mean reward 1.140, eps 0.02, speed 146.52 f/s
Best mean reward updated 0.780 -> 1.140, model saved
370022: done 204 games, mean reward 1.470, eps 0.02, speed 150.73 f/s
Best mean reward updated 1.140 -> 1.470, model saved
372160: done 205 games, mean reward 1.810, eps 0.02, speed 151.34 f/s
Best mean reward updated 1.470 -> 1.810, model saved
374154: done 206 games, mean reward 2.150, eps 0.02, speed 149.60 f/s

Best mean reward updated 1.810 -> 2.150, model saved
376045: done 207 games, mean reward 2.510, eps 0.02, speed 147.68 f/s
Best mean reward updated 2.150 -> 2.510, model saved
378293: done 208 games, mean reward 2.820, eps 0.02, speed 148.84 f/s
Best mean reward updated 2.510 -> 2.820, model saved
379994: done 209 games, mean reward 3.180, eps 0.02, speed 145.57 f/s
Best mean reward updated 2.820 -> 3.180, model saved
381757: done 210 games, mean reward 3.560, eps 0.02, speed 146.06 f/s
Best mean reward updated 3.180 -> 3.560, model saved
384020: done 211 games, mean reward 3.880, eps 0.02, speed 147.53 f/s
Best mean reward updated 3.560 -> 3.880, model saved
385872: done 212 games, mean reward 4.200, eps 0.02, speed 144.56 f/s
Best mean reward updated 3.880 -> 4.200, model saved
387802: done 213 games, mean reward 4.500, eps 0.02, speed 146.16 f/s
Best mean reward updated 4.200 -> 4.500, model saved
389501: done 214 games, mean reward 4.830, eps 0.02, speed 144.97 f/s
Best mean reward updated 4.500 -> 4.830, model saved
391775: done 215 games, mean reward 5.120, eps 0.02, speed 146.91 f/s
Best mean reward updated 4.830 -> 5.120, model saved
393877: done 216 games, mean reward 5.440, eps 0.02, speed 140.15 f/s
Best mean reward updated 5.120 -> 5.440, model saved
395786: done 217 games, mean reward 5.690, eps 0.02, speed 143.36 f/s
Best mean reward updated 5.440 -> 5.690, model saved
397744: done 218 games, mean reward 6.010, eps 0.02, speed 142.23 f/s
Best mean reward updated 5.690 -> 6.010, model saved
399478: done 219 games, mean reward 6.330, eps 0.02, speed 144.34 f/s
Best mean reward updated 6.010 -> 6.330, model saved
401178: done 220 games, mean reward 6.650, eps 0.02, speed 135.17 f/s
Best mean reward updated 6.330 -> 6.650, model saved
403300: done 221 games, mean reward 6.950, eps 0.02, speed 135.83 f/s
Best mean reward updated 6.650 -> 6.950, model saved
405995: done 222 games, mean reward 7.140, eps 0.02, speed 142.26 f/s
Best mean reward updated 6.950 -> 7.140, model saved
408404: done 223 games, mean reward 7.410, eps 0.02, speed 142.82 f/s
Best mean reward updated 7.140 -> 7.410, model saved
410173: done 224 games, mean reward 7.790, eps 0.02, speed 144.30 f/s
Best mean reward updated 7.410 -> 7.790, model saved
411842: done 225 games, mean reward 8.120, eps 0.02, speed 150.20 f/s
Best mean reward updated 7.790 -> 8.120, model saved
413605: done 226 games, mean reward 8.470, eps 0.02, speed 147.37 f/s
Best mean reward updated 8.120 -> 8.470, model saved
415335: done 227 games, mean reward 8.800, eps 0.02, speed 145.98 f/s
Best mean reward updated 8.470 -> 8.800, model saved
417027: done 228 games, mean reward 9.150, eps 0.02, speed 145.69 f/s

Best mean reward updated 8.800 -> 9.150, model saved
419164: done 229 games, mean reward 9.400, eps 0.02, speed 148.72 f/s
Best mean reward updated 9.150 -> 9.400, model saved
420892: done 230 games, mean reward 9.730, eps 0.02, speed 146.49 f/s
Best mean reward updated 9.400 -> 9.730, model saved
422593: done 231 games, mean reward 10.040, eps 0.02, speed 146.46 f/s
Best mean reward updated 9.730 -> 10.040, model saved
424311: done 232 games, mean reward 10.350, eps 0.02, speed 146.09 f/s
Best mean reward updated 10.040 -> 10.350, model saved
426419: done 233 games, mean reward 10.630, eps 0.02, speed 143.62 f/s
Best mean reward updated 10.350 -> 10.630, model saved
428757: done 234 games, mean reward 10.860, eps 0.02, speed 144.67 f/s
Best mean reward updated 10.630 -> 10.860, model saved
431279: done 235 games, mean reward 11.130, eps 0.02, speed 140.78 f/s
Best mean reward updated 10.860 -> 11.130, model saved
433109: done 236 games, mean reward 11.360, eps 0.02, speed 143.53 f/s
Best mean reward updated 11.130 -> 11.360, model saved
434827: done 237 games, mean reward 11.620, eps 0.02, speed 143.71 f/s
Best mean reward updated 11.360 -> 11.620, model saved
436699: done 238 games, mean reward 11.990, eps 0.02, speed 145.05 f/s
Best mean reward updated 11.620 -> 11.990, model saved
438600: done 239 games, mean reward 12.240, eps 0.02, speed 145.68 f/s
Best mean reward updated 11.990 -> 12.240, model saved
440600: done 240 games, mean reward 12.490, eps 0.02, speed 147.15 f/s
Best mean reward updated 12.240 -> 12.490, model saved
442387: done 241 games, mean reward 12.840, eps 0.02, speed 145.50 f/s
Best mean reward updated 12.490 -> 12.840, model saved
444337: done 242 games, mean reward 13.130, eps 0.02, speed 141.78 f/s
Best mean reward updated 12.840 -> 13.130, model saved
446142: done 243 games, mean reward 13.440, eps 0.02, speed 142.20 f/s
Best mean reward updated 13.130 -> 13.440, model saved
448418: done 244 games, mean reward 13.590, eps 0.02, speed 139.54 f/s
Best mean reward updated 13.440 -> 13.590, model saved
450114: done 245 games, mean reward 13.900, eps 0.02, speed 136.71 f/s
Best mean reward updated 13.590 -> 13.900, model saved
452107: done 246 games, mean reward 14.160, eps 0.02, speed 140.51 f/s
Best mean reward updated 13.900 -> 14.160, model saved
453901: done 247 games, mean reward 14.380, eps 0.02, speed 142.52 f/s
Best mean reward updated 14.160 -> 14.380, model saved
455653: done 248 games, mean reward 14.660, eps 0.02, speed 143.99 f/s
Best mean reward updated 14.380 -> 14.660, model saved
457636: done 249 games, mean reward 14.800, eps 0.02, speed 144.91 f/s
Best mean reward updated 14.660 -> 14.800, model saved
459436: done 250 games, mean reward 15.050, eps 0.02, speed 144.78 f/s

Best mean reward updated 14.800 -> 15.050, model saved
461135: done 251 games, mean reward 15.220, eps 0.02, speed 142.91 f/s
Best mean reward updated 15.050 -> 15.220, model saved
462805: done 252 games, mean reward 15.580, eps 0.02, speed 143.44 f/s
Best mean reward updated 15.220 -> 15.580, model saved
464627: done 253 games, mean reward 15.750, eps 0.02, speed 142.00 f/s
Best mean reward updated 15.580 -> 15.750, model saved
466771: done 254 games, mean reward 15.830, eps 0.02, speed 142.54 f/s
Best mean reward updated 15.750 -> 15.830, model saved
470029: done 255 games, mean reward 15.790, eps 0.02, speed 143.53 f/s
471725: done 256 games, mean reward 16.000, eps 0.02, speed 143.10 f/s
Best mean reward updated 15.830 -> 16.000, model saved
473964: done 257 games, mean reward 16.080, eps 0.02, speed 145.92 f/s
Best mean reward updated 16.000 -> 16.080, model saved
475661: done 258 games, mean reward 16.200, eps 0.02, speed 143.90 f/s
Best mean reward updated 16.080 -> 16.200, model saved
477620: done 259 games, mean reward 16.300, eps 0.02, speed 146.21 f/s
Best mean reward updated 16.200 -> 16.300, model saved
479789: done 260 games, mean reward 16.380, eps 0.02, speed 143.25 f/s
Best mean reward updated 16.300 -> 16.380, model saved
481870: done 261 games, mean reward 16.330, eps 0.02, speed 145.81 f/s
483901: done 262 games, mean reward 16.320, eps 0.02, speed 146.31 f/s
485628: done 263 games, mean reward 16.340, eps 0.02, speed 147.40 f/s
487323: done 264 games, mean reward 16.330, eps 0.02, speed 145.64 f/s
489990: done 265 games, mean reward 16.290, eps 0.02, speed 146.07 f/s
492365: done 266 games, mean reward 16.230, eps 0.02, speed 142.19 f/s
494320: done 267 games, mean reward 16.210, eps 0.02, speed 147.85 f/s
496478: done 268 games, mean reward 16.470, eps 0.02, speed 148.55 f/s
Best mean reward updated 16.380 -> 16.470, model saved
498271: done 269 games, mean reward 16.590, eps 0.02, speed 147.19 f/s
Best mean reward updated 16.470 -> 16.590, model saved
500047: done 270 games, mean reward 16.620, eps 0.02, speed 145.95 f/s
Best mean reward updated 16.590 -> 16.620, model saved
502411: done 271 games, mean reward 16.590, eps 0.02, speed 147.08 f/s
504193: done 272 games, mean reward 16.610, eps 0.02, speed 147.57 f/s
506133: done 273 games, mean reward 16.610, eps 0.02, speed 145.75 f/s
508241: done 274 games, mean reward 16.740, eps 0.02, speed 148.05 f/s
Best mean reward updated 16.620 -> 16.740, model saved
511226: done 275 games, mean reward 16.630, eps 0.02, speed 145.00 f/s
513539: done 276 games, mean reward 16.610, eps 0.02, speed 148.75 f/s
515206: done 277 games, mean reward 16.650, eps 0.02, speed 147.23 f/s
517659: done 278 games, mean reward 16.590, eps 0.02, speed 150.26 f/s
519718: done 279 games, mean reward 16.560, eps 0.02, speed 148.78 f/s
521800: done 280 games, mean reward 16.560, eps 0.02, speed 149.48 f/s

523907: done 281 games, mean reward 16.540, eps 0.02, speed 144.73 f/s
525599: done 282 games, mean reward 16.550, eps 0.02, speed 149.28 f/s
529003: done 283 games, mean reward 16.460, eps 0.02, speed 145.35 f/s
531524: done 284 games, mean reward 16.400, eps 0.02, speed 143.76 f/s
533619: done 285 games, mean reward 16.420, eps 0.02, speed 145.56 f/s
535787: done 286 games, mean reward 16.420, eps 0.02, speed 148.08 f/s
537892: done 287 games, mean reward 16.410, eps 0.02, speed 146.64 f/s
539971: done 288 games, mean reward 16.400, eps 0.02, speed 148.14 f/s
542264: done 289 games, mean reward 16.370, eps 0.02, speed 146.63 f/s
544251: done 290 games, mean reward 16.360, eps 0.02, speed 148.30 f/s
546699: done 291 games, mean reward 16.240, eps 0.02, speed 139.98 f/s
548491: done 292 games, mean reward 16.260, eps 0.02, speed 140.31 f/s
550487: done 293 games, mean reward 16.230, eps 0.02, speed 144.92 f/s
552244: done 294 games, mean reward 16.250, eps 0.02, speed 145.50 f/s
554218: done 295 games, mean reward 16.270, eps 0.02, speed 145.33 f/s
555943: done 296 games, mean reward 16.300, eps 0.02, speed 145.02 f/s
557737: done 297 games, mean reward 16.290, eps 0.02, speed 146.02 f/s
559494: done 298 games, mean reward 16.320, eps 0.02, speed 148.21 f/s
561264: done 299 games, mean reward 16.370, eps 0.02, speed 145.34 f/s
562931: done 300 games, mean reward 16.390, eps 0.02, speed 149.76 f/s
565250: done 301 games, mean reward 16.460, eps 0.02, speed 137.72 f/s
567208: done 302 games, mean reward 16.540, eps 0.02, speed 142.70 f/s
569329: done 303 games, mean reward 16.490, eps 0.02, speed 141.74 f/s
571297: done 304 games, mean reward 16.500, eps 0.02, speed 146.13 f/s
573845: done 305 games, mean reward 16.420, eps 0.02, speed 149.83 f/s
575631: done 306 games, mean reward 16.470, eps 0.02, speed 148.10 f/s
578056: done 307 games, mean reward 16.390, eps 0.02, speed 145.86 f/s
580322: done 308 games, mean reward 16.370, eps 0.02, speed 145.06 f/s
582460: done 309 games, mean reward 16.330, eps 0.02, speed 144.14 f/s
584265: done 310 games, mean reward 16.340, eps 0.02, speed 145.24 f/s
586174: done 311 games, mean reward 16.370, eps 0.02, speed 146.29 f/s
588431: done 312 games, mean reward 16.320, eps 0.02, speed 146.58 f/s
590439: done 313 games, mean reward 16.310, eps 0.02, speed 149.68 f/s
592118: done 314 games, mean reward 16.310, eps 0.02, speed 145.22 f/s
593785: done 315 games, mean reward 16.390, eps 0.02, speed 145.99 f/s
595724: done 316 games, mean reward 16.450, eps 0.02, speed 147.14 f/s
597614: done 317 games, mean reward 16.450, eps 0.02, speed 146.44 f/s
599954: done 318 games, mean reward 16.420, eps 0.02, speed 148.44 f/s
602655: done 319 games, mean reward 16.270, eps 0.02, speed 142.36 f/s
604938: done 320 games, mean reward 16.220, eps 0.02, speed 140.26 f/s
607303: done 321 games, mean reward 16.220, eps 0.02, speed 142.72 f/s
609489: done 322 games, mean reward 16.320, eps 0.02, speed 143.17 f/s
611906: done 323 games, mean reward 16.130, eps 0.02, speed 136.35 f/s
614460: done 324 games, mean reward 16.030, eps 0.02, speed 143.55 f/s

616728: done 325 games, mean reward 15.970, eps 0.02, speed 141.60 f/s
618789: done 326 games, mean reward 15.950, eps 0.02, speed 137.40 f/s
620638: done 327 games, mean reward 15.920, eps 0.02, speed 138.56 f/s
622550: done 328 games, mean reward 15.900, eps 0.02, speed 139.80 f/s
624526: done 329 games, mean reward 15.930, eps 0.02, speed 132.70 f/s
626597: done 330 games, mean reward 15.900, eps 0.02, speed 137.90 f/s
628592: done 331 games, mean reward 15.860, eps 0.02, speed 135.49 f/s
630414: done 332 games, mean reward 15.850, eps 0.02, speed 137.06 f/s
632665: done 333 games, mean reward 15.810, eps 0.02, speed 144.19 f/s
634361: done 334 games, mean reward 15.900, eps 0.02, speed 139.06 f/s
636346: done 335 games, mean reward 15.940, eps 0.02, speed 134.76 f/s
638130: done 336 games, mean reward 15.960, eps 0.02, speed 129.65 f/s
639788: done 337 games, mean reward 15.960, eps 0.02, speed 135.57 f/s
641480: done 338 games, mean reward 15.970, eps 0.02, speed 136.58 f/s
643730: done 339 games, mean reward 15.930, eps 0.02, speed 136.20 f/s
645595: done 340 games, mean reward 15.940, eps 0.02, speed 130.97 f/s
647580: done 341 games, mean reward 15.920, eps 0.02, speed 129.85 f/s
649710: done 342 games, mean reward 15.890, eps 0.02, speed 141.08 f/s
651339: done 343 games, mean reward 15.910, eps 0.02, speed 135.90 f/s
653420: done 344 games, mean reward 15.910, eps 0.02, speed 135.42 f/s
655087: done 345 games, mean reward 15.910, eps 0.02, speed 138.51 f/s
656909: done 346 games, mean reward 15.910, eps 0.02, speed 141.38 f/s
658980: done 347 games, mean reward 15.870, eps 0.02, speed 139.71 f/s
661345: done 348 games, mean reward 15.780, eps 0.02, speed 143.30 f/s
663066: done 349 games, mean reward 15.810, eps 0.02, speed 143.10 f/s
664733: done 350 games, mean reward 15.830, eps 0.02, speed 141.71 f/s
666858: done 351 games, mean reward 15.780, eps 0.02, speed 138.66 f/s
668555: done 352 games, mean reward 15.780, eps 0.02, speed 142.91 f/s
670607: done 353 games, mean reward 15.770, eps 0.02, speed 134.83 f/s
672478: done 354 games, mean reward 15.790, eps 0.02, speed 141.71 f/s
674176: done 355 games, mean reward 15.950, eps 0.02, speed 134.80 f/s
676029: done 356 games, mean reward 15.940, eps 0.02, speed 132.57 f/s
677761: done 357 games, mean reward 15.970, eps 0.02, speed 141.63 f/s
679579: done 358 games, mean reward 15.960, eps 0.02, speed 141.21 f/s
681847: done 359 games, mean reward 15.890, eps 0.02, speed 143.98 f/s
683835: done 360 games, mean reward 15.900, eps 0.02, speed 143.19 f/s
686059: done 361 games, mean reward 15.890, eps 0.02, speed 143.73 f/s
688078: done 362 games, mean reward 15.880, eps 0.02, speed 143.93 f/s
689799: done 363 games, mean reward 15.870, eps 0.02, speed 141.69 f/s
691498: done 364 games, mean reward 15.870, eps 0.02, speed 141.41 f/s
693248: done 365 games, mean reward 15.960, eps 0.02, speed 136.58 f/s
695106: done 366 games, mean reward 16.030, eps 0.02, speed 140.48 f/s
697402: done 367 games, mean reward 15.970, eps 0.02, speed 141.57 f/s
699155: done 368 games, mean reward 16.040, eps 0.02, speed 136.37 f/s

701139: done 369 games, mean reward 16.010, eps 0.02, speed 141.85 f/s
702806: done 370 games, mean reward 16.010, eps 0.02, speed 135.30 f/s
704804: done 371 games, mean reward 16.050, eps 0.02, speed 137.83 f/s
706532: done 372 games, mean reward 16.070, eps 0.02, speed 148.43 f/s
708552: done 373 games, mean reward 16.050, eps 0.02, speed 143.64 f/s
710176: done 374 games, mean reward 16.100, eps 0.02, speed 148.78 f/s
711964: done 375 games, mean reward 16.250, eps 0.02, speed 143.99 f/s
713870: done 376 games, mean reward 16.260, eps 0.02, speed 144.68 f/s
715498: done 377 games, mean reward 16.270, eps 0.02, speed 147.46 f/s
717448: done 378 games, mean reward 16.300, eps 0.02, speed 151.44 f/s
719725: done 379 games, mean reward 16.220, eps 0.02, speed 150.36 f/s
721918: done 380 games, mean reward 16.200, eps 0.02, speed 143.56 f/s
723844: done 381 games, mean reward 16.190, eps 0.02, speed 147.94 f/s
725987: done 382 games, mean reward 16.100, eps 0.02, speed 144.55 f/s
727867: done 383 games, mean reward 16.230, eps 0.02, speed 147.48 f/s
729562: done 384 games, mean reward 16.320, eps 0.02, speed 149.10 f/s
731637: done 385 games, mean reward 16.270, eps 0.02, speed 140.10 f/s
733649: done 386 games, mean reward 16.270, eps 0.02, speed 145.02 f/s
735643: done 387 games, mean reward 16.240, eps 0.02, speed 151.74 f/s
737805: done 388 games, mean reward 16.210, eps 0.02, speed 149.25 f/s
740123: done 389 games, mean reward 16.180, eps 0.02, speed 148.36 f/s
742269: done 390 games, mean reward 16.160, eps 0.02, speed 146.37 f/s
744000: done 391 games, mean reward 16.280, eps 0.02, speed 148.85 f/s
746145: done 392 games, mean reward 16.260, eps 0.02, speed 148.02 f/s
748696: done 393 games, mean reward 16.240, eps 0.02, speed 144.72 f/s
750638: done 394 games, mean reward 16.220, eps 0.02, speed 147.25 f/s
752939: done 395 games, mean reward 16.180, eps 0.02, speed 149.87 f/s
754997: done 396 games, mean reward 16.140, eps 0.02, speed 149.04 f/s
757079: done 397 games, mean reward 16.120, eps 0.02, speed 142.95 f/s
759461: done 398 games, mean reward 16.070, eps 0.02, speed 126.90 f/s
761588: done 399 games, mean reward 16.020, eps 0.02, speed 139.31 f/s
763554: done 400 games, mean reward 16.000, eps 0.02, speed 139.64 f/s

Traceback (most recent call last):

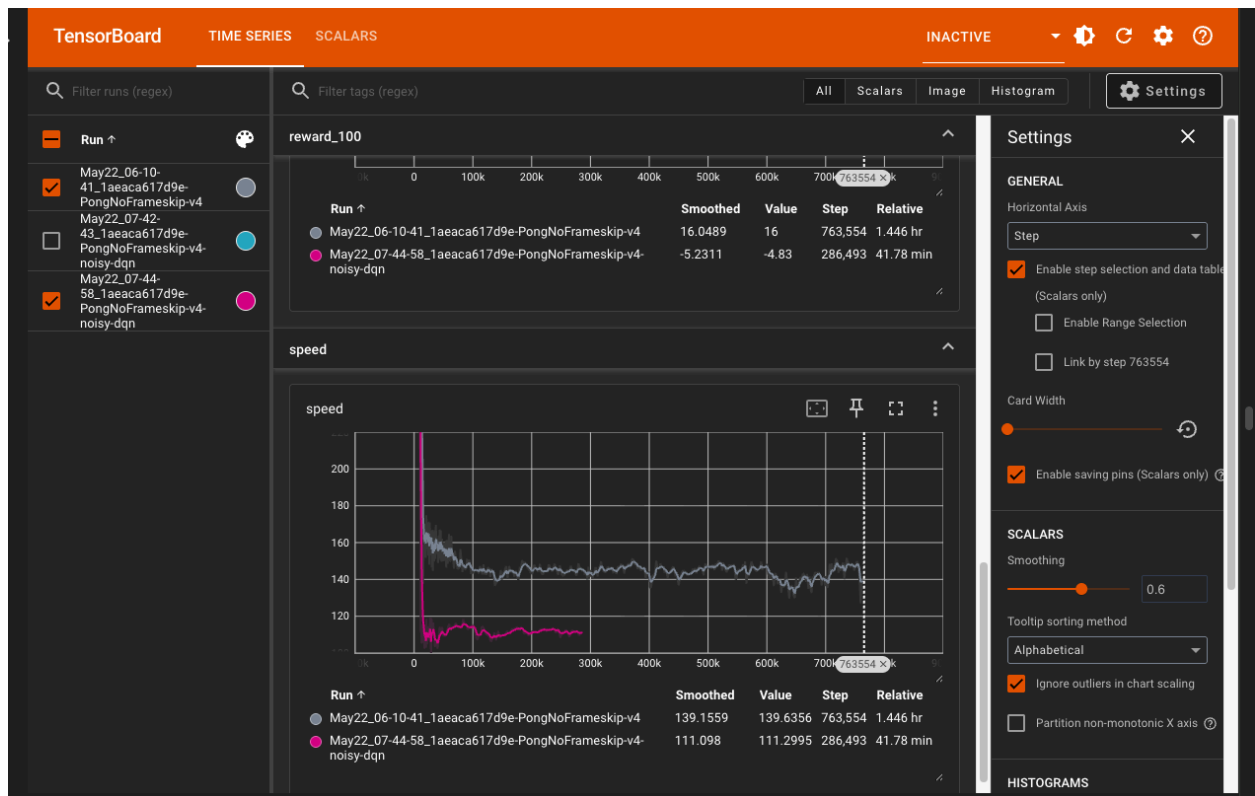
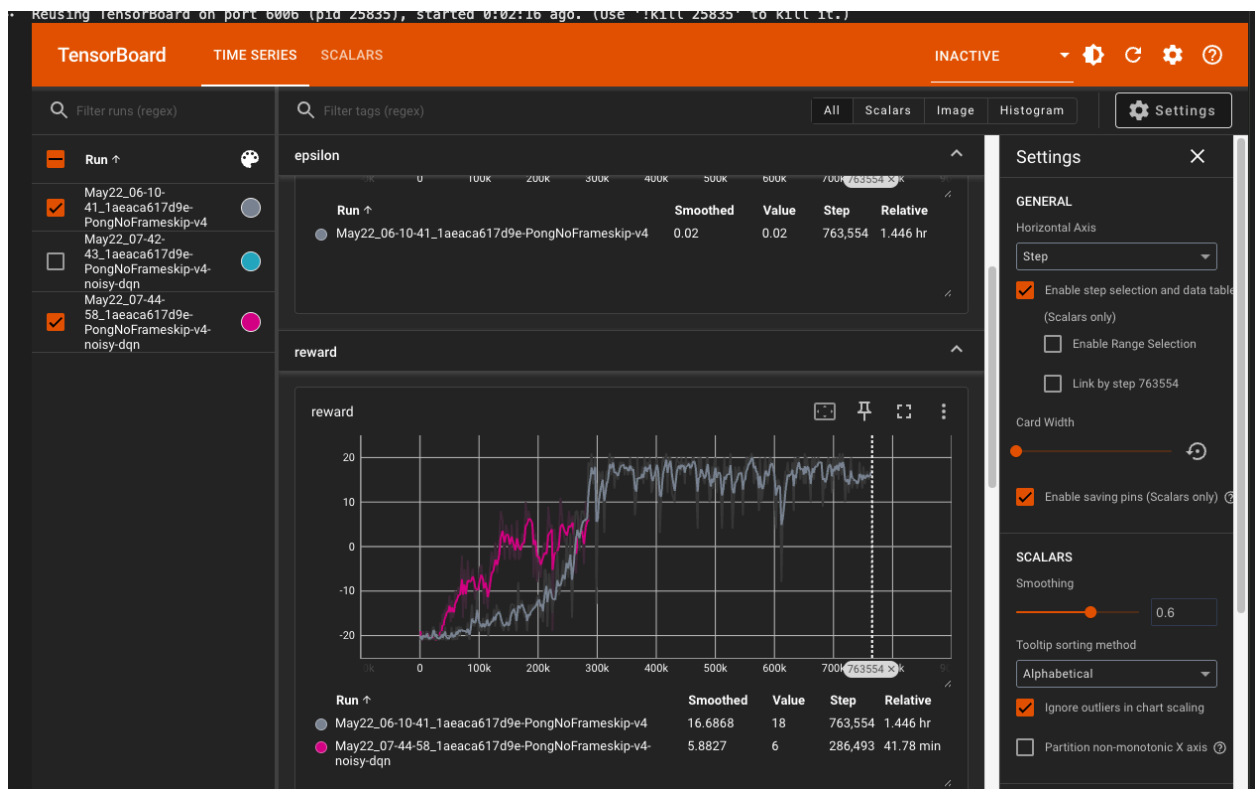
File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/02_dqn_pong.py", line 174,
in <module>

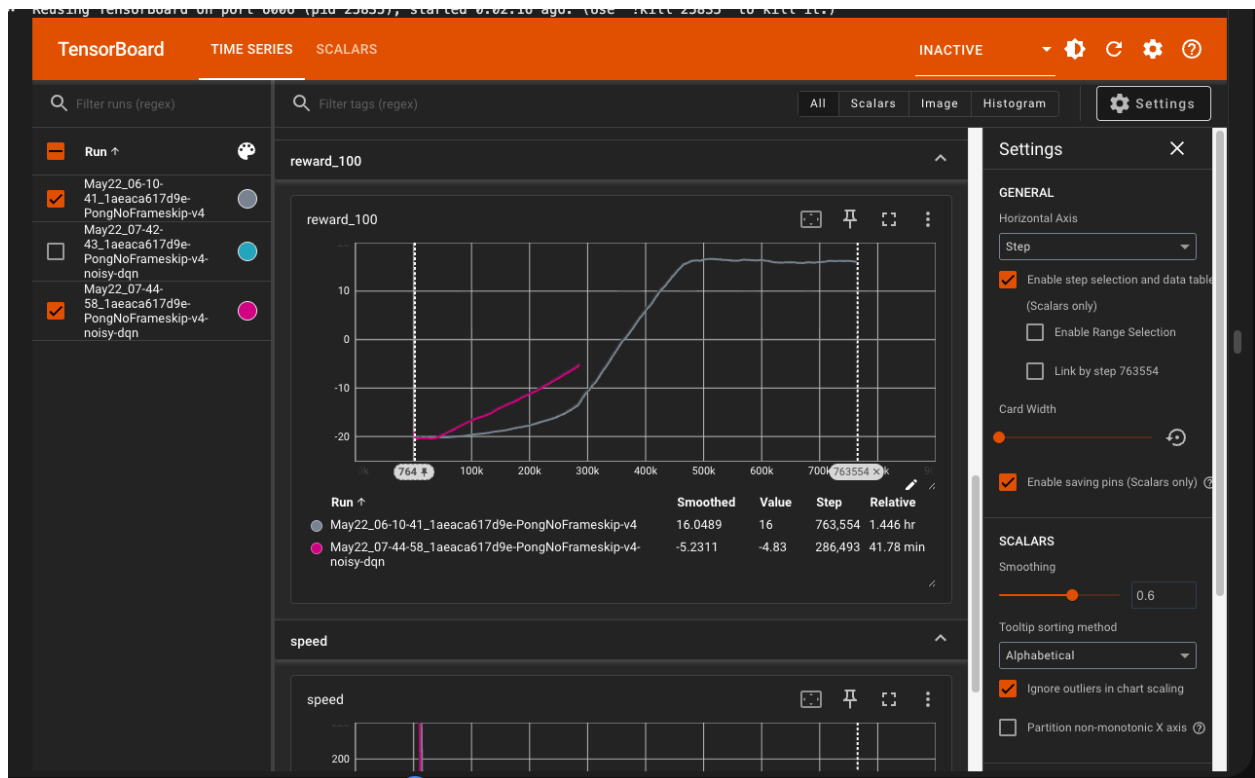
```
loss_t = calc_loss(batch, net, tgt_net, device=device)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/02_dqn_pong.py", line 96,
in calc_loss

```
next_states_v = torch.tensor(next_states).to(device)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

KeyboardInterrupt





A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)

[Powered by Stella]

NoisyDQN(

(conv): Sequential(

(0): Conv2d(1, 32, kernel_size=(8, 8), stride=(4, 4))

(1): ReLU()

(2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))

(3): ReLU()

(4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))

(5): ReLU()

)

(fc): Sequential(

(0): NoisyLinear()

(1): ReLU()

(2): NoisyLinear()

)

)

1008: done 1 games, mean reward -19.000, speed 307.32 f/s

Best mean reward updated -99.000 -> -19.000, model saved

1772: done 2 games, mean reward -20.000, speed 380.57 f/s

2536: done 3 games, mean reward -20.333, speed 386.87 f/s

3300: done 4 games, mean reward -20.500, speed 394.65 f/s

4064: done 5 games, mean reward -20.600, speed 399.02 f/s

4856: done 6 games, mean reward -20.667, speed 288.27 f/s
5864: done 7 games, mean reward -20.429, speed 356.97 f/s
6813: done 8 games, mean reward -20.375, speed 388.81 f/s
7577: done 9 games, mean reward -20.444, speed 391.63 f/s
8558: done 10 games, mean reward -20.400, speed 370.23 f/s
9507: done 11 games, mean reward -20.364, speed 298.56 f/s
10349: done 12 games, mean reward -20.333, speed 175.72 f/s
11173: done 13 games, mean reward -20.385, speed 111.37 f/s
12257: done 14 games, mean reward -20.357, speed 111.13 f/s
13077: done 15 games, mean reward -20.400, speed 105.05 f/s
13901: done 16 games, mean reward -20.438, speed 115.61 f/s
14838: done 17 games, mean reward -20.353, speed 105.93 f/s
15770: done 18 games, mean reward -20.333, speed 106.64 f/s
16553: done 19 games, mean reward -20.368, speed 113.38 f/s
17503: done 20 games, mean reward -20.350, speed 102.35 f/s
18344: done 21 games, mean reward -20.381, speed 109.05 f/s
19262: done 22 games, mean reward -20.364, speed 108.14 f/s
20188: done 23 games, mean reward -20.348, speed 102.88 f/s
21031: done 24 games, mean reward -20.375, speed 118.67 f/s
21901: done 25 games, mean reward -20.360, speed 104.69 f/s
22712: done 26 games, mean reward -20.385, speed 114.63 f/s
23554: done 27 games, mean reward -20.370, speed 106.92 f/s
24378: done 28 games, mean reward -20.393, speed 106.75 f/s
25352: done 29 games, mean reward -20.414, speed 114.18 f/s
26255: done 30 games, mean reward -20.433, speed 107.26 f/s
27094: done 31 games, mean reward -20.452, speed 119.11 f/s
27977: done 32 games, mean reward -20.469, speed 108.39 f/s
28958: done 33 games, mean reward -20.455, speed 98.07 f/s
29741: done 34 games, mean reward -20.471, speed 119.06 f/s
30619: done 35 games, mean reward -20.457, speed 109.11 f/s
31854: done 36 games, mean reward -20.444, speed 110.85 f/s
32817: done 37 games, mean reward -20.459, speed 116.62 f/s
34348: done 38 games, mean reward -20.368, speed 107.70 f/s
35419: done 39 games, mean reward -20.359, speed 103.27 f/s
36669: done 40 games, mean reward -20.325, speed 105.50 f/s
38293: done 41 games, mean reward -20.195, speed 107.65 f/s
39999: done 42 games, mean reward -20.143, speed 102.35 f/s
41827: done 43 games, mean reward -20.000, speed 109.29 f/s
43621: done 44 games, mean reward -19.886, speed 114.25 f/s
45839: done 45 games, mean reward -19.733, speed 110.67 f/s
47780: done 46 games, mean reward -19.630, speed 114.15 f/s
49960: done 47 games, mean reward -19.447, speed 108.95 f/s
52030: done 48 games, mean reward -19.396, speed 109.49 f/s
54128: done 49 games, mean reward -19.245, speed 109.27 f/s

55877: done 50 games, mean reward -19.180, speed 111.74 f/s
57509: done 51 games, mean reward -19.098, speed 114.00 f/s
59638: done 52 games, mean reward -18.923, speed 115.90 f/s
Best mean reward updated -19.000 -> -18.923, model saved
62172: done 53 games, mean reward -18.774, speed 112.26 f/s
Best mean reward updated -18.923 -> -18.774, model saved
65204: done 54 games, mean reward -18.593, speed 114.70 f/s
Best mean reward updated -18.774 -> -18.593, model saved
67828: done 55 games, mean reward -18.455, speed 114.13 f/s
Best mean reward updated -18.593 -> -18.455, model saved
71159: done 56 games, mean reward -18.161, speed 116.41 f/s
Best mean reward updated -18.455 -> -18.161, model saved
72580: done 57 games, mean reward -18.123, speed 114.61 f/s
Best mean reward updated -18.161 -> -18.123, model saved
75336: done 58 games, mean reward -17.897, speed 113.83 f/s
Best mean reward updated -18.123 -> -17.897, model saved
77674: done 59 games, mean reward -17.831, speed 114.16 f/s
Best mean reward updated -17.897 -> -17.831, model saved
80792: done 60 games, mean reward -17.583, speed 113.49 f/s
Best mean reward updated -17.831 -> -17.583, model saved
83335: done 61 games, mean reward -17.426, speed 118.42 f/s
Best mean reward updated -17.583 -> -17.426, model saved
86264: done 62 games, mean reward -17.274, speed 115.65 f/s
Best mean reward updated -17.426 -> -17.274, model saved
89101: done 63 games, mean reward -17.079, speed 114.56 f/s
Best mean reward updated -17.274 -> -17.079, model saved
91527: done 64 games, mean reward -16.953, speed 112.60 f/s
Best mean reward updated -17.079 -> -16.953, model saved
93857: done 65 games, mean reward -16.877, speed 114.29 f/s
Best mean reward updated -16.953 -> -16.877, model saved
96144: done 66 games, mean reward -16.773, speed 113.38 f/s
Best mean reward updated -16.877 -> -16.773, model saved
98296: done 67 games, mean reward -16.672, speed 114.68 f/s
Best mean reward updated -16.773 -> -16.672, model saved
101062: done 68 games, mean reward -16.500, speed 115.15 f/s
Best mean reward updated -16.672 -> -16.500, model saved
103773: done 69 games, mean reward -16.319, speed 111.68 f/s
Best mean reward updated -16.500 -> -16.319, model saved
106222: done 70 games, mean reward -16.200, speed 107.64 f/s
Best mean reward updated -16.319 -> -16.200, model saved
108329: done 71 games, mean reward -16.155, speed 107.13 f/s
Best mean reward updated -16.200 -> -16.155, model saved
110936: done 72 games, mean reward -16.014, speed 110.62 f/s
Best mean reward updated -16.155 -> -16.014, model saved

113057: done 73 games, mean reward -15.959, speed 109.35 f/s
Best mean reward updated -16.014 -> -15.959, model saved
114826: done 74 games, mean reward -15.932, speed 114.45 f/s
Best mean reward updated -15.959 -> -15.932, model saved
117920: done 75 games, mean reward -15.800, speed 114.18 f/s
Best mean reward updated -15.932 -> -15.800, model saved
121221: done 76 games, mean reward -15.605, speed 110.56 f/s
Best mean reward updated -15.800 -> -15.605, model saved
124162: done 77 games, mean reward -15.468, speed 109.87 f/s
Best mean reward updated -15.605 -> -15.468, model saved
127865: done 78 games, mean reward -15.256, speed 108.84 f/s
Best mean reward updated -15.468 -> -15.256, model saved
130624: done 79 games, mean reward -15.139, speed 106.01 f/s
Best mean reward updated -15.256 -> -15.139, model saved
133978: done 80 games, mean reward -14.938, speed 108.12 f/s
Best mean reward updated -15.139 -> -14.938, model saved
136692: done 81 games, mean reward -14.654, speed 111.25 f/s
Best mean reward updated -14.938 -> -14.654, model saved
139756: done 82 games, mean reward -14.402, speed 106.81 f/s
Best mean reward updated -14.654 -> -14.402, model saved
143022: done 83 games, mean reward -14.265, speed 112.05 f/s
Best mean reward updated -14.402 -> -14.265, model saved
145631: done 84 games, mean reward -14.036, speed 108.22 f/s
Best mean reward updated -14.265 -> -14.036, model saved
149042: done 85 games, mean reward -13.835, speed 110.05 f/s
Best mean reward updated -14.036 -> -13.835, model saved
152231: done 86 games, mean reward -13.709, speed 109.79 f/s
Best mean reward updated -13.835 -> -13.709, model saved
155202: done 87 games, mean reward -13.494, speed 109.97 f/s
Best mean reward updated -13.709 -> -13.494, model saved
158393: done 88 games, mean reward -13.364, speed 112.03 f/s
Best mean reward updated -13.494 -> -13.364, model saved
161338: done 89 games, mean reward -13.247, speed 110.75 f/s
Best mean reward updated -13.364 -> -13.247, model saved
164247: done 90 games, mean reward -13.033, speed 111.77 f/s
Best mean reward updated -13.247 -> -13.033, model saved
167309: done 91 games, mean reward -12.923, speed 113.34 f/s
Best mean reward updated -13.033 -> -12.923, model saved
170322: done 92 games, mean reward -12.804, speed 112.91 f/s
Best mean reward updated -12.923 -> -12.804, model saved
173303: done 93 games, mean reward -12.634, speed 113.76 f/s
Best mean reward updated -12.804 -> -12.634, model saved
176349: done 94 games, mean reward -12.489, speed 114.11 f/s
Best mean reward updated -12.634 -> -12.489, model saved

178870: done 95 games, mean reward -12.253, speed 112.25 f/s
Best mean reward updated -12.489 -> -12.253, model saved
181491: done 96 games, mean reward -12.062, speed 114.20 f/s
Best mean reward updated -12.253 -> -12.062, model saved
184192: done 97 games, mean reward -11.856, speed 113.38 f/s
Best mean reward updated -12.062 -> -11.856, model saved
186915: done 98 games, mean reward -11.673, speed 110.35 f/s
Best mean reward updated -11.856 -> -11.673, model saved
190154: done 99 games, mean reward -11.515, speed 110.83 f/s
Best mean reward updated -11.673 -> -11.515, model saved
192369: done 100 games, mean reward -11.470, speed 110.42 f/s
Best mean reward updated -11.515 -> -11.470, model saved
195490: done 101 games, mean reward -11.250, speed 109.39 f/s
Best mean reward updated -11.470 -> -11.250, model saved
198101: done 102 games, mean reward -11.120, speed 110.88 f/s
Best mean reward updated -11.250 -> -11.120, model saved
201432: done 103 games, mean reward -10.930, speed 112.05 f/s
Best mean reward updated -11.120 -> -10.930, model saved
204324: done 104 games, mean reward -10.790, speed 111.75 f/s
Best mean reward updated -10.930 -> -10.790, model saved
207443: done 105 games, mean reward -10.590, speed 111.19 f/s
Best mean reward updated -10.790 -> -10.590, model saved
211133: done 106 games, mean reward -10.370, speed 111.17 f/s
Best mean reward updated -10.590 -> -10.370, model saved
214424: done 107 games, mean reward -10.200, speed 113.09 f/s
Best mean reward updated -10.370 -> -10.200, model saved
217214: done 108 games, mean reward -9.910, speed 112.97 f/s
Best mean reward updated -10.200 -> -9.910, model saved
220433: done 109 games, mean reward -9.660, speed 113.01 f/s
Best mean reward updated -9.910 -> -9.660, model saved
223277: done 110 games, mean reward -9.410, speed 113.50 f/s
Best mean reward updated -9.660 -> -9.410, model saved
224408: done 111 games, mean reward -9.400, speed 111.58 f/s
Best mean reward updated -9.410 -> -9.400, model saved
227764: done 112 games, mean reward -9.160, speed 112.91 f/s
Best mean reward updated -9.400 -> -9.160, model saved
231608: done 113 games, mean reward -8.960, speed 112.35 f/s
Best mean reward updated -9.160 -> -8.960, model saved
235015: done 114 games, mean reward -8.740, speed 113.55 f/s
Best mean reward updated -8.960 -> -8.740, model saved
237930: done 115 games, mean reward -8.420, speed 113.45 f/s
Best mean reward updated -8.740 -> -8.420, model saved
241071: done 116 games, mean reward -8.150, speed 110.81 f/s
Best mean reward updated -8.420 -> -8.150, model saved

245114: done 117 games, mean reward -7.950, speed 113.15 f/s
Best mean reward updated -8.150 -> -7.950, model saved

248694: done 118 games, mean reward -7.740, speed 108.18 f/s
Best mean reward updated -7.950 -> -7.740, model saved

252191: done 119 games, mean reward -7.450, speed 110.79 f/s
Best mean reward updated -7.740 -> -7.450, model saved

255491: done 120 games, mean reward -7.210, speed 111.39 f/s
Best mean reward updated -7.450 -> -7.210, model saved

258422: done 121 games, mean reward -6.930, speed 112.82 f/s
Best mean reward updated -7.210 -> -6.930, model saved

261670: done 122 games, mean reward -6.780, speed 110.72 f/s
Best mean reward updated -6.930 -> -6.780, model saved

264407: done 123 games, mean reward -6.640, speed 108.57 f/s
Best mean reward updated -6.780 -> -6.640, model saved

267724: done 124 games, mean reward -6.360, speed 112.46 f/s
Best mean reward updated -6.640 -> -6.360, model saved

271019: done 125 games, mean reward -6.120, speed 111.92 f/s
Best mean reward updated -6.360 -> -6.120, model saved

274387: done 126 games, mean reward -5.860, speed 111.52 f/s
Best mean reward updated -6.120 -> -5.860, model saved

277565: done 127 games, mean reward -5.700, speed 112.26 f/s
Best mean reward updated -5.860 -> -5.700, model saved

280625: done 128 games, mean reward -5.390, speed 109.40 f/s
Best mean reward updated -5.700 -> -5.390, model saved

283443: done 129 games, mean reward -5.100, speed 111.16 f/s
Best mean reward updated -5.390 -> -5.100, model saved

286493: done 130 games, mean reward -4.830, speed 111.30 f/s
Best mean reward updated -5.100 -> -4.830, model saved

289924: done 131 games, mean reward -4.600, speed 111.01 f/s
Best mean reward updated -4.830 -> -4.600, model saved

292815: done 132 games, mean reward -4.290, speed 112.12 f/s
Best mean reward updated -4.600 -> -4.290, model saved

295939: done 133 games, mean reward -4.030, speed 112.02 f/s
Best mean reward updated -4.290 -> -4.030, model saved

298815: done 134 games, mean reward -3.730, speed 111.16 f/s
Best mean reward updated -4.030 -> -3.730, model saved

301587: done 135 games, mean reward -3.440, speed 110.46 f/s
Best mean reward updated -3.730 -> -3.440, model saved

304018: done 136 games, mean reward -3.130, speed 115.52 f/s
Best mean reward updated -3.440 -> -3.130, model saved

306357: done 137 games, mean reward -2.790, speed 110.04 f/s
Best mean reward updated -3.130 -> -2.790, model saved

308539: done 138 games, mean reward -2.490, speed 110.60 f/s
Best mean reward updated -2.790 -> -2.490, model saved

310906: done 139 games, mean reward -2.160, speed 110.46 f/s
Best mean reward updated -2.490 -> -2.160, model saved

312892: done 140 games, mean reward -1.810, speed 109.98 f/s
Best mean reward updated -2.160 -> -1.810, model saved

315037: done 141 games, mean reward -1.540, speed 110.23 f/s
Best mean reward updated -1.810 -> -1.540, model saved

317199: done 142 games, mean reward -1.220, speed 108.84 f/s
Best mean reward updated -1.540 -> -1.220, model saved

319196: done 143 games, mean reward -0.920, speed 110.65 f/s
Best mean reward updated -1.220 -> -0.920, model saved

321042: done 144 games, mean reward -0.590, speed 111.48 f/s
Best mean reward updated -0.920 -> -0.590, model saved

322988: done 145 games, mean reward -0.300, speed 106.82 f/s
Best mean reward updated -0.590 -> -0.300, model saved

324819: done 146 games, mean reward 0.030, speed 111.20 f/s
Best mean reward updated -0.300 -> 0.030, model saved

326637: done 147 games, mean reward 0.330, speed 111.89 f/s
Best mean reward updated 0.030 -> 0.330, model saved

328669: done 148 games, mean reward 0.670, speed 105.77 f/s
Best mean reward updated 0.330 -> 0.670, model saved

330659: done 149 games, mean reward 0.960, speed 109.84 f/s
Best mean reward updated 0.670 -> 0.960, model saved

332723: done 150 games, mean reward 1.280, speed 109.65 f/s
Best mean reward updated 0.960 -> 1.280, model saved

334431: done 151 games, mean reward 1.630, speed 106.55 f/s
Best mean reward updated 1.280 -> 1.630, model saved

336130: done 152 games, mean reward 1.930, speed 112.03 f/s
Best mean reward updated 1.630 -> 1.930, model saved

337913: done 153 games, mean reward 2.230, speed 112.58 f/s
Best mean reward updated 1.930 -> 2.230, model saved

339772: done 154 games, mean reward 2.510, speed 113.79 f/s
Best mean reward updated 2.230 -> 2.510, model saved

341579: done 155 games, mean reward 2.820, speed 107.79 f/s
Best mean reward updated 2.510 -> 2.820, model saved

343215: done 156 games, mean reward 3.050, speed 111.44 f/s
Best mean reward updated 2.820 -> 3.050, model saved

344923: done 157 games, mean reward 3.410, speed 112.90 f/s
Best mean reward updated 3.050 -> 3.410, model saved

346559: done 158 games, mean reward 3.670, speed 112.86 f/s
Best mean reward updated 3.410 -> 3.670, model saved

348484: done 159 games, mean reward 3.980, speed 112.07 f/s
Best mean reward updated 3.670 -> 3.980, model saved

350677: done 160 games, mean reward 4.110, speed 110.08 f/s
Best mean reward updated 3.980 -> 4.110, model saved

352412: done 161 games, mean reward 4.380, speed 113.62 f/s
Best mean reward updated 4.110 -> 4.380, model saved

354148: done 162 games, mean reward 4.650, speed 112.28 f/s
Best mean reward updated 4.380 -> 4.650, model saved

355957: done 163 games, mean reward 4.900, speed 107.61 f/s
Best mean reward updated 4.650 -> 4.900, model saved

357628: done 164 games, mean reward 5.190, speed 113.65 f/s
Best mean reward updated 4.900 -> 5.190, model saved

359399: done 165 games, mean reward 5.490, speed 110.79 f/s
Best mean reward updated 5.190 -> 5.490, model saved

361070: done 166 games, mean reward 5.790, speed 110.26 f/s
Best mean reward updated 5.490 -> 5.790, model saved

363008: done 167 games, mean reward 6.050, speed 110.69 f/s
Best mean reward updated 5.790 -> 6.050, model saved

365258: done 168 games, mean reward 6.150, speed 106.03 f/s
Best mean reward updated 6.050 -> 6.150, model saved

366993: done 169 games, mean reward 6.380, speed 107.25 f/s
Best mean reward updated 6.150 -> 6.380, model saved

369070: done 170 games, mean reward 6.580, speed 103.16 f/s
Best mean reward updated 6.380 -> 6.580, model saved

371373: done 171 games, mean reward 6.760, speed 109.27 f/s
Best mean reward updated 6.580 -> 6.760, model saved

373748: done 172 games, mean reward 6.860, speed 111.18 f/s
Best mean reward updated 6.760 -> 6.860, model saved

376336: done 173 games, mean reward 6.990, speed 109.51 f/s
Best mean reward updated 6.860 -> 6.990, model saved

378491: done 174 games, mean reward 7.280, speed 110.63 f/s
Best mean reward updated 6.990 -> 7.280, model saved

380162: done 175 games, mean reward 7.540, speed 108.75 f/s
Best mean reward updated 7.280 -> 7.540, model saved

382042: done 176 games, mean reward 7.720, speed 112.51 f/s
Best mean reward updated 7.540 -> 7.720, model saved

383890: done 177 games, mean reward 7.950, speed 112.33 f/s
Best mean reward updated 7.720 -> 7.950, model saved

385657: done 178 games, mean reward 8.130, speed 109.87 f/s
Best mean reward updated 7.950 -> 8.130, model saved

387293: done 179 games, mean reward 8.400, speed 106.49 f/s
Best mean reward updated 8.130 -> 8.400, model saved

389088: done 180 games, mean reward 8.580, speed 112.81 f/s
Best mean reward updated 8.400 -> 8.580, model saved

390926: done 181 games, mean reward 8.680, speed 111.61 f/s
Best mean reward updated 8.580 -> 8.680, model saved

392829: done 182 games, mean reward 8.770, speed 111.34 f/s
Best mean reward updated 8.680 -> 8.770, model saved

394545: done 183 games, mean reward 9.000, speed 107.49 f/s
Best mean reward updated 8.770 -> 9.000, model saved

396627: done 184 games, mean reward 9.080, speed 104.36 f/s
Best mean reward updated 9.000 -> 9.080, model saved

398699: done 185 games, mean reward 9.210, speed 107.82 f/s
Best mean reward updated 9.080 -> 9.210, model saved

400398: done 186 games, mean reward 9.440, speed 109.58 f/s
Best mean reward updated 9.210 -> 9.440, model saved

402160: done 187 games, mean reward 9.580, speed 105.68 f/s
Best mean reward updated 9.440 -> 9.580, model saved

403859: done 188 games, mean reward 9.800, speed 107.57 f/s
Best mean reward updated 9.580 -> 9.800, model saved

406000: done 189 games, mean reward 9.950, speed 107.62 f/s
Best mean reward updated 9.800 -> 9.950, model saved

407636: done 190 games, mean reward 10.100, speed 108.54 f/s
Best mean reward updated 9.950 -> 10.100, model saved

409902: done 191 games, mean reward 10.230, speed 107.43 f/s
Best mean reward updated 10.100 -> 10.230, model saved

411771: done 192 games, mean reward 10.410, speed 111.41 f/s
Best mean reward updated 10.230 -> 10.410, model saved

414524: done 193 games, mean reward 10.420, speed 109.40 f/s
Best mean reward updated 10.410 -> 10.420, model saved

416259: done 194 games, mean reward 10.600, speed 110.31 f/s
Best mean reward updated 10.420 -> 10.600, model saved

418048: done 195 games, mean reward 10.680, speed 108.58 f/s
Best mean reward updated 10.600 -> 10.680, model saved

419810: done 196 games, mean reward 10.810, speed 107.28 f/s
Best mean reward updated 10.680 -> 10.810, model saved

421446: done 197 games, mean reward 10.940, speed 112.49 f/s
Best mean reward updated 10.810 -> 10.940, model saved

423316: done 198 games, mean reward 11.060, speed 110.18 f/s
Best mean reward updated 10.940 -> 11.060, model saved

425352: done 199 games, mean reward 11.170, speed 107.96 f/s
Best mean reward updated 11.060 -> 11.170, model saved

427086: done 200 games, mean reward 11.430, speed 107.99 f/s
Best mean reward updated 11.170 -> 11.430, model saved

429003: done 201 games, mean reward 11.560, speed 111.31 f/s
Best mean reward updated 11.430 -> 11.560, model saved

430860: done 202 games, mean reward 11.810, speed 113.02 f/s
Best mean reward updated 11.560 -> 11.810, model saved

432638: done 203 games, mean reward 12.020, speed 107.17 f/s
Best mean reward updated 11.810 -> 12.020, model saved

434714: done 204 games, mean reward 12.250, speed 112.77 f/s
Best mean reward updated 12.020 -> 12.250, model saved

436475: done 205 games, mean reward 12.450, speed 111.11 f/s
Best mean reward updated 12.250 -> 12.450, model saved

438204: done 206 games, mean reward 12.650, speed 110.89 f/s
Best mean reward updated 12.450 -> 12.650, model saved

439952: done 207 games, mean reward 12.870, speed 109.12 f/s
Best mean reward updated 12.650 -> 12.870, model saved

441650: done 208 games, mean reward 12.980, speed 112.89 f/s
Best mean reward updated 12.870 -> 12.980, model saved

443410: done 209 games, mean reward 13.140, speed 112.94 f/s
Best mean reward updated 12.980 -> 13.140, model saved

445340: done 210 games, mean reward 13.240, speed 111.85 f/s
Best mean reward updated 13.140 -> 13.240, model saved

447242: done 211 games, mean reward 13.620, speed 107.75 f/s
Best mean reward updated 13.240 -> 13.620, model saved

448941: done 212 games, mean reward 13.780, speed 107.15 f/s
Best mean reward updated 13.620 -> 13.780, model saved

451795: done 213 games, mean reward 13.830, speed 105.67 f/s
Best mean reward updated 13.780 -> 13.830, model saved

453530: done 214 games, mean reward 14.010, speed 107.62 f/s
Best mean reward updated 13.830 -> 14.010, model saved

455520: done 215 games, mean reward 14.070, speed 106.26 f/s
Best mean reward updated 14.010 -> 14.070, model saved

457479: done 216 games, mean reward 14.170, speed 112.69 f/s
Best mean reward updated 14.070 -> 14.170, model saved

459573: done 217 games, mean reward 14.310, speed 109.51 f/s
Best mean reward updated 14.170 -> 14.310, model saved

461322: done 218 games, mean reward 14.500, speed 108.24 f/s
Best mean reward updated 14.310 -> 14.500, model saved

462958: done 219 games, mean reward 14.630, speed 109.40 f/s
Best mean reward updated 14.500 -> 14.630, model saved

464657: done 220 games, mean reward 14.790, speed 112.13 f/s
Best mean reward updated 14.630 -> 14.790, model saved

466477: done 221 games, mean reward 14.900, speed 111.16 f/s
Best mean reward updated 14.790 -> 14.900, model saved

468340: done 222 games, mean reward 15.100, speed 108.03 f/s
Best mean reward updated 14.900 -> 15.100, model saved

470075: done 223 games, mean reward 15.350, speed 109.83 f/s
Best mean reward updated 15.100 -> 15.350, model saved

472264: done 224 games, mean reward 15.430, speed 111.54 f/s
Best mean reward updated 15.350 -> 15.430, model saved

473963: done 225 games, mean reward 15.590, speed 106.27 f/s
Best mean reward updated 15.430 -> 15.590, model saved

475696: done 226 games, mean reward 15.740, speed 110.15 f/s
Best mean reward updated 15.590 -> 15.740, model saved

477331: done 227 games, mean reward 15.990, speed 108.66 f/s
Best mean reward updated 15.740 -> 15.990, model saved
479410: done 228 games, mean reward 16.050, speed 108.27 f/s
Best mean reward updated 15.990 -> 16.050, model saved
481433: done 229 games, mean reward 16.130, speed 108.75 f/s
Best mean reward updated 16.050 -> 16.130, model saved
483805: done 230 games, mean reward 16.210, speed 109.32 f/s
Best mean reward updated 16.130 -> 16.210, model saved
485441: done 231 games, mean reward 16.400, speed 111.46 f/s
Best mean reward updated 16.210 -> 16.400, model saved
487334: done 232 games, mean reward 16.450, speed 112.60 f/s
Best mean reward updated 16.400 -> 16.450, model saved
489066: done 233 games, mean reward 16.590, speed 111.94 f/s
Best mean reward updated 16.450 -> 16.590, model saved
490736: done 234 games, mean reward 16.700, speed 109.65 f/s
Best mean reward updated 16.590 -> 16.700, model saved
492463: done 235 games, mean reward 16.810, speed 108.76 f/s
Best mean reward updated 16.700 -> 16.810, model saved
494116: done 236 games, mean reward 16.910, speed 111.39 f/s
Best mean reward updated 16.810 -> 16.910, model saved
495752: done 237 games, mean reward 16.990, speed 109.46 f/s
Best mean reward updated 16.910 -> 16.990, model saved
497451: done 238 games, mean reward 17.060, speed 111.33 f/s
Best mean reward updated 16.990 -> 17.060, model saved
499424: done 239 games, mean reward 17.100, speed 109.71 f/s
Best mean reward updated 17.060 -> 17.100, model saved
501059: done 240 games, mean reward 17.150, speed 110.02 f/s
Best mean reward updated 17.100 -> 17.150, model saved
502758: done 241 games, mean reward 17.230, speed 112.16 f/s
Best mean reward updated 17.150 -> 17.230, model saved
504575: done 242 games, mean reward 17.280, speed 112.07 f/s
Best mean reward updated 17.230 -> 17.280, model saved
506631: done 243 games, mean reward 17.260, speed 109.83 f/s
508382: done 244 games, mean reward 17.270, speed 114.07 f/s
510335: done 245 games, mean reward 17.280, speed 114.45 f/s
512222: done 246 games, mean reward 17.260, speed 112.85 f/s
514048: done 247 games, mean reward 17.260, speed 109.98 f/s
516358: done 248 games, mean reward 17.180, speed 114.96 f/s
517994: done 249 games, mean reward 17.220, speed 113.12 f/s
519748: done 250 games, mean reward 17.240, speed 110.17 f/s
521417: done 251 games, mean reward 17.250, speed 112.77 f/s
523053: done 252 games, mean reward 17.260, speed 113.19 f/s
524849: done 253 games, mean reward 17.250, speed 113.84 f/s
526485: done 254 games, mean reward 17.270, speed 113.44 f/s

528279: done 255 games, mean reward 17.260, speed 111.14 f/s
530227: done 256 games, mean reward 17.240, speed 113.64 f/s
531863: done 257 games, mean reward 17.250, speed 113.05 f/s
533499: done 258 games, mean reward 17.250, speed 114.13 f/s
535135: done 259 games, mean reward 17.290, speed 114.45 f/s
Best mean reward updated 17.280 -> 17.290, model saved
536771: done 260 games, mean reward 17.400, speed 113.19 f/s
Best mean reward updated 17.290 -> 17.400, model saved
538598: done 261 games, mean reward 17.400, speed 108.63 f/s
540234: done 262 games, mean reward 17.420, speed 112.40 f/s
Best mean reward updated 17.400 -> 17.420, model saved
541870: done 263 games, mean reward 17.430, speed 115.31 f/s
Best mean reward updated 17.420 -> 17.430, model saved
543522: done 264 games, mean reward 17.440, speed 112.73 f/s
Best mean reward updated 17.430 -> 17.440, model saved
545352: done 265 games, mean reward 17.450, speed 115.32 f/s
Best mean reward updated 17.440 -> 17.450, model saved
546988: done 266 games, mean reward 17.460, speed 111.83 f/s
Best mean reward updated 17.450 -> 17.460, model saved
548686: done 267 games, mean reward 17.500, speed 113.03 f/s
Best mean reward updated 17.460 -> 17.500, model saved
550323: done 268 games, mean reward 17.660, speed 114.41 f/s
Best mean reward updated 17.500 -> 17.660, model saved
552054: done 269 games, mean reward 17.670, speed 115.15 f/s
Best mean reward updated 17.660 -> 17.670, model saved
554165: done 270 games, mean reward 17.720, speed 115.53 f/s
Best mean reward updated 17.670 -> 17.720, model saved
555863: done 271 games, mean reward 17.870, speed 113.35 f/s
Best mean reward updated 17.720 -> 17.870, model saved
557563: done 272 games, mean reward 18.030, speed 113.24 f/s
Best mean reward updated 17.870 -> 18.030, model saved
559328: done 273 games, mean reward 18.210, speed 113.37 f/s
Best mean reward updated 18.030 -> 18.210, model saved
561090: done 274 games, mean reward 18.250, speed 114.93 f/s
Best mean reward updated 18.210 -> 18.250, model saved
562726: done 275 games, mean reward 18.260, speed 113.02 f/s
Best mean reward updated 18.250 -> 18.260, model saved
564649: done 276 games, mean reward 18.240, speed 109.72 f/s
566615: done 277 games, mean reward 18.210, speed 114.79 f/s
568303: done 278 games, mean reward 18.220, speed 113.17 f/s
569939: done 279 games, mean reward 18.220, speed 109.16 f/s
571638: done 280 games, mean reward 18.230, speed 109.42 f/s
573337: done 281 games, mean reward 18.250, speed 108.58 f/s
575158: done 282 games, mean reward 18.280, speed 111.93 f/s

Best mean reward updated 18.260 -> 18.280, model saved
576794: done 283 games, mean reward 18.290, speed 113.44 f/s
Best mean reward updated 18.280 -> 18.290, model saved
578430: done 284 games, mean reward 18.370, speed 113.84 f/s
Best mean reward updated 18.290 -> 18.370, model saved
580222: done 285 games, mean reward 18.410, speed 111.77 f/s
Best mean reward updated 18.370 -> 18.410, model saved
582670: done 286 games, mean reward 18.290, speed 112.89 f/s
584556: done 287 games, mean reward 18.290, speed 112.66 f/s
586718: done 288 games, mean reward 18.240, speed 112.08 f/s
588503: done 289 games, mean reward 18.310, speed 112.61 f/s
590771: done 290 games, mean reward 18.210, speed 110.15 f/s
592408: done 291 games, mean reward 18.320, speed 113.87 f/s
594159: done 292 games, mean reward 18.360, speed 113.65 f/s
595795: done 293 games, mean reward 18.530, speed 112.79 f/s
Best mean reward updated 18.410 -> 18.530, model saved
597495: done 294 games, mean reward 18.540, speed 114.04 f/s
Best mean reward updated 18.530 -> 18.540, model saved
599245: done 295 games, mean reward 18.550, speed 111.00 f/s
Best mean reward updated 18.540 -> 18.550, model saved
601416: done 296 games, mean reward 18.470, speed 112.79 f/s
603053: done 297 games, mean reward 18.470, speed 112.91 f/s
604690: done 298 games, mean reward 18.500, speed 113.08 f/s
606561: done 299 games, mean reward 18.510, speed 109.63 f/s
608197: done 300 games, mean reward 18.530, speed 114.17 f/s
610142: done 301 games, mean reward 18.560, speed 113.23 f/s
Best mean reward updated 18.550 -> 18.560, model saved
612030: done 302 games, mean reward 18.560, speed 113.73 f/s
613730: done 303 games, mean reward 18.570, speed 107.87 f/s
Best mean reward updated 18.560 -> 18.570, model saved
615462: done 304 games, mean reward 18.610, speed 112.13 f/s
Best mean reward updated 18.570 -> 18.610, model saved
617098: done 305 games, mean reward 18.630, speed 110.27 f/s
Best mean reward updated 18.610 -> 18.630, model saved
618734: done 306 games, mean reward 18.630, speed 111.23 f/s
620466: done 307 games, mean reward 18.630, speed 111.61 f/s
622256: done 308 games, mean reward 18.630, speed 109.38 f/s

Traceback (most recent call last):

File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/03_noisy_dqn_pong.py",
line 211, in <module>

 reward = agent.play_step(net, device=device)

 ^^

File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/03_noisy_dqn_pong.py",
line 147, in play_step

```
q_vals_v = net(state_t)
^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1776, in
_wrapped_call_impl

```
return self._call_impl(*args, **kwargs)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1787, in
_call_impl

File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/03_noisy_dqn_pong.py",
line 106, in forward

```
conv_out = self.conv(x).view(x.size(0), -1)
^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1776, in
_wrapped_call_impl

```
return self._call_impl(*args, **kwargs)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1787, in
_call_impl

```
return forward_call(*args, **kwargs)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/container.py", line 253, in
forward

```
input = module(input)
^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1776, in
_wrapped_call_impl

```
return self._call_impl(*args, **kwargs)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

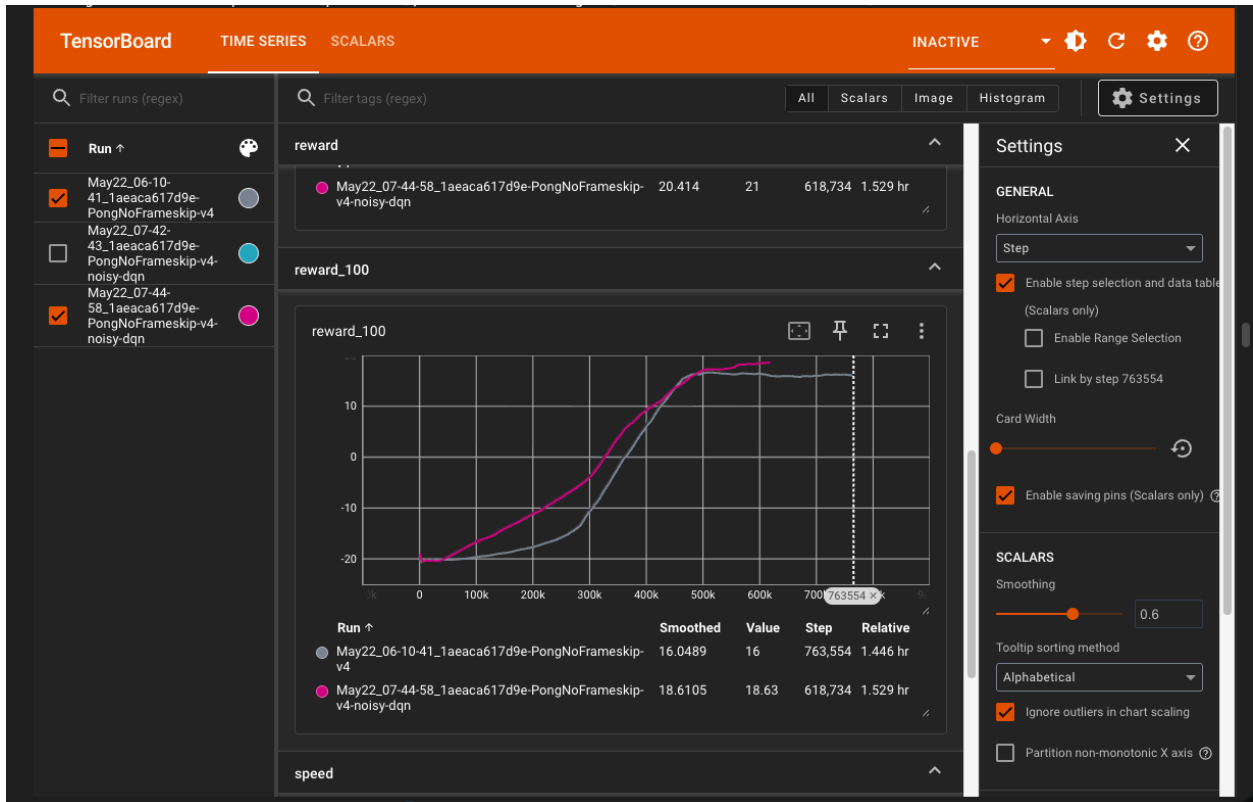
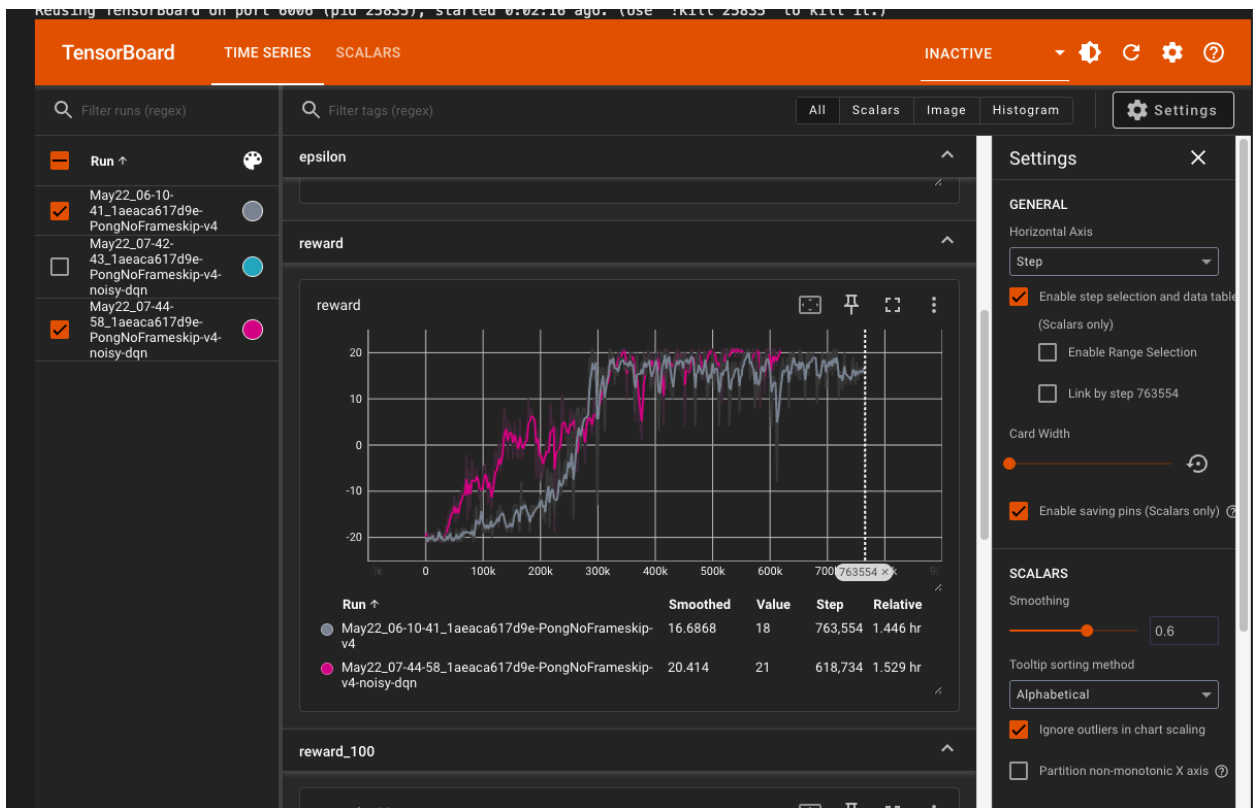
File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1787, in
_call_impl

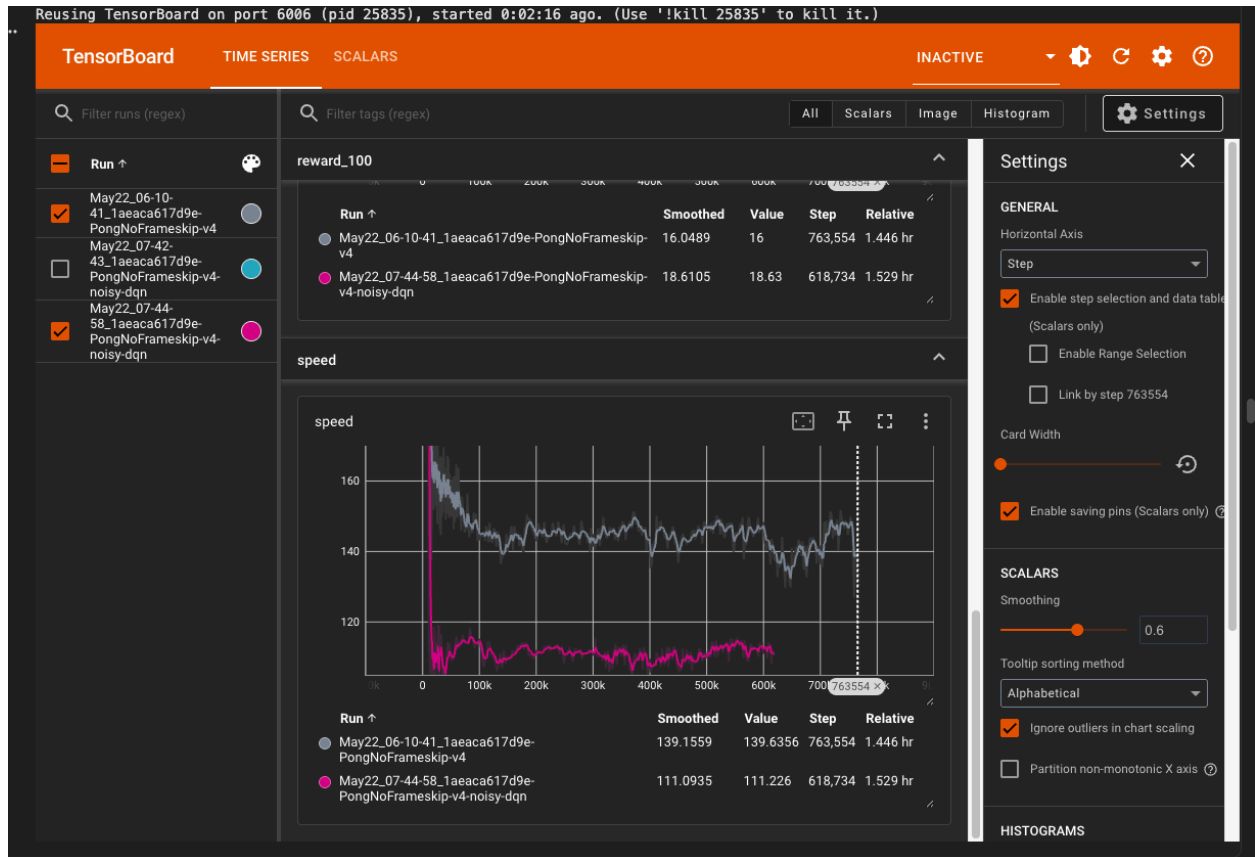
```
return forward_call(*args, **kwargs)
^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/activation.py", line 139, in
forward

```
def forward(self, input: Tensor) -> Tensor:
```

KeyboardInterrupt





A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)

[Powered by Stella]

CategoricalDQN(

(conv): Sequential(

(0): Conv2d(1, 32, kernel_size=(8, 8), stride=(4, 4))

(1): ReLU()

(2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))

(3): ReLU()

(4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))

(5): ReLU()

)

(fc): Sequential(

(0): Linear(in_features=3136, out_features=512, bias=True)

(1): ReLU()

(2): Linear(in_features=512, out_features=306, bias=True)

)

(softmax): Softmax(dim=-1)

)

973: done 1 games, mean reward -19.000, eps 0.99, speed 467.48 f/s

Best mean reward updated -99.000 -> -19.000, model saved

1798: done 2 games, mean reward -20.000, eps 0.97, speed 751.34 f/s

3011: done 3 games, mean reward -19.000, eps 0.96, speed 784.76 f/s

3803: done 4 games, mean reward -19.500, eps 0.95, speed 746.97 f/s
4567: done 5 games, mean reward -19.800, eps 0.93, speed 767.02 f/s
5564: done 6 games, mean reward -19.667, eps 0.92, speed 712.80 f/s
6453: done 7 games, mean reward -19.714, eps 0.91, speed 480.97 f/s
7217: done 8 games, mean reward -19.875, eps 0.90, speed 552.88 f/s
8060: done 9 games, mean reward -20.000, eps 0.88, speed 751.22 f/s
8824: done 10 games, mean reward -20.100, eps 0.87, speed 726.35 f/s
9622: done 11 games, mean reward -20.182, eps 0.86, speed 748.45 f/s
10492: done 12 games, mean reward -20.167, eps 0.85, speed 102.87 f/s
11669: done 13 games, mean reward -20.077, eps 0.83, speed 63.66 f/s
12521: done 14 games, mean reward -20.143, eps 0.82, speed 65.55 f/s
13419: done 15 games, mean reward -20.133, eps 0.81, speed 65.53 f/s
14504: done 16 games, mean reward -20.062, eps 0.79, speed 66.09 f/s
15390: done 17 games, mean reward -20.118, eps 0.78, speed 63.04 f/s
16154: done 18 games, mean reward -20.167, eps 0.77, speed 64.13 f/s
16937: done 19 games, mean reward -20.211, eps 0.76, speed 63.43 f/s
17779: done 20 games, mean reward -20.200, eps 0.75, speed 67.04 f/s
18604: done 21 games, mean reward -20.238, eps 0.73, speed 65.43 f/s
19719: done 22 games, mean reward -20.182, eps 0.72, speed 65.85 f/s
20603: done 23 games, mean reward -20.217, eps 0.71, speed 61.05 f/s
21446: done 24 games, mean reward -20.208, eps 0.69, speed 64.52 f/s
22330: done 25 games, mean reward -20.240, eps 0.68, speed 61.58 f/s
23236: done 26 games, mean reward -20.231, eps 0.67, speed 64.45 f/s
24231: done 27 games, mean reward -20.185, eps 0.65, speed 64.69 f/s
25085: done 28 games, mean reward -20.214, eps 0.64, speed 63.30 f/s
26135: done 29 games, mean reward -20.207, eps 0.63, speed 64.66 f/s
26959: done 30 games, mean reward -20.233, eps 0.61, speed 64.90 f/s
27843: done 31 games, mean reward -20.258, eps 0.60, speed 62.88 f/s
28805: done 32 games, mean reward -20.250, eps 0.59, speed 64.85 f/s
29789: done 33 games, mean reward -20.242, eps 0.57, speed 60.92 f/s
31002: done 34 games, mean reward -20.176, eps 0.56, speed 64.57 f/s
31935: done 35 games, mean reward -20.200, eps 0.54, speed 63.34 f/s
32761: done 36 games, mean reward -20.222, eps 0.53, speed 64.64 f/s
33692: done 37 games, mean reward -20.216, eps 0.52, speed 66.07 f/s
34590: done 38 games, mean reward -20.237, eps 0.51, speed 65.81 f/s
35471: done 39 games, mean reward -20.256, eps 0.49, speed 66.23 f/s
36355: done 40 games, mean reward -20.275, eps 0.48, speed 64.69 f/s
37352: done 41 games, mean reward -20.244, eps 0.47, speed 61.55 f/s
38370: done 42 games, mean reward -20.238, eps 0.45, speed 64.08 f/s
39351: done 43 games, mean reward -20.233, eps 0.44, speed 64.30 f/s
40283: done 44 games, mean reward -20.227, eps 0.42, speed 64.02 f/s
41107: done 45 games, mean reward -20.244, eps 0.41, speed 63.89 f/s
41998: done 46 games, mean reward -20.261, eps 0.40, speed 62.84 f/s
42840: done 47 games, mean reward -20.255, eps 0.39, speed 63.62 f/s

43775: done 48 games, mean reward -20.250, eps 0.37, speed 62.84 f/s
44645: done 49 games, mean reward -20.245, eps 0.36, speed 62.80 f/s
45496: done 50 games, mean reward -20.260, eps 0.35, speed 63.76 f/s
46417: done 51 games, mean reward -20.255, eps 0.34, speed 60.35 f/s
47520: done 52 games, mean reward -20.250, eps 0.32, speed 63.37 f/s
48520: done 53 games, mean reward -20.245, eps 0.31, speed 64.30 f/s
49470: done 54 games, mean reward -20.259, eps 0.29, speed 64.31 f/s
50438: done 55 games, mean reward -20.236, eps 0.28, speed 60.97 f/s
51403: done 56 games, mean reward -20.214, eps 0.27, speed 60.74 f/s
52413: done 57 games, mean reward -20.193, eps 0.25, speed 64.13 f/s
53237: done 58 games, mean reward -20.207, eps 0.24, speed 63.12 f/s
54139: done 59 games, mean reward -20.203, eps 0.23, speed 64.45 f/s
55010: done 60 games, mean reward -20.217, eps 0.21, speed 64.31 f/s
55864: done 61 games, mean reward -20.230, eps 0.20, speed 63.27 f/s
56628: done 62 games, mean reward -20.242, eps 0.19, speed 62.62 f/s
57454: done 63 games, mean reward -20.254, eps 0.18, speed 63.22 f/s
58477: done 64 games, mean reward -20.250, eps 0.16, speed 64.84 f/s
59301: done 65 games, mean reward -20.262, eps 0.15, speed 63.28 f/s
60294: done 66 games, mean reward -20.258, eps 0.14, speed 60.87 f/s
61166: done 67 games, mean reward -20.269, eps 0.13, speed 60.92 f/s
61958: done 68 games, mean reward -20.279, eps 0.11, speed 62.78 f/s
62812: done 69 games, mean reward -20.290, eps 0.10, speed 61.47 f/s
63595: done 70 games, mean reward -20.300, eps 0.09, speed 63.03 f/s
64406: done 71 games, mean reward -20.310, eps 0.08, speed 61.90 f/s
65170: done 72 games, mean reward -20.319, eps 0.07, speed 61.76 f/s
65995: done 73 games, mean reward -20.329, eps 0.06, speed 62.53 f/s
66974: done 74 games, mean reward -20.324, eps 0.04, speed 60.71 f/s
67862: done 75 games, mean reward -20.333, eps 0.03, speed 60.49 f/s
68688: done 76 games, mean reward -20.342, eps 0.02, speed 60.71 f/s
69602: done 77 games, mean reward -20.351, eps 0.02, speed 61.22 f/s
70366: done 78 games, mean reward -20.359, eps 0.02, speed 60.82 f/s
71342: done 79 games, mean reward -20.367, eps 0.02, speed 63.81 f/s
72106: done 80 games, mean reward -20.375, eps 0.02, speed 61.45 f/s
73068: done 81 games, mean reward -20.370, eps 0.02, speed 58.17 f/s
73954: done 82 games, mean reward -20.378, eps 0.02, speed 63.00 f/s
75094: done 83 games, mean reward -20.373, eps 0.02, speed 63.78 f/s
75886: done 84 games, mean reward -20.381, eps 0.02, speed 62.07 f/s
76669: done 85 games, mean reward -20.388, eps 0.02, speed 61.40 f/s
77572: done 86 games, mean reward -20.395, eps 0.02, speed 62.27 f/s
78457: done 87 games, mean reward -20.402, eps 0.02, speed 62.92 f/s
79419: done 88 games, mean reward -20.398, eps 0.02, speed 60.50 f/s
80340: done 89 games, mean reward -20.393, eps 0.02, speed 60.87 f/s
81243: done 90 games, mean reward -20.389, eps 0.02, speed 62.53 f/s
82067: done 91 games, mean reward -20.396, eps 0.02, speed 62.74 f/s

82831: done 92 games, mean reward -20.402, eps 0.02, speed 62.03 f/s
83673: done 93 games, mean reward -20.398, eps 0.02, speed 62.17 f/s
84525: done 94 games, mean reward -20.404, eps 0.02, speed 63.30 f/s
85444: done 95 games, mean reward -20.400, eps 0.02, speed 62.70 f/s
86270: done 96 games, mean reward -20.406, eps 0.02, speed 62.59 f/s
87232: done 97 games, mean reward -20.402, eps 0.02, speed 63.85 f/s
88086: done 98 games, mean reward -20.408, eps 0.02, speed 63.37 f/s
88976: done 99 games, mean reward -20.404, eps 0.02, speed 62.85 f/s
89779: done 100 games, mean reward -20.410, eps 0.02, speed 63.17 f/s
90621: done 101 games, mean reward -20.420, eps 0.02, speed 63.59 f/s
91503: done 102 games, mean reward -20.420, eps 0.02, speed 60.51 f/s
92346: done 103 games, mean reward -20.460, eps 0.02, speed 61.88 f/s
93110: done 104 games, mean reward -20.460, eps 0.02, speed 63.98 f/s
94101: done 105 games, mean reward -20.460, eps 0.02, speed 61.78 f/s
94865: done 106 games, mean reward -20.480, eps 0.02, speed 62.58 f/s
95708: done 107 games, mean reward -20.490, eps 0.02, speed 59.51 f/s
96472: done 108 games, mean reward -20.490, eps 0.02, speed 55.51 f/s
97326: done 109 games, mean reward -20.490, eps 0.02, speed 60.22 f/s
98109: done 110 games, mean reward -20.490, eps 0.02, speed 63.32 f/s
98979: done 111 games, mean reward -20.480, eps 0.02, speed 63.70 f/s
99898: done 112 games, mean reward -20.480, eps 0.02, speed 62.34 f/s
100768: done 113 games, mean reward -20.490, eps 0.02, speed 63.82 f/s
101828: done 114 games, mean reward -20.470, eps 0.02, speed 63.85 f/s
102592: done 115 games, mean reward -20.480, eps 0.02, speed 62.44 f/s
103384: done 116 games, mean reward -20.500, eps 0.02, speed 62.89 f/s
104208: done 117 games, mean reward -20.500, eps 0.02, speed 63.12 f/s
104972: done 118 games, mean reward -20.500, eps 0.02, speed 61.12 f/s
105783: done 119 games, mean reward -20.500, eps 0.02, speed 62.63 f/s
106607: done 120 games, mean reward -20.510, eps 0.02, speed 62.82 f/s
107491: done 121 games, mean reward -20.510, eps 0.02, speed 61.96 f/s
108352: done 122 games, mean reward -20.520, eps 0.02, speed 59.94 f/s
109236: done 123 games, mean reward -20.520, eps 0.02, speed 62.28 f/s
110028: done 124 games, mean reward -20.530, eps 0.02, speed 62.53 f/s
110820: done 125 games, mean reward -20.530, eps 0.02, speed 62.71 f/s
111612: done 126 games, mean reward -20.540, eps 0.02, speed 64.21 f/s
112404: done 127 games, mean reward -20.560, eps 0.02, speed 64.06 f/s
113258: done 128 games, mean reward -20.560, eps 0.02, speed 62.21 f/s
114107: done 129 games, mean reward -20.570, eps 0.02, speed 63.57 f/s
114931: done 130 games, mean reward -20.570, eps 0.02, speed 63.83 f/s
115783: done 131 games, mean reward -20.570, eps 0.02, speed 63.32 f/s
116656: done 132 games, mean reward -20.580, eps 0.02, speed 63.73 f/s
117559: done 133 games, mean reward -20.580, eps 0.02, speed 63.92 f/s
118413: done 134 games, mean reward -20.610, eps 0.02, speed 63.66 f/s
119255: done 135 games, mean reward -20.610, eps 0.02, speed 63.27 f/s

120140: done 136 games, mean reward -20.610, eps 0.02, speed 63.50 f/s
121026: done 137 games, mean reward -20.620, eps 0.02, speed 62.99 f/s
121850: done 138 games, mean reward -20.620, eps 0.02, speed 62.38 f/s
122752: done 139 games, mean reward -20.620, eps 0.02, speed 64.54 f/s
123516: done 140 games, mean reward -20.620, eps 0.02, speed 63.50 f/s
124280: done 141 games, mean reward -20.640, eps 0.02, speed 62.91 f/s
125044: done 142 games, mean reward -20.650, eps 0.02, speed 62.99 f/s
125836: done 143 games, mean reward -20.660, eps 0.02, speed 61.49 f/s
126619: done 144 games, mean reward -20.670, eps 0.02, speed 62.22 f/s
127383: done 145 games, mean reward -20.670, eps 0.02, speed 62.12 f/s
128147: done 146 games, mean reward -20.670, eps 0.02, speed 60.93 f/s
128967: done 147 games, mean reward -20.680, eps 0.02, speed 62.34 f/s
129731: done 148 games, mean reward -20.690, eps 0.02, speed 61.00 f/s
130661: done 149 games, mean reward -20.690, eps 0.02, speed 62.65 f/s
131444: done 150 games, mean reward -20.690, eps 0.02, speed 61.55 f/s
132264: done 151 games, mean reward -20.700, eps 0.02, speed 61.71 f/s
133047: done 152 games, mean reward -20.710, eps 0.02, speed 60.67 f/s
133873: done 153 games, mean reward -20.720, eps 0.02, speed 62.27 f/s
134725: done 154 games, mean reward -20.720, eps 0.02, speed 62.19 f/s
135551: done 155 games, mean reward -20.740, eps 0.02, speed 62.49 f/s
136315: done 156 games, mean reward -20.760, eps 0.02, speed 61.86 f/s
137167: done 157 games, mean reward -20.780, eps 0.02, speed 61.77 f/s
137931: done 158 games, mean reward -20.780, eps 0.02, speed 61.41 f/s
138695: done 159 games, mean reward -20.790, eps 0.02, speed 60.61 f/s
139459: done 160 games, mean reward -20.790, eps 0.02, speed 62.07 f/s
140223: done 161 games, mean reward -20.790, eps 0.02, speed 62.26 f/s
140987: done 162 games, mean reward -20.790, eps 0.02, speed 61.75 f/s
141770: done 163 games, mean reward -20.790, eps 0.02, speed 62.23 f/s
142596: done 164 games, mean reward -20.800, eps 0.02, speed 62.12 f/s
143595: done 165 games, mean reward -20.780, eps 0.02, speed 61.83 f/s
144378: done 166 games, mean reward -20.790, eps 0.02, speed 61.17 f/s
145142: done 167 games, mean reward -20.790, eps 0.02, speed 62.11 f/s
145934: done 168 games, mean reward -20.790, eps 0.02, speed 62.06 f/s
146854: done 169 games, mean reward -20.770, eps 0.02, speed 62.50 f/s
147618: done 170 games, mean reward -20.770, eps 0.02, speed 63.47 f/s
148502: done 171 games, mean reward -20.770, eps 0.02, speed 61.26 f/s
149354: done 172 games, mean reward -20.770, eps 0.02, speed 61.13 f/s
150224: done 173 games, mean reward -20.760, eps 0.02, speed 61.45 f/s
151048: done 174 games, mean reward -20.770, eps 0.02, speed 63.26 f/s
151872: done 175 games, mean reward -20.770, eps 0.02, speed 62.82 f/s
152757: done 176 games, mean reward -20.770, eps 0.02, speed 62.64 f/s
153521: done 177 games, mean reward -20.770, eps 0.02, speed 62.35 f/s
154285: done 178 games, mean reward -20.770, eps 0.02, speed 62.73 f/s
155077: done 179 games, mean reward -20.770, eps 0.02, speed 61.57 f/s

155991: done 180 games, mean reward -20.770, eps 0.02, speed 62.83 f/s
156817: done 181 games, mean reward -20.780, eps 0.02, speed 62.90 f/s
157581: done 182 games, mean reward -20.780, eps 0.02, speed 59.49 f/s
158407: done 183 games, mean reward -20.790, eps 0.02, speed 60.24 f/s
159249: done 184 games, mean reward -20.780, eps 0.02, speed 61.78 f/s
160137: done 185 games, mean reward -20.780, eps 0.02, speed 64.73 f/s
160963: done 186 games, mean reward -20.780, eps 0.02, speed 63.54 f/s
161850: done 187 games, mean reward -20.770, eps 0.02, speed 62.81 f/s
162693: done 188 games, mean reward -20.780, eps 0.02, speed 63.51 f/s
163519: done 189 games, mean reward -20.790, eps 0.02, speed 63.07 f/s
164344: done 190 games, mean reward -20.800, eps 0.02, speed 63.22 f/s
165108: done 191 games, mean reward -20.800, eps 0.02, speed 62.77 f/s
165932: done 192 games, mean reward -20.800, eps 0.02, speed 64.12 f/s
166743: done 193 games, mean reward -20.810, eps 0.02, speed 64.26 f/s
167507: done 194 games, mean reward -20.810, eps 0.02, speed 63.48 f/s
168271: done 195 games, mean reward -20.820, eps 0.02, speed 62.65 f/s
169035: done 196 games, mean reward -20.820, eps 0.02, speed 63.43 f/s
169799: done 197 games, mean reward -20.830, eps 0.02, speed 63.20 f/s
170599: done 198 games, mean reward -20.830, eps 0.02, speed 64.09 f/s
171502: done 199 games, mean reward -20.830, eps 0.02, speed 64.57 f/s
172328: done 200 games, mean reward -20.830, eps 0.02, speed 63.96 f/s
173092: done 201 games, mean reward -20.840, eps 0.02, speed 63.72 f/s
173856: done 202 games, mean reward -20.840, eps 0.02, speed 63.31 f/s
174620: done 203 games, mean reward -20.840, eps 0.02, speed 63.97 f/s
175384: done 204 games, mean reward -20.840, eps 0.02, speed 63.03 f/s
176148: done 205 games, mean reward -20.840, eps 0.02, speed 63.43 f/s
176912: done 206 games, mean reward -20.840, eps 0.02, speed 65.01 f/s
177919: done 207 games, mean reward -20.830, eps 0.02, speed 62.72 f/s
178683: done 208 games, mean reward -20.830, eps 0.02, speed 62.84 f/s
179567: done 209 games, mean reward -20.830, eps 0.02, speed 65.19 f/s
180419: done 210 games, mean reward -20.830, eps 0.02, speed 64.55 f/s
181183: done 211 games, mean reward -20.840, eps 0.02, speed 62.03 f/s
182102: done 212 games, mean reward -20.840, eps 0.02, speed 61.19 f/s
182988: done 213 games, mean reward -20.850, eps 0.02, speed 62.86 f/s
183892: done 214 games, mean reward -20.860, eps 0.02, speed 62.82 f/s
184776: done 215 games, mean reward -20.860, eps 0.02, speed 62.82 f/s
185540: done 216 games, mean reward -20.860, eps 0.02, speed 62.53 f/s
186394: done 217 games, mean reward -20.860, eps 0.02, speed 61.96 f/s
187158: done 218 games, mean reward -20.860, eps 0.02, speed 61.33 f/s
188042: done 219 games, mean reward -20.860, eps 0.02, speed 62.47 f/s
188963: done 220 games, mean reward -20.850, eps 0.02, speed 61.75 f/s
189789: done 221 games, mean reward -20.850, eps 0.02, speed 62.45 f/s
190615: done 222 games, mean reward -20.860, eps 0.02, speed 65.06 f/s
191379: done 223 games, mean reward -20.860, eps 0.02, speed 65.38 f/s

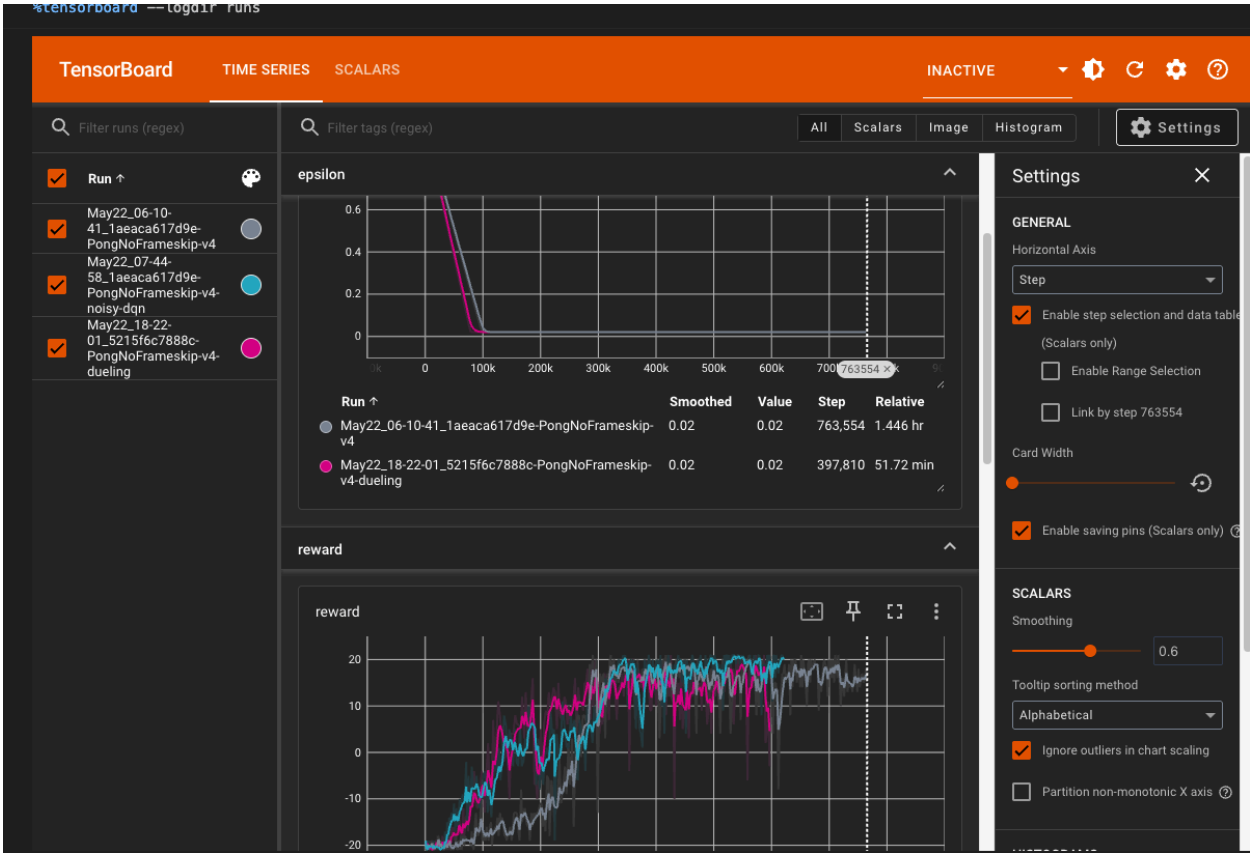
192143: done 224 games, mean reward -20.860, eps 0.02, speed 63.56 f/s
193124: done 225 games, mean reward -20.850, eps 0.02, speed 60.91 f/s
193976: done 226 games, mean reward -20.850, eps 0.02, speed 63.04 f/s
194837: done 227 games, mean reward -20.840, eps 0.02, speed 64.11 f/s
195601: done 228 games, mean reward -20.840, eps 0.02, speed 63.19 f/s
196444: done 229 games, mean reward -20.840, eps 0.02, speed 63.88 f/s
197208: done 230 games, mean reward -20.840, eps 0.02, speed 62.60 f/s
197972: done 231 games, mean reward -20.840, eps 0.02, speed 66.97 f/s
198736: done 232 games, mean reward -20.840, eps 0.02, speed 65.43 f/s
199562: done 233 games, mean reward -20.850, eps 0.02, speed 63.81 f/s
200326: done 234 games, mean reward -20.850, eps 0.02, speed 65.18 f/s
201090: done 235 games, mean reward -20.850, eps 0.02, speed 63.39 f/s
201914: done 236 games, mean reward -20.850, eps 0.02, speed 63.92 f/s
202800: done 237 games, mean reward -20.850, eps 0.02, speed 64.53 f/s
203564: done 238 games, mean reward -20.850, eps 0.02, speed 64.00 f/s
204376: done 239 games, mean reward -20.850, eps 0.02, speed 64.57 f/s
205140: done 240 games, mean reward -20.850, eps 0.02, speed 63.57 f/s
205904: done 241 games, mean reward -20.850, eps 0.02, speed 63.09 f/s
206687: done 242 games, mean reward -20.850, eps 0.02, speed 63.85 f/s
207470: done 243 games, mean reward -20.850, eps 0.02, speed 63.69 f/s
208252: done 244 games, mean reward -20.850, eps 0.02, speed 63.84 f/s
209106: done 245 games, mean reward -20.850, eps 0.02, speed 64.44 f/s
209932: done 246 games, mean reward -20.850, eps 0.02, speed 64.53 f/s
210931: done 247 games, mean reward -20.830, eps 0.02, speed 65.34 f/s
211723: done 248 games, mean reward -20.830, eps 0.02, speed 63.93 f/s
212610: done 249 games, mean reward -20.830, eps 0.02, speed 64.34 f/s
213514: done 250 games, mean reward -20.820, eps 0.02, speed 64.46 f/s
214340: done 251 games, mean reward -20.820, eps 0.02, speed 63.67 f/s
215123: done 252 games, mean reward -20.820, eps 0.02, speed 64.45 f/s
215915: done 253 games, mean reward -20.820, eps 0.02, speed 63.74 f/s
216679: done 254 games, mean reward -20.820, eps 0.02, speed 63.38 f/s
217505: done 255 games, mean reward -20.820, eps 0.02, speed 63.57 f/s
218269: done 256 games, mean reward -20.820, eps 0.02, speed 64.59 f/s
219190: done 257 games, mean reward -20.810, eps 0.02, speed 65.09 f/s
220016: done 258 games, mean reward -20.810, eps 0.02, speed 63.96 f/s
220840: done 259 games, mean reward -20.810, eps 0.02, speed 64.69 f/s
221604: done 260 games, mean reward -20.810, eps 0.02, speed 63.69 f/s
222368: done 261 games, mean reward -20.810, eps 0.02, speed 62.21 f/s
223132: done 262 games, mean reward -20.810, eps 0.02, speed 62.70 f/s
223896: done 263 games, mean reward -20.810, eps 0.02, speed 63.41 f/s
224660: done 264 games, mean reward -20.810, eps 0.02, speed 62.76 f/s
225424: done 265 games, mean reward -20.830, eps 0.02, speed 64.35 f/s
226207: done 266 games, mean reward -20.830, eps 0.02, speed 64.55 f/s
226990: done 267 games, mean reward -20.830, eps 0.02, speed 63.73 f/s

227782: done 268 games, mean reward -20.830, eps 0.02, speed 63.71 f/s
228546: done 269 games, mean reward -20.850, eps 0.02, speed 63.05 f/s
229370: done 270 games, mean reward -20.850, eps 0.02, speed 63.82 f/s
230212: done 271 games, mean reward -20.840, eps 0.02, speed 64.63 f/s
231054: done 272 games, mean reward -20.830, eps 0.02, speed 64.21 f/s
231818: done 273 games, mean reward -20.840, eps 0.02, speed 63.25 f/s
232582: done 274 games, mean reward -20.840, eps 0.02, speed 63.64 f/s
233346: done 275 games, mean reward -20.840, eps 0.02, speed 63.30 f/s
234188: done 276 games, mean reward -20.830, eps 0.02, speed 63.35 f/s
235030: done 277 games, mean reward -20.820, eps 0.02, speed 63.46 f/s
235794: done 278 games, mean reward -20.820, eps 0.02, speed 63.19 f/s
236577: done 279 games, mean reward -20.820, eps 0.02, speed 63.26 f/s
237449: done 280 games, mean reward -20.820, eps 0.02, speed 64.21 f/s
238291: done 281 games, mean reward -20.810, eps 0.02, speed 63.87 f/s
239083: done 282 games, mean reward -20.810, eps 0.02, speed 63.55 f/s
239847: done 283 games, mean reward -20.810, eps 0.02, speed 63.09 f/s
240671: done 284 games, mean reward -20.820, eps 0.02, speed 62.54 f/s
241541: done 285 games, mean reward -20.810, eps 0.02, speed 62.76 f/s
242305: done 286 games, mean reward -20.810, eps 0.02, speed 62.63 f/s
243097: done 287 games, mean reward -20.820, eps 0.02, speed 63.61 f/s
243921: done 288 games, mean reward -20.820, eps 0.02, speed 64.22 f/s
244805: done 289 games, mean reward -20.820, eps 0.02, speed 64.89 f/s
245569: done 290 games, mean reward -20.820, eps 0.02, speed 63.73 f/s
246352: done 291 games, mean reward -20.820, eps 0.02, speed 63.93 f/s
247176: done 292 games, mean reward -20.820, eps 0.02, speed 64.66 f/s
248120: done 293 games, mean reward -20.800, eps 0.02, speed 63.70 f/s
248884: done 294 games, mean reward -20.800, eps 0.02, speed 63.64 f/s
249648: done 295 games, mean reward -20.800, eps 0.02, speed 64.28 f/s
250412: done 296 games, mean reward -20.800, eps 0.02, speed 65.95 f/s
251223: done 297 games, mean reward -20.800, eps 0.02, speed 63.24 f/s
252174: done 298 games, mean reward -20.790, eps 0.02, speed 63.15 f/s
253016: done 299 games, mean reward -20.790, eps 0.02, speed 64.23 f/s
253840: done 300 games, mean reward -20.790, eps 0.02, speed 63.94 f/s
254623: done 301 games, mean reward -20.790, eps 0.02, speed 63.84 f/s
255493: done 302 games, mean reward -20.780, eps 0.02, speed 64.73 f/s
256319: done 303 games, mean reward -20.780, eps 0.02, speed 63.70 f/s
257145: done 304 games, mean reward -20.780, eps 0.02, speed 62.81 f/s
257969: done 305 games, mean reward -20.780, eps 0.02, speed 63.23 f/s
258793: done 306 games, mean reward -20.780, eps 0.02, speed 63.23 f/s
259557: done 307 games, mean reward -20.790, eps 0.02, speed 62.98 f/s
260368: done 308 games, mean reward -20.790, eps 0.02, speed 63.08 f/s
261151: done 309 games, mean reward -20.790, eps 0.02, speed 63.62 f/s
261915: done 310 games, mean reward -20.790, eps 0.02, speed 63.93 f/s
262679: done 311 games, mean reward -20.790, eps 0.02, speed 65.84 f/s

263531: done 312 games, mean reward -20.800, eps 0.02, speed 62.09 f/s
264314: done 313 games, mean reward -20.800, eps 0.02, speed 66.08 f/s
265078: done 314 games, mean reward -20.810, eps 0.02, speed 66.06 f/s
265956: done 315 games, mean reward -20.800, eps 0.02, speed 62.49 f/s
266739: done 316 games, mean reward -20.800, eps 0.02, speed 65.48 f/s
267591: done 317 games, mean reward -20.800, eps 0.02, speed 62.67 f/s
268383: done 318 games, mean reward -20.800, eps 0.02, speed 64.45 f/s
269147: done 319 games, mean reward -20.800, eps 0.02, speed 64.19 f/s
269930: done 320 games, mean reward -20.810, eps 0.02, speed 63.00 f/s
270800: done 321 games, mean reward -20.800, eps 0.02, speed 64.54 f/s
271564: done 322 games, mean reward -20.800, eps 0.02, speed 62.93 f/s
272388: done 323 games, mean reward -20.800, eps 0.02, speed 63.60 f/s
273379: done 324 games, mean reward -20.790, eps 0.02, speed 59.91 f/s
274143: done 325 games, mean reward -20.800, eps 0.02, speed 65.59 f/s
274935: done 326 games, mean reward -20.800, eps 0.02, speed 65.03 f/s
275699: done 327 games, mean reward -20.810, eps 0.02, speed 66.21 f/s
276523: done 328 games, mean reward -20.810, eps 0.02, speed 63.59 f/s
277347: done 329 games, mean reward -20.810, eps 0.02, speed 63.59 f/s
278289: done 330 games, mean reward -20.810, eps 0.02, speed 59.85 f/s
279173: done 331 games, mean reward -20.810, eps 0.02, speed 63.85 f/s
279937: done 332 games, mean reward -20.810, eps 0.02, speed 62.65 f/s
280761: done 333 games, mean reward -20.810, eps 0.02, speed 63.18 f/s
281525: done 334 games, mean reward -20.810, eps 0.02, speed 62.11 f/s
282317: done 335 games, mean reward -20.810, eps 0.02, speed 64.18 f/s
283081: done 336 games, mean reward -20.810, eps 0.02, speed 65.80 f/s
283905: done 337 games, mean reward -20.810, eps 0.02, speed 63.33 f/s
284747: done 338 games, mean reward -20.800, eps 0.02, speed 61.79 f/s
285511: done 339 games, mean reward -20.800, eps 0.02, speed 64.44 f/s
286275: done 340 games, mean reward -20.800, eps 0.02, speed 65.75 f/s
287039: done 341 games, mean reward -20.800, eps 0.02, speed 62.54 f/s
287863: done 342 games, mean reward -20.800, eps 0.02, speed 63.72 f/s
288627: done 343 games, mean reward -20.800, eps 0.02, speed 62.71 f/s
289451: done 344 games, mean reward -20.800, eps 0.02, speed 62.44 f/s
290215: done 345 games, mean reward -20.800, eps 0.02, speed 63.36 f/s
291057: done 346 games, mean reward -20.790, eps 0.02, speed 63.37 f/s
291881: done 347 games, mean reward -20.810, eps 0.02, speed 62.55 f/s
292723: done 348 games, mean reward -20.800, eps 0.02, speed 62.47 f/s
293609: done 349 games, mean reward -20.810, eps 0.02, speed 62.61 f/s
294433: done 350 games, mean reward -20.820, eps 0.02, speed 62.54 f/s
295197: done 351 games, mean reward -20.820, eps 0.02, speed 65.12 f/s
295961: done 352 games, mean reward -20.820, eps 0.02, speed 62.78 f/s
296772: done 353 games, mean reward -20.820, eps 0.02, speed 62.99 f/s
297624: done 354 games, mean reward -20.820, eps 0.02, speed 63.84 f/s
298388: done 355 games, mean reward -20.820, eps 0.02, speed 61.93 f/s

299212: done 356 games, mean reward -20.820, eps 0.02, speed 63.86 f/s
300055: done 357 games, mean reward -20.830, eps 0.02, speed 63.34 f/s
300866: done 358 games, mean reward -20.830, eps 0.02, speed 63.39 f/s
301690: done 359 games, mean reward -20.830, eps 0.02, speed 63.89 f/s
302514: done 360 games, mean reward -20.830, eps 0.02, speed 63.92 f/s
303416: done 361 games, mean reward -20.820, eps 0.02, speed 61.22 f/s
304303: done 362 games, mean reward -20.820, eps 0.02, speed 61.66 f/s
305129: done 363 games, mean reward -20.820, eps 0.02, speed 64.66 f/s
305911: done 364 games, mean reward -20.820, eps 0.02, speed 65.51 f/s
306721: done 365 games, mean reward -20.820, eps 0.02, speed 64.02 f/s
307541: done 366 games, mean reward -20.820, eps 0.02, speed 63.23 f/s
308333: done 367 games, mean reward -20.820, eps 0.02, speed 61.99 f/s
309157: done 368 games, mean reward -20.820, eps 0.02, speed 63.18 f/s
309949: done 369 games, mean reward -20.820, eps 0.02, speed 65.30 f/s
310769: done 370 games, mean reward -20.820, eps 0.02, speed 64.49 f/s
311533: done 371 games, mean reward -20.830, eps 0.02, speed 65.27 f/s
312375: done 372 games, mean reward -20.830, eps 0.02, speed 62.65 f/s
313201: done 373 games, mean reward -20.830, eps 0.02, speed 63.83 f/s
313965: done 374 games, mean reward -20.830, eps 0.02, speed 65.09 f/s
314789: done 375 games, mean reward -20.830, eps 0.02, speed 63.38 f/s
315659: done 376 games, mean reward -20.830, eps 0.02, speed 60.30 f/s
316451: done 377 games, mean reward -20.840, eps 0.02, speed 64.30 f/s
317215: done 378 games, mean reward -20.840, eps 0.02, speed 64.74 f/s
317979: done 379 games, mean reward -20.840, eps 0.02, speed 63.63 f/s
318771: done 380 games, mean reward -20.840, eps 0.02, speed 63.27 f/s
319641: done 381 games, mean reward -20.840, eps 0.02, speed 63.51 f/s
320467: done 382 games, mean reward -20.840, eps 0.02, speed 63.21 f/s
321291: done 383 games, mean reward -20.840, eps 0.02, speed 61.41 f/s
322115: done 384 games, mean reward -20.840, eps 0.02, speed 62.80 f/s
322879: done 385 games, mean reward -20.850, eps 0.02, speed 63.60 f/s
323643: done 386 games, mean reward -20.850, eps 0.02, speed 61.83 f/s
324407: done 387 games, mean reward -20.850, eps 0.02, speed 62.90 f/s
325291: done 388 games, mean reward -20.850, eps 0.02, speed 62.66 f/s
326055: done 389 games, mean reward -20.850, eps 0.02, speed 61.72 f/s
326879: done 390 games, mean reward -20.850, eps 0.02, speed 63.33 f/s
327705: done 391 games, mean reward -20.850, eps 0.02, speed 63.53 f/s
328531: done 392 games, mean reward -20.850, eps 0.02, speed 63.34 f/s
329295: done 393 games, mean reward -20.870, eps 0.02, speed 63.28 f/s
330121: done 394 games, mean reward -20.870, eps 0.02, speed 63.25 f/s
331007: done 395 games, mean reward -20.870, eps 0.02, speed 63.26 f/s
331771: done 396 games, mean reward -20.870, eps 0.02, speed 62.85 f/s
332535: done 397 games, mean reward -20.870, eps 0.02, speed 63.35 f/s
333437: done 398 games, mean reward -20.870, eps 0.02, speed 64.49 f/s
334279: done 399 games, mean reward -20.870, eps 0.02, speed 63.76 f/s

335043: done 400 games, mean reward -20.870, eps 0.02, speed 63.67 f/s
335863: done 401 games, mean reward -20.870, eps 0.02, speed 63.33 f/s
336627: done 402 games, mean reward -20.880, eps 0.02, speed 63.58 f/s
337391: done 403 games, mean reward -20.880, eps 0.02, speed 63.29 f/s
338244: done 404 games, mean reward -20.880, eps 0.02, speed 63.62 f/s
339036: done 405 games, mean reward -20.880, eps 0.02, speed 62.34 f/s
339828: done 406 games, mean reward -20.880, eps 0.02, speed 63.44 f/s
340620: done 407 games, mean reward -20.880, eps 0.02, speed 62.74 f/s
341403: done 408 games, mean reward -20.880, eps 0.02, speed 63.50 f/s
342227: done 409 games, mean reward -20.880, eps 0.02, speed 63.58 f/s
342991: done 410 games, mean reward -20.880, eps 0.02, speed 62.62 f/s
343817: done 411 games, mean reward -20.880, eps 0.02, speed 63.29 f/s
344609: done 412 games, mean reward -20.880, eps 0.02, speed 64.00 f/s
345401: done 413 games, mean reward -20.880, eps 0.02, speed 63.59 f/s
346193: done 414 games, mean reward -20.880, eps 0.02, speed 62.34 f/s
346985: done 415 games, mean reward -20.890, eps 0.02, speed 63.20 f/s
347827: done 416 games, mean reward -20.880, eps 0.02, speed 63.86 f/s
348610: done 417 games, mean reward -20.880, eps 0.02, speed 63.29 f/s
349402: done 418 games, mean reward -20.880, eps 0.02, speed 63.69 f/s
350366: done 419 games, mean reward -20.870, eps 0.02, speed 63.63 f/s
351130: done 420 games, mean reward -20.870, eps 0.02, speed 62.66 f/s
351894: done 421 games, mean reward -20.880, eps 0.02, speed 63.57 f/s
352768: done 422 games, mean reward -20.880, eps 0.02, speed 63.82 f/s
353532: done 423 games, mean reward -20.880, eps 0.02, speed 62.55 f/s
354403: done 424 games, mean reward -20.880, eps 0.02, speed 63.77 f/s
355287: done 425 games, mean reward -20.880, eps 0.02, speed 62.99 f/s
356051: done 426 games, mean reward -20.880, eps 0.02, speed 63.58 f/s
356875: done 427 games, mean reward -20.880, eps 0.02, speed 63.43 f/s
357699: done 428 games, mean reward -20.880, eps 0.02, speed 62.90 f/s
358463: done 429 games, mean reward -20.880, eps 0.02, speed 62.45 f/s
359227: done 430 games, mean reward -20.880, eps 0.02, speed 63.30 f/s
359991: done 431 games, mean reward -20.880, eps 0.02, speed 63.14 f/s
360833: done 432 games, mean reward -20.870, eps 0.02, speed 63.09 f/s
361625: done 433 games, mean reward -20.870, eps 0.02, speed 62.82 f/s
362598: done 434 games, mean reward -20.870, eps 0.02, speed 64.35 f/s



```
%load_ext tensorboard
%tensorboard --logdir runs
```

TensorBoard

TIME SERIES

SCALARS

INACTIVE

Filter runs (regex)

Run ↑

May22_06-10-41_1aeaca617d9e-PongNoFrameskip-v4

May22_07-44-58_1aeaca617d9e-PongNoFrameskip-v4-noisy-dqn

May22_18-22-01_5215f6c7888c-PongNoFrameskip-v4-dueling

Filter tags (regex)

All

Scalars

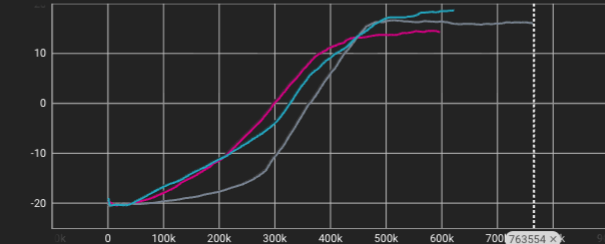
Image

Histogram

Settings

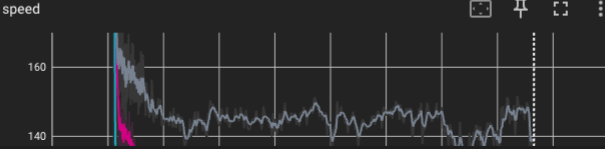
reward_100

reward_100



Run ↑	Smoothed	Value	Step	Relative
May22_06-10-41_1aeaca617d9e-PongNoFrameskip-v4	16.0489	16	763,554	1.446 hr
May22_07-44-58_1aeaca617d9e-PongNoFrameskip-v4-noisy-dqn	18.623	18.63	622,256	1.538 hr

speed



Settings

GENERAL

Horizontal Axis

Step

☒ Enable step selection and data table (Scalars only)

☐ Enable Range Selection

☐ Link by step 763554

Card Width

☒ Enable saving pins (Scalars only)

SCALARS

Smoothing

0.6

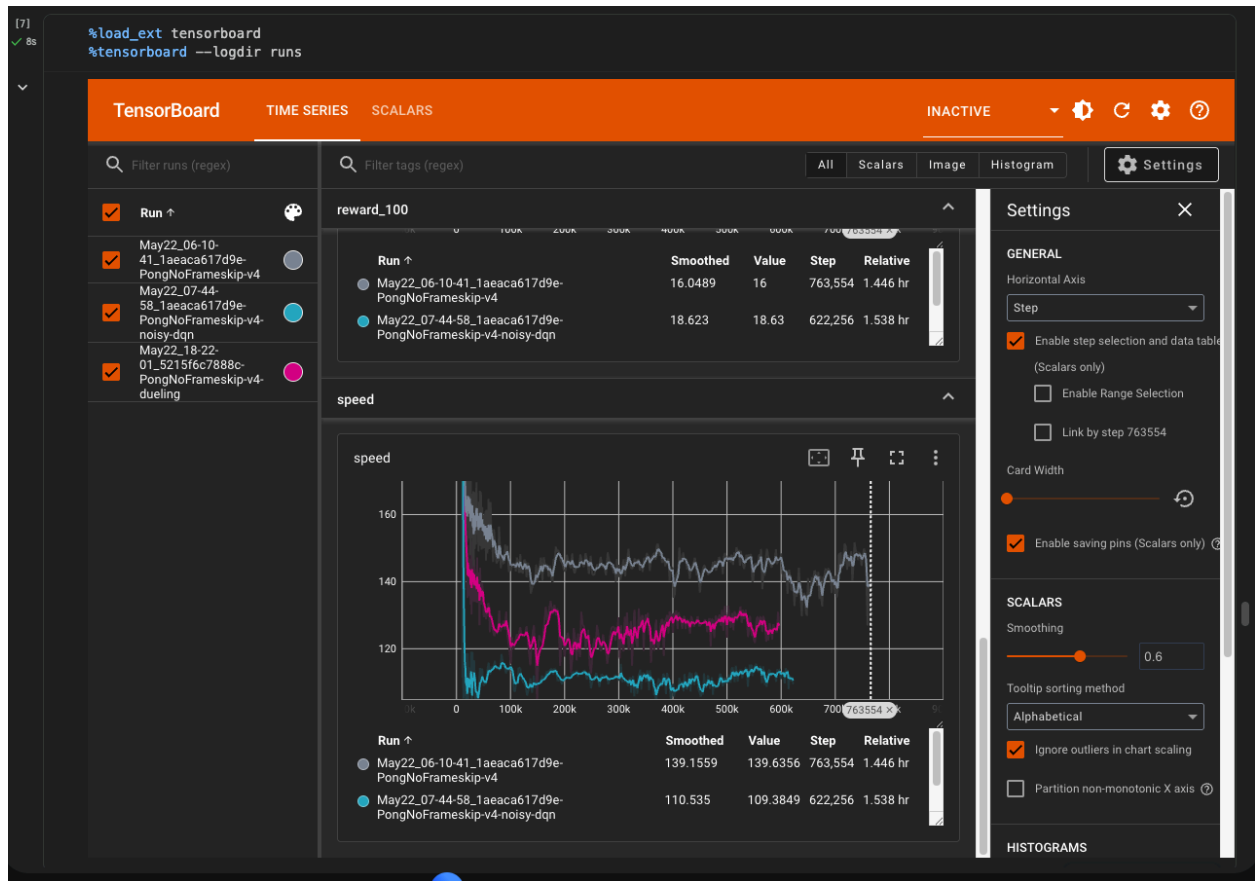
Tooltip sorting method

Alphabetical

☒ Ignore outliers in chart scaling

☐ Partition non-monotonic X axis

HISTOGRAMS



A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)

[Powered by Stella]

DuelingDQN(

(conv): Sequential(

(0): Conv2d(1, 32, kernel_size=(8, 8), stride=(4, 4))

(1): ReLU()

(2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))

(3): ReLU()

(4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))

(5): ReLU()

)

(value_stream): Sequential(

(0): Linear(in_features=3136, out_features=256, bias=True)

(1): ReLU()

(2): Linear(in_features=256, out_features=1, bias=True)

)

(advantage_stream): Sequential(

(0): Linear(in_features=3136, out_features=256, bias=True)

(1): ReLU()

```
(2): Linear(in_features=256, out_features=6, bias=True)
)
)
921: done 1 games, mean reward -20.000, eps 0.99, speed 468.59 f/s
1937: done 2 games, mean reward -19.500, eps 0.98, speed 794.63 f/s
2761: done 3 games, mean reward -20.000, eps 0.97, speed 772.59 f/s
3681: done 4 games, mean reward -19.750, eps 0.95, speed 810.83 f/s
4716: done 5 games, mean reward -19.800, eps 0.94, speed 808.14 f/s
5540: done 6 games, mean reward -20.000, eps 0.93, speed 818.79 f/s
6364: done 7 games, mean reward -20.143, eps 0.92, speed 627.00 f/s
7128: done 8 games, mean reward -20.250, eps 0.91, speed 497.30 f/s
7892: done 9 games, mean reward -20.333, eps 0.90, speed 598.94 f/s
8813: done 10 games, mean reward -20.200, eps 0.89, speed 756.16 f/s
9637: done 11 games, mean reward -20.273, eps 0.88, speed 765.05 f/s
10459: done 12 games, mean reward -20.333, eps 0.87, speed 204.40 f/s
11345: done 13 games, mean reward -20.385, eps 0.86, speed 139.12 f/s
12156: done 14 games, mean reward -20.429, eps 0.85, speed 156.07 f/s
13181: done 15 games, mean reward -20.467, eps 0.84, speed 142.25 f/s
14035: done 16 games, mean reward -20.500, eps 0.82, speed 153.50 f/s
14966: done 17 games, mean reward -20.471, eps 0.81, speed 144.38 f/s
15848: done 18 games, mean reward -20.500, eps 0.80, speed 143.02 f/s
16749: done 19 games, mean reward -20.526, eps 0.79, speed 148.74 f/s
17693: done 20 games, mean reward -20.450, eps 0.78, speed 147.50 f/s
18701: done 21 games, mean reward -20.381, eps 0.77, speed 143.42 f/s
19717: done 22 games, mean reward -20.318, eps 0.75, speed 142.23 f/s
20743: done 23 games, mean reward -20.304, eps 0.74, speed 135.11 f/s
21567: done 24 games, mean reward -20.333, eps 0.73, speed 150.95 f/s
22486: done 25 games, mean reward -20.320, eps 0.72, speed 135.65 f/s
23420: done 26 games, mean reward -20.346, eps 0.71, speed 146.90 f/s
24445: done 27 games, mean reward -20.333, eps 0.69, speed 137.41 f/s
25410: done 28 games, mean reward -20.286, eps 0.68, speed 145.67 f/s
26251: done 29 games, mean reward -20.310, eps 0.67, speed 133.13 f/s
27241: done 30 games, mean reward -20.300, eps 0.66, speed 149.98 f/s
28083: done 31 games, mean reward -20.290, eps 0.65, speed 130.15 f/s
28925: done 32 games, mean reward -20.281, eps 0.64, speed 152.10 f/s
30047: done 33 games, mean reward -20.212, eps 0.62, speed 134.07 f/s
31080: done 34 games, mean reward -20.235, eps 0.61, speed 146.69 f/s
32112: done 35 games, mean reward -20.200, eps 0.60, speed 133.92 f/s
33091: done 36 games, mean reward -20.222, eps 0.59, speed 146.67 f/s
34110: done 37 games, mean reward -20.243, eps 0.57, speed 133.92 f/s
35268: done 38 games, mean reward -20.211, eps 0.56, speed 139.05 f/s
36407: done 39 games, mean reward -20.154, eps 0.54, speed 142.58 f/s
37452: done 40 games, mean reward -20.125, eps 0.53, speed 133.70 f/s
38413: done 41 games, mean reward -20.146, eps 0.52, speed 147.19 f/s
```

39572: done 42 games, mean reward -20.119, eps 0.51, speed 133.66 f/s
40726: done 43 games, mean reward -20.070, eps 0.49, speed 134.76 f/s
41987: done 44 games, mean reward -20.068, eps 0.48, speed 142.80 f/s
42751: done 45 games, mean reward -20.089, eps 0.47, speed 126.95 f/s
43790: done 46 games, mean reward -20.065, eps 0.45, speed 144.18 f/s
45154: done 47 games, mean reward -19.979, eps 0.44, speed 135.44 f/s
46241: done 48 games, mean reward -19.938, eps 0.42, speed 133.14 f/s
47671: done 49 games, mean reward -19.898, eps 0.40, speed 139.53 f/s
49189: done 50 games, mean reward -19.820, eps 0.39, speed 136.72 f/s
50505: done 51 games, mean reward -19.804, eps 0.37, speed 129.41 f/s
51860: done 52 games, mean reward -19.750, eps 0.35, speed 131.79 f/s
53522: done 53 games, mean reward -19.717, eps 0.33, speed 130.49 f/s
54987: done 54 games, mean reward -19.685, eps 0.31, speed 128.74 f/s
56844: done 55 games, mean reward -19.636, eps 0.29, speed 131.07 f/s
58545: done 56 games, mean reward -19.518, eps 0.27, speed 133.65 f/s
60403: done 57 games, mean reward -19.456, eps 0.24, speed 132.81 f/s
61771: done 58 games, mean reward -19.448, eps 0.23, speed 124.92 f/s
63789: done 59 games, mean reward -19.424, eps 0.20, speed 130.34 f/s
66072: done 60 games, mean reward -19.300, eps 0.17, speed 126.31 f/s
67785: done 61 games, mean reward -19.279, eps 0.15, speed 125.20 f/s
70027: done 62 games, mean reward -19.210, eps 0.12, speed 123.73 f/s
72225: done 63 games, mean reward -19.111, eps 0.10, speed 125.20 f/s
74166: done 64 games, mean reward -19.078, eps 0.07, speed 124.48 f/s
76361: done 65 games, mean reward -18.985, eps 0.05, speed 124.64 f/s
79181: done 66 games, mean reward -18.803, eps 0.02, speed 131.39 f/s
81619: done 67 games, mean reward -18.701, eps 0.02, speed 125.14 f/s
84719: done 68 games, mean reward -18.485, eps 0.02, speed 126.61 f/s
87151: done 69 games, mean reward -18.362, eps 0.02, speed 128.41 f/s
89697: done 70 games, mean reward -18.271, eps 0.02, speed 125.70 f/s
92014: done 71 games, mean reward -18.183, eps 0.02, speed 131.12 f/s
94445: done 72 games, mean reward -18.056, eps 0.02, speed 123.74 f/s
97737: done 73 games, mean reward -17.863, eps 0.02, speed 120.02 f/s
100592: done 74 games, mean reward -17.703, eps 0.02, speed 115.50 f/s
102992: done 75 games, mean reward -17.640, eps 0.02, speed 123.45 f/s
105817: done 76 games, mean reward -17.474, eps 0.02, speed 119.80 f/s
109040: done 77 games, mean reward -17.221, eps 0.02, speed 122.46 f/s
111893: done 78 games, mean reward -17.103, eps 0.02, speed 124.33 f/s
114630: done 79 games, mean reward -17.025, eps 0.02, speed 123.38 f/s
117814: done 80 games, mean reward -16.725, eps 0.02, speed 126.41 f/s
120923: done 81 games, mean reward -16.568, eps 0.02, speed 123.99 f/s
123963: done 82 games, mean reward -16.317, eps 0.02, speed 123.80 f/s
126577: done 83 games, mean reward -15.988, eps 0.02, speed 123.90 f/s
129635: done 84 games, mean reward -15.702, eps 0.02, speed 126.77 f/s
132924: done 85 games, mean reward -15.482, eps 0.02, speed 120.80 f/s

136471: done 86 games, mean reward -15.326, eps 0.02, speed 115.00 f/s
139903: done 87 games, mean reward -15.138, eps 0.02, speed 123.47 f/s
143224: done 88 games, mean reward -14.989, eps 0.02, speed 127.08 f/s
146111: done 89 games, mean reward -14.933, eps 0.02, speed 118.35 f/s
149226: done 90 games, mean reward -14.722, eps 0.02, speed 104.99 f/s
152663: done 91 games, mean reward -14.516, eps 0.02, speed 124.05 f/s
155914: done 92 games, mean reward -14.272, eps 0.02, speed 122.48 f/s
159214: done 93 games, mean reward -14.054, eps 0.02, speed 121.69 f/s
162170: done 94 games, mean reward -13.840, eps 0.02, speed 127.54 f/s
165275: done 95 games, mean reward -13.642, eps 0.02, speed 132.58 f/s
168141: done 96 games, mean reward -13.375, eps 0.02, speed 127.01 f/s
171620: done 97 games, mean reward -13.206, eps 0.02, speed 132.13 f/s
174227: done 98 games, mean reward -12.939, eps 0.02, speed 134.23 f/s
177645: done 99 games, mean reward -12.798, eps 0.02, speed 130.27 f/s
181181: done 100 games, mean reward -12.610, eps 0.02, speed 133.13 f/s
184052: done 101 games, mean reward -12.290, eps 0.02, speed 129.90 f/s
186468: done 102 games, mean reward -11.940, eps 0.02, speed 132.04 f/s
189248: done 103 games, mean reward -11.800, eps 0.02, speed 125.66 f/s
191777: done 104 games, mean reward -11.670, eps 0.02, speed 114.87 f/s
194092: done 105 games, mean reward -11.570, eps 0.02, speed 123.86 f/s
196988: done 106 games, mean reward -11.420, eps 0.02, speed 122.26 f/s
200172: done 107 games, mean reward -11.160, eps 0.02, speed 117.59 f/s
202976: done 108 games, mean reward -11.010, eps 0.02, speed 123.00 f/s
205988: done 109 games, mean reward -10.740, eps 0.02, speed 124.81 f/s
208806: done 110 games, mean reward -10.460, eps 0.02, speed 122.11 f/s
211742: done 111 games, mean reward -10.170, eps 0.02, speed 112.79 f/s
214315: done 112 games, mean reward -9.830, eps 0.02, speed 117.27 f/s
216741: done 113 games, mean reward -9.520, eps 0.02, speed 126.14 f/s
219443: done 114 games, mean reward -9.200, eps 0.02, speed 123.05 f/s
222260: done 115 games, mean reward -8.850, eps 0.02, speed 116.45 f/s
224855: done 116 games, mean reward -8.510, eps 0.02, speed 124.55 f/s
228319: done 117 games, mean reward -8.260, eps 0.02, speed 117.50 f/s
230703: done 118 games, mean reward -7.900, eps 0.02, speed 131.03 f/s
233475: done 119 games, mean reward -7.610, eps 0.02, speed 127.24 f/s
236116: done 120 games, mean reward -7.350, eps 0.02, speed 126.52 f/s
238692: done 121 games, mean reward -7.090, eps 0.02, speed 119.97 f/s
241081: done 122 games, mean reward -6.780, eps 0.02, speed 122.14 f/s
243783: done 123 games, mean reward -6.470, eps 0.02, speed 125.65 f/s
246400: done 124 games, mean reward -6.200, eps 0.02, speed 115.05 f/s
248918: done 125 games, mean reward -5.900, eps 0.02, speed 112.66 f/s
251825: done 126 games, mean reward -5.630, eps 0.02, speed 119.75 f/s
254641: done 127 games, mean reward -5.330, eps 0.02, speed 123.12 f/s
257337: done 128 games, mean reward -5.020, eps 0.02, speed 119.89 f/s
259682: done 129 games, mean reward -4.650, eps 0.02, speed 127.12 f/s

262664: done 130 games, mean reward -4.410, eps 0.02, speed 123.98 f/s
265295: done 131 games, mean reward -4.070, eps 0.02, speed 119.66 f/s
268121: done 132 games, mean reward -3.770, eps 0.02, speed 119.52 f/s
270958: done 133 games, mean reward -3.480, eps 0.02, speed 124.28 f/s
273897: done 134 games, mean reward -3.180, eps 0.02, speed 121.03 f/s
276560: done 135 games, mean reward -2.840, eps 0.02, speed 119.03 f/s
278850: done 136 games, mean reward -2.480, eps 0.02, speed 115.53 f/s
281244: done 137 games, mean reward -2.090, eps 0.02, speed 118.36 f/s
283745: done 138 games, mean reward -1.770, eps 0.02, speed 123.71 f/s
286441: done 139 games, mean reward -1.520, eps 0.02, speed 126.06 f/s
288468: done 140 games, mean reward -1.140, eps 0.02, speed 126.61 f/s
291304: done 141 games, mean reward -0.830, eps 0.02, speed 126.52 f/s
293407: done 142 games, mean reward -0.500, eps 0.02, speed 122.35 f/s
295115: done 143 games, mean reward -0.120, eps 0.02, speed 126.39 f/s
297060: done 144 games, mean reward 0.250, eps 0.02, speed 124.74 f/s
299320: done 145 games, mean reward 0.540, eps 0.02, speed 129.40 f/s
301787: done 146 games, mean reward 0.770, eps 0.02, speed 117.09 f/s
304310: done 147 games, mean reward 1.000, eps 0.02, speed 127.01 f/s
307028: done 148 games, mean reward 1.240, eps 0.02, speed 120.12 f/s
308970: done 149 games, mean reward 1.590, eps 0.02, speed 119.15 f/s
311082: done 150 games, mean reward 1.920, eps 0.02, speed 129.59 f/s
313258: done 151 games, mean reward 2.260, eps 0.02, speed 123.23 f/s
315292: done 152 games, mean reward 2.580, eps 0.02, speed 130.83 f/s
318370: done 153 games, mean reward 2.830, eps 0.02, speed 119.68 f/s
320385: done 154 games, mean reward 3.160, eps 0.02, speed 124.56 f/s
322326: done 155 games, mean reward 3.520, eps 0.02, speed 128.29 f/s
324556: done 156 games, mean reward 3.810, eps 0.02, speed 114.95 f/s
326799: done 157 games, mean reward 4.120, eps 0.02, speed 125.63 f/s
329049: done 158 games, mean reward 4.440, eps 0.02, speed 131.64 f/s
331915: done 159 games, mean reward 4.690, eps 0.02, speed 122.18 f/s
334571: done 160 games, mean reward 4.950, eps 0.02, speed 120.45 f/s
337014: done 161 games, mean reward 5.270, eps 0.02, speed 132.04 f/s
339435: done 162 games, mean reward 5.570, eps 0.02, speed 125.63 f/s
341459: done 163 games, mean reward 5.870, eps 0.02, speed 129.49 f/s
343997: done 164 games, mean reward 6.160, eps 0.02, speed 130.02 f/s
346034: done 165 games, mean reward 6.460, eps 0.02, speed 119.36 f/s
347817: done 166 games, mean reward 6.720, eps 0.02, speed 128.06 f/s
349767: done 167 games, mean reward 7.010, eps 0.02, speed 130.21 f/s
351512: done 168 games, mean reward 7.250, eps 0.02, speed 128.57 f/s
353773: done 169 games, mean reward 7.310, eps 0.02, speed 130.75 f/s
355891: done 170 games, mean reward 7.590, eps 0.02, speed 124.07 f/s
357932: done 171 games, mean reward 7.870, eps 0.02, speed 125.44 f/s
359599: done 172 games, mean reward 8.160, eps 0.02, speed 115.83 f/s
361294: done 173 games, mean reward 8.400, eps 0.02, speed 123.50 f/s

363440: done 174 games, mean reward 8.580, eps 0.02, speed 122.66 f/s
365340: done 175 games, mean reward 8.880, eps 0.02, speed 119.33 f/s
367255: done 176 games, mean reward 9.090, eps 0.02, speed 126.26 f/s
369273: done 177 games, mean reward 9.250, eps 0.02, speed 128.68 f/s
371105: done 178 games, mean reward 9.500, eps 0.02, speed 121.42 f/s
373138: done 179 games, mean reward 9.750, eps 0.02, speed 119.73 f/s
375192: done 180 games, mean reward 9.820, eps 0.02, speed 122.76 f/s
376993: done 181 games, mean reward 10.030, eps 0.02, speed 127.15 f/s
379486: done 182 games, mean reward 10.080, eps 0.02, speed 128.82 f/s
381242: done 183 games, mean reward 10.170, eps 0.02, speed 128.72 f/s
383246: done 184 games, mean reward 10.250, eps 0.02, speed 117.84 f/s
385008: done 185 games, mean reward 10.410, eps 0.02, speed 127.60 f/s
386926: done 186 games, mean reward 10.590, eps 0.02, speed 128.79 f/s
389537: done 187 games, mean reward 10.670, eps 0.02, speed 130.06 f/s
391841: done 188 games, mean reward 10.780, eps 0.02, speed 124.55 f/s
393943: done 189 games, mean reward 11.040, eps 0.02, speed 131.19 f/s
395890: done 190 games, mean reward 11.180, eps 0.02, speed 129.39 f/s
397810: done 191 games, mean reward 11.310, eps 0.02, speed 130.56 f/s
400103: done 192 games, mean reward 11.330, eps 0.02, speed 124.76 f/s
402056: done 193 games, mean reward 11.430, eps 0.02, speed 130.75 f/s
404044: done 194 games, mean reward 11.530, eps 0.02, speed 129.35 f/s
406261: done 195 games, mean reward 11.640, eps 0.02, speed 131.46 f/s
408810: done 196 games, mean reward 11.650, eps 0.02, speed 125.82 f/s
410675: done 197 games, mean reward 11.810, eps 0.02, speed 130.89 f/s
412724: done 198 games, mean reward 11.840, eps 0.02, speed 129.17 f/s
414427: done 199 games, mean reward 12.030, eps 0.02, speed 128.84 f/s
417226: done 200 games, mean reward 12.050, eps 0.02, speed 125.89 f/s
419075: done 201 games, mean reward 12.110, eps 0.02, speed 128.61 f/s
421155: done 202 games, mean reward 12.100, eps 0.02, speed 129.82 f/s
423560: done 203 games, mean reward 12.290, eps 0.02, speed 125.85 f/s
425450: done 204 games, mean reward 12.560, eps 0.02, speed 128.48 f/s
427577: done 205 games, mean reward 12.790, eps 0.02, speed 128.71 f/s
429450: done 206 games, mean reward 13.020, eps 0.02, speed 126.82 f/s
431732: done 207 games, mean reward 12.870, eps 0.02, speed 124.52 f/s
433724: done 208 games, mean reward 13.110, eps 0.02, speed 129.79 f/s
436389: done 209 games, mean reward 13.160, eps 0.02, speed 128.47 f/s
438708: done 210 games, mean reward 13.220, eps 0.02, speed 126.56 f/s
441383: done 211 games, mean reward 13.210, eps 0.02, speed 127.73 f/s
443263: done 212 games, mean reward 13.270, eps 0.02, speed 128.32 f/s
446065: done 213 games, mean reward 13.220, eps 0.02, speed 129.68 f/s
448746: done 214 games, mean reward 13.200, eps 0.02, speed 126.66 f/s
451694: done 215 games, mean reward 13.140, eps 0.02, speed 127.05 f/s
453486: done 216 games, mean reward 13.210, eps 0.02, speed 127.80 f/s
456807: done 217 games, mean reward 13.190, eps 0.02, speed 127.51 f/s

459015: done 218 games, mean reward 13.210, eps 0.02, speed 128.85 f/s
461137: done 219 games, mean reward 13.270, eps 0.02, speed 128.21 f/s
463526: done 220 games, mean reward 13.320, eps 0.02, speed 124.87 f/s
465349: done 221 games, mean reward 13.440, eps 0.02, speed 128.18 f/s
467719: done 222 games, mean reward 13.450, eps 0.02, speed 129.12 f/s
469860: done 223 games, mean reward 13.450, eps 0.02, speed 120.58 f/s
472038: done 224 games, mean reward 13.540, eps 0.02, speed 128.96 f/s
474235: done 225 games, mean reward 13.570, eps 0.02, speed 128.82 f/s
476445: done 226 games, mean reward 13.660, eps 0.02, speed 123.15 f/s
479000: done 227 games, mean reward 13.650, eps 0.02, speed 129.85 f/s
481168: done 228 games, mean reward 13.700, eps 0.02, speed 129.12 f/s
483364: done 229 games, mean reward 13.710, eps 0.02, speed 127.84 f/s
485823: done 230 games, mean reward 13.800, eps 0.02, speed 130.24 f/s
488542: done 231 games, mean reward 13.720, eps 0.02, speed 126.86 f/s
490935: done 232 games, mean reward 13.720, eps 0.02, speed 131.64 f/s
493176: done 233 games, mean reward 13.740, eps 0.02, speed 125.30 f/s
495865: done 234 games, mean reward 13.690, eps 0.02, speed 132.80 f/s
497893: done 235 games, mean reward 13.690, eps 0.02, speed 129.37 f/s
499856: done 236 games, mean reward 13.710, eps 0.02, speed 123.25 f/s
502311: done 237 games, mean reward 13.670, eps 0.02, speed 132.81 f/s
504340: done 238 games, mean reward 13.720, eps 0.02, speed 129.65 f/s
506697: done 239 games, mean reward 13.780, eps 0.02, speed 126.20 f/s
509488: done 240 games, mean reward 13.690, eps 0.02, speed 130.69 f/s
511769: done 241 games, mean reward 13.740, eps 0.02, speed 125.46 f/s
513984: done 242 games, mean reward 13.740, eps 0.02, speed 131.18 f/s
515907: done 243 games, mean reward 13.730, eps 0.02, speed 130.23 f/s
518574: done 244 games, mean reward 13.690, eps 0.02, speed 128.37 f/s
520626: done 245 games, mean reward 13.800, eps 0.02, speed 130.63 f/s
522901: done 246 games, mean reward 13.940, eps 0.02, speed 131.18 f/s
524833: done 247 games, mean reward 14.050, eps 0.02, speed 132.61 f/s
527129: done 248 games, mean reward 14.160, eps 0.02, speed 127.68 f/s
529445: done 249 games, mean reward 14.180, eps 0.02, speed 131.59 f/s
531724: done 250 games, mean reward 14.170, eps 0.02, speed 128.70 f/s
534392: done 251 games, mean reward 14.080, eps 0.02, speed 131.69 f/s
536713: done 252 games, mean reward 14.050, eps 0.02, speed 128.27 f/s
538910: done 253 games, mean reward 14.180, eps 0.02, speed 123.42 f/s
540607: done 254 games, mean reward 14.240, eps 0.02, speed 125.61 f/s
542464: done 255 games, mean reward 14.220, eps 0.02, speed 129.80 f/s
544225: done 256 games, mean reward 14.240, eps 0.02, speed 126.35 f/s
545982: done 257 games, mean reward 14.280, eps 0.02, speed 123.72 f/s
547667: done 258 games, mean reward 14.350, eps 0.02, speed 125.33 f/s
549556: done 259 games, mean reward 14.470, eps 0.02, speed 122.64 f/s
551682: done 260 games, mean reward 14.500, eps 0.02, speed 121.67 f/s
553941: done 261 games, mean reward 14.520, eps 0.02, speed 128.27 f/s

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1776, in
_wrapped_call_impl

```
    return self._call_impl(*args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1787, in
_call_impl

```
    return forward_call(*args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/container.py", line 253, in
forward

```
    input = module(input)
    ^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1776, in
_wrapped_call_impl

```
    return self._call_impl(*args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

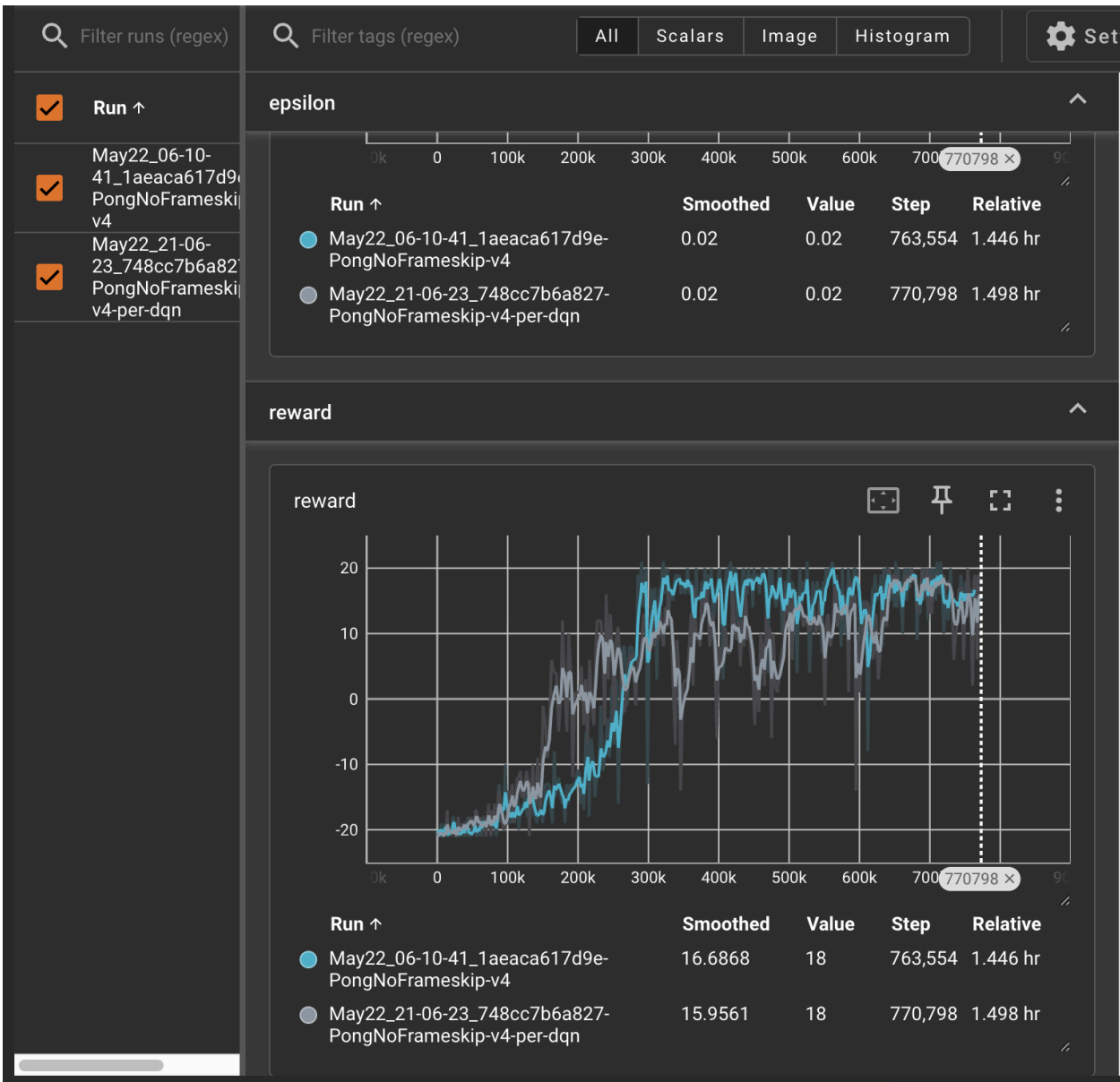
File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/module.py", line 1787, in
_call_impl

```
    return forward_call(*args, **kwargs)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

File "/usr/local/lib/python3.12/dist-packages/torch/nn/modules/linear.py", line 134, in forward

```
    return F.linear(input, self.weight, self.bias)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

KeyboardInterrupt



🔍 Filter runs (regex)

🔍 Filter tags (regex)

All

Scalars

Image

Histogram

⚙️ Settings

☑ Run ↑

☑ May22_06-10-41_1aeaca617d9e-PongNoFrameskip-v4

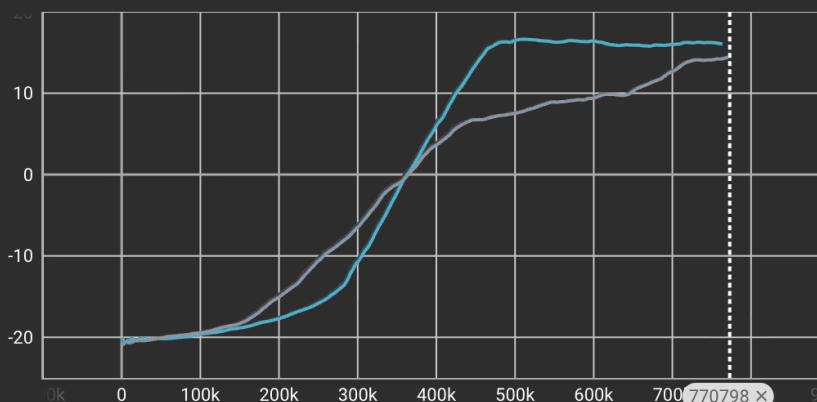
☑ May22_21-06-23_748cc7b6a827-PongNoFrameskip-v4-per-dqn

reward

● May22_06-10-41_1aeaca617d9e-PongNoFrameskip-v4	16.6868	18	763,554	1.446 hr
● May22_21-06-23_748cc7b6a827-PongNoFrameskip-v4-per-dqn	15.9561	18	770,798	1.498 hr

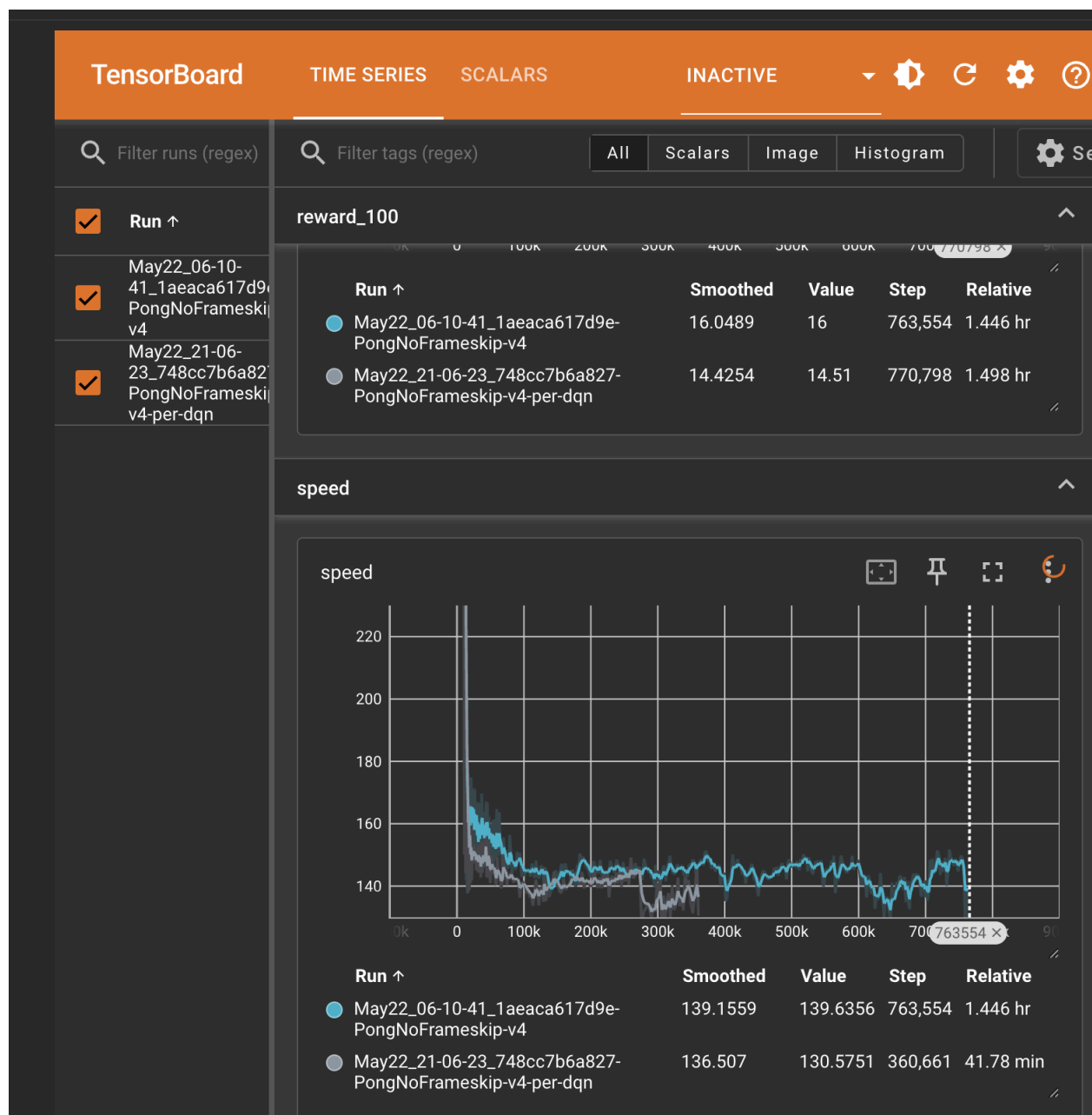
reward_100

reward_100



Run ↑	Smoothed	Value	Step	Relative
● May22_06-10-41_1aeaca617d9e-PongNoFrameskip-v4	16.0489	16	763,554	1.446 hr
● May22_21-06-23_748cc7b6a827-PongNoFrameskip-v4-per-dqn	14.4254	14.51	770,798	1.498 hr

speed



A.L.E: Arcade Learning Environment (version 0.11.2+ecc1138)

[Powered by Stella]

DQN(

(conv): Sequential(

(0): Conv2d(1, 32, kernel_size=(8, 8), stride=(4, 4))

(1): ReLU()

(2): Conv2d(32, 64, kernel_size=(4, 4), stride=(2, 2))

(3): ReLU()

(4): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1))

(5): ReLU())

```

)
(fc): Sequential(
  (0): Linear(in_features=3136, out_features=512, bias=True)
  (1): ReLU()
  (2): Linear(in_features=512, out_features=6, bias=True)
)
)
861: done 1 games, mean reward -20.000, eps 0.99, beta 0.41, speed 699.51 f/s
1866: done 2 games, mean reward -20.500, eps 0.97, beta 0.42, speed 738.11 f/s
2768: done 3 games, mean reward -20.333, eps 0.96, beta 0.42, speed 508.90 f/s
3594: done 4 games, mean reward -20.500, eps 0.95, beta 0.43, speed 587.23 f/s
4531: done 5 games, mean reward -20.200, eps 0.94, beta 0.44, speed 736.89 f/s
5722: done 6 games, mean reward -20.000, eps 0.92, beta 0.45, speed 797.09 f/s
6606: done 7 games, mean reward -20.143, eps 0.91, beta 0.46, speed 784.04 f/s
7389: done 8 games, mean reward -20.250, eps 0.89, beta 0.46, speed 780.43 f/s
8393: done 9 games, mean reward -20.333, eps 0.88, beta 0.47, speed 724.28 f/s
9235: done 10 games, mean reward -20.300, eps 0.87, beta 0.48, speed 789.80 f/s
10274: done 11 games, mean reward -20.273, eps 0.85, beta 0.49, speed 324.63 f/s
11100: done 12 games, mean reward -20.333, eps 0.84, beta 0.50, speed 144.38 f/s
12203: done 13 games, mean reward -20.154, eps 0.83, beta 0.50, speed 153.41 f/s
13073: done 14 games, mean reward -20.143, eps 0.81, beta 0.51, speed 143.60 f/s
13837: done 15 games, mean reward -20.200, eps 0.80, beta 0.52, speed 154.52 f/s
14680: done 16 games, mean reward -20.250, eps 0.79, beta 0.53, speed 136.67 f/s
15550: done 17 games, mean reward -20.235, eps 0.78, beta 0.53, speed 157.93 f/s
16566: done 18 games, mean reward -20.167, eps 0.76, beta 0.54, speed 140.05 f/s
17330: done 19 games, mean reward -20.211, eps 0.75, beta 0.55, speed 157.59 f/s
18173: done 20 games, mean reward -20.250, eps 0.74, beta 0.56, speed 136.09 f/s
19227: done 21 games, mean reward -20.238, eps 0.73, beta 0.56, speed 155.61 f/s
20010: done 22 games, mean reward -20.273, eps 0.71, beta 0.57, speed 135.27 f/s
20802: done 23 games, mean reward -20.304, eps 0.70, beta 0.58, speed 155.78 f/s
21612: done 24 games, mean reward -20.333, eps 0.69, beta 0.59, speed 152.82 f/s
22574: done 25 games, mean reward -20.320, eps 0.68, beta 0.59, speed 141.90 f/s
23495: done 26 games, mean reward -20.308, eps 0.66, beta 0.60, speed 147.41 f/s
24591: done 27 games, mean reward -20.259, eps 0.65, beta 0.61, speed 145.47 f/s
25537: done 28 games, mean reward -20.286, eps 0.64, beta 0.62, speed 139.97 f/s
26838: done 29 games, mean reward -20.207, eps 0.62, beta 0.63, speed 151.88 f/s
27701: done 30 games, mean reward -20.233, eps 0.60, beta 0.64, speed 136.64 f/s
28925: done 31 games, mean reward -20.194, eps 0.59, beta 0.65, speed 156.40 f/s
29990: done 32 games, mean reward -20.219, eps 0.57, beta 0.66, speed 141.29 f/s
30921: done 33 games, mean reward -20.242, eps 0.56, beta 0.67, speed 156.50 f/s
31851: done 34 games, mean reward -20.235, eps 0.54, beta 0.67, speed 138.69 f/s
33142: done 35 games, mean reward -20.171, eps 0.53, beta 0.68, speed 141.49 f/s
34289: done 36 games, mean reward -20.167, eps 0.51, beta 0.69, speed 153.89 f/s
35319: done 37 games, mean reward -20.162, eps 0.50, beta 0.70, speed 137.35 f/s

```

36277: done 38 games, mean reward -20.184, eps 0.48, beta 0.71, speed 150.00 f/s
37521: done 39 games, mean reward -20.179, eps 0.46, beta 0.72, speed 139.14 f/s
38469: done 40 games, mean reward -20.200, eps 0.45, beta 0.73, speed 141.05 f/s
39628: done 41 games, mean reward -20.195, eps 0.43, beta 0.74, speed 146.50 f/s
41293: done 42 games, mean reward -20.190, eps 0.41, beta 0.75, speed 141.58 f/s
42595: done 43 games, mean reward -20.186, eps 0.39, beta 0.77, speed 138.45 f/s
44188: done 44 games, mean reward -20.114, eps 0.37, beta 0.78, speed 140.00 f/s
45928: done 45 games, mean reward -20.089, eps 0.34, beta 0.79, speed 140.65 f/s
47013: done 46 games, mean reward -20.087, eps 0.33, beta 0.80, speed 150.82 f/s
48712: done 47 games, mean reward -20.021, eps 0.30, beta 0.82, speed 141.11 f/s
50172: done 48 games, mean reward -20.021, eps 0.28, beta 0.83, speed 134.51 f/s
51642: done 49 games, mean reward -19.980, eps 0.26, beta 0.84, speed 131.43 f/s
52671: done 50 games, mean reward -19.960, eps 0.25, beta 0.85, speed 130.25 f/s
53872: done 51 games, mean reward -19.941, eps 0.23, beta 0.86, speed 138.05 f/s
56100: done 52 games, mean reward -19.750, eps 0.20, beta 0.88, speed 135.22 f/s
57864: done 53 games, mean reward -19.736, eps 0.17, beta 0.90, speed 135.01 f/s
59289: done 54 games, mean reward -19.722, eps 0.15, beta 0.91, speed 130.73 f/s
60948: done 55 games, mean reward -19.691, eps 0.13, beta 0.92, speed 121.34 f/s
63220: done 56 games, mean reward -19.571, eps 0.10, beta 0.94, speed 126.54 f/s
65135: done 57 games, mean reward -19.491, eps 0.07, beta 0.96, speed 128.74 f/s
67183: done 58 games, mean reward -19.483, eps 0.04, beta 0.98, speed 125.27 f/s
69319: done 59 games, mean reward -19.390, eps 0.02, beta 0.99, speed 130.25 f/s
71378: done 60 games, mean reward -19.283, eps 0.02, beta 1.00, speed 127.33 f/s
73092: done 61 games, mean reward -19.295, eps 0.02, beta 1.00, speed 126.09 f/s
75891: done 62 games, mean reward -19.161, eps 0.02, beta 1.00, speed 134.91 f/s
78848: done 63 games, mean reward -19.032, eps 0.02, beta 1.00, speed 127.34 f/s
81029: done 64 games, mean reward -18.984, eps 0.02, beta 1.00, speed 125.98 f/s
83431: done 65 games, mean reward -18.923, eps 0.02, beta 1.00, speed 130.64 f/s
85750: done 66 games, mean reward -18.864, eps 0.02, beta 1.00, speed 134.12 f/s
87687: done 67 games, mean reward -18.866, eps 0.02, beta 1.00, speed 131.41 f/s
90665: done 68 games, mean reward -18.750, eps 0.02, beta 1.00, speed 128.40 f/s
92668: done 69 games, mean reward -18.739, eps 0.02, beta 1.00, speed 119.72 f/s
95319: done 70 games, mean reward -18.657, eps 0.02, beta 1.00, speed 131.00 f/s
97248: done 71 games, mean reward -18.634, eps 0.02, beta 1.00, speed 130.45 f/s
99169: done 72 games, mean reward -18.625, eps 0.02, beta 1.00, speed 134.39 f/s
101566: done 73 games, mean reward -18.493, eps 0.02, beta 1.00, speed 129.71 f/s
103216: done 74 games, mean reward -18.446, eps 0.02, beta 1.00, speed 127.96 f/s
105676: done 75 games, mean reward -18.347, eps 0.02, beta 1.00, speed 134.73 f/s
108209: done 76 games, mean reward -18.263, eps 0.02, beta 1.00, speed 129.28 f/s
110601: done 77 games, mean reward -18.182, eps 0.02, beta 1.00, speed 136.42 f/s
112857: done 78 games, mean reward -18.103, eps 0.02, beta 1.00, speed 133.48 f/s
115309: done 79 games, mean reward -18.000, eps 0.02, beta 1.00, speed 123.61 f/s
117890: done 80 games, mean reward -17.863, eps 0.02, beta 1.00, speed 128.08 f/s
119999: done 81 games, mean reward -17.802, eps 0.02, beta 1.00, speed 125.85 f/s

122289: done 82 games, mean reward -17.756, eps 0.02, beta 1.00, speed 136.71 f/s
124913: done 83 games, mean reward -17.699, eps 0.02, beta 1.00, speed 132.97 f/s
127215: done 84 games, mean reward -17.655, eps 0.02, beta 1.00, speed 128.66 f/s
129579: done 85 games, mean reward -17.588, eps 0.02, beta 1.00, speed 136.26 f/s
131696: done 86 games, mean reward -17.512, eps 0.02, beta 1.00, speed 131.36 f/s
133751: done 87 games, mean reward -17.460, eps 0.02, beta 1.00, speed 132.44 f/s
136182: done 88 games, mean reward -17.364, eps 0.02, beta 1.00, speed 136.53 f/s
138058: done 89 games, mean reward -17.337, eps 0.02, beta 1.00, speed 134.22 f/s
139623: done 90 games, mean reward -17.322, eps 0.02, beta 1.00, speed 134.40 f/s
142080: done 91 games, mean reward -17.242, eps 0.02, beta 1.00, speed 133.00 f/s
144412: done 92 games, mean reward -17.163, eps 0.02, beta 1.00, speed 122.93 f/s
146773: done 93 games, mean reward -17.065, eps 0.02, beta 1.00, speed 138.08 f/s
148946: done 94 games, mean reward -17.011, eps 0.02, beta 1.00, speed 123.89 f/s
151134: done 95 games, mean reward -16.947, eps 0.02, beta 1.00, speed 136.17 f/s
153206: done 96 games, mean reward -16.885, eps 0.02, beta 1.00, speed 136.38 f/s
155638: done 97 games, mean reward -16.804, eps 0.02, beta 1.00, speed 133.79 f/s
157916: done 98 games, mean reward -16.735, eps 0.02, beta 1.00, speed 134.16 f/s
159805: done 99 games, mean reward -16.697, eps 0.02, beta 1.00, speed 134.96 f/s
161787: done 100 games, mean reward -16.650, eps 0.02, beta 1.00, speed 120.10 f/s
164415: done 101 games, mean reward -16.500, eps 0.02, beta 1.00, speed 130.95 f/s
167077: done 102 games, mean reward -16.340, eps 0.02, beta 1.00, speed 136.95 f/s
169507: done 103 games, mean reward -16.230, eps 0.02, beta 1.00, speed 129.64 f/s
172037: done 104 games, mean reward -16.100, eps 0.02, beta 1.00, speed 134.42 f/s
174232: done 105 games, mean reward -16.030, eps 0.02, beta 1.00, speed 127.83 f/s
176272: done 106 games, mean reward -15.990, eps 0.02, beta 1.00, speed 130.38 f/s
178801: done 107 games, mean reward -15.860, eps 0.02, beta 1.00, speed 134.17 f/s
181376: done 108 games, mean reward -15.750, eps 0.02, beta 1.00, speed 133.05 f/s
183261: done 109 games, mean reward -15.690, eps 0.02, beta 1.00, speed 137.05 f/s
185519: done 110 games, mean reward -15.610, eps 0.02, beta 1.00, speed 137.44 f/s
187965: done 111 games, mean reward -15.500, eps 0.02, beta 1.00, speed 138.24 f/s
190151: done 112 games, mean reward -15.410, eps 0.02, beta 1.00, speed 132.95 f/s
192442: done 113 games, mean reward -15.330, eps 0.02, beta 1.00, speed 138.57 f/s
194572: done 114 games, mean reward -15.240, eps 0.02, beta 1.00, speed 139.09 f/s
197844: done 115 games, mean reward -15.020, eps 0.02, beta 1.00, speed 136.58 f/s
200729: done 116 games, mean reward -14.860, eps 0.02, beta 1.00, speed 135.13 f/s
203031: done 117 games, mean reward -14.780, eps 0.02, beta 1.00, speed 139.78 f/s
205950: done 118 games, mean reward -14.610, eps 0.02, beta 1.00, speed 136.04 f/s
208371: done 119 games, mean reward -14.520, eps 0.02, beta 1.00, speed 137.98 f/s
211423: done 120 games, mean reward -14.260, eps 0.02, beta 1.00, speed 134.92 f/s
214137: done 121 games, mean reward -14.110, eps 0.02, beta 1.00, speed 141.16 f/s
215964: done 122 games, mean reward -14.060, eps 0.02, beta 1.00, speed 136.05 f/s
218659: done 123 games, mean reward -13.890, eps 0.02, beta 1.00, speed 134.14 f/s
221957: done 124 games, mean reward -13.770, eps 0.02, beta 1.00, speed 136.22 f/s
224458: done 125 games, mean reward -13.620, eps 0.02, beta 1.00, speed 139.57 f/s

226471: done 126 games, mean reward -13.540, eps 0.02, beta 1.00, speed 137.14 f/s
229220: done 127 games, mean reward -13.430, eps 0.02, beta 1.00, speed 135.58 f/s
231257: done 128 games, mean reward -13.310, eps 0.02, beta 1.00, speed 137.49 f/s
233931: done 129 games, mean reward -13.070, eps 0.02, beta 1.00, speed 132.78 f/s
236892: done 130 games, mean reward -12.900, eps 0.02, beta 1.00, speed 138.86 f/s
239010: done 131 games, mean reward -12.820, eps 0.02, beta 1.00, speed 129.34 f/s
242172: done 132 games, mean reward -12.630, eps 0.02, beta 1.00, speed 137.43 f/s
245107: done 133 games, mean reward -12.380, eps 0.02, beta 1.00, speed 133.87 f/s
247852: done 134 games, mean reward -12.130, eps 0.02, beta 1.00, speed 133.17 f/s
250135: done 135 games, mean reward -12.060, eps 0.02, beta 1.00, speed 137.45 f/s
252902: done 136 games, mean reward -11.900, eps 0.02, beta 1.00, speed 132.07 f/s
256006: done 137 games, mean reward -11.690, eps 0.02, beta 1.00, speed 139.17 f/s
258800: done 138 games, mean reward -11.450, eps 0.02, beta 1.00, speed 132.13 f/s
261281: done 139 games, mean reward -11.150, eps 0.02, beta 1.00, speed 137.60 f/s
264108: done 140 games, mean reward -10.970, eps 0.02, beta 1.00, speed 133.23 f/s
266772: done 141 games, mean reward -10.810, eps 0.02, beta 1.00, speed 133.89 f/s
269002: done 142 games, mean reward -10.740, eps 0.02, beta 1.00, speed 140.76 f/s
271774: done 143 games, mean reward -10.520, eps 0.02, beta 1.00, speed 140.76 f/s
274383: done 144 games, mean reward -10.310, eps 0.02, beta 1.00, speed 136.13 f/s
277088: done 145 games, mean reward -10.040, eps 0.02, beta 1.00, speed 140.48 f/s
279650: done 146 games, mean reward -9.760, eps 0.02, beta 1.00, speed 135.46 f/s
281827: done 147 games, mean reward -9.490, eps 0.02, beta 1.00, speed 141.03 f/s
283551: done 148 games, mean reward -9.100, eps 0.02, beta 1.00, speed 136.37 f/s
285921: done 149 games, mean reward -8.880, eps 0.02, beta 1.00, speed 138.28 f/s
288463: done 150 games, mean reward -8.600, eps 0.02, beta 1.00, speed 134.76 f/s
291308: done 151 games, mean reward -8.350, eps 0.02, beta 1.00, speed 138.35 f/s
294299: done 152 games, mean reward -8.190, eps 0.02, beta 1.00, speed 136.23 f/s
296117: done 153 games, mean reward -7.830, eps 0.02, beta 1.00, speed 136.30 f/s
298546: done 154 games, mean reward -7.520, eps 0.02, beta 1.00, speed 131.10 f/s
300197: done 155 games, mean reward -7.150, eps 0.02, beta 1.00, speed 136.24 f/s
303075: done 156 games, mean reward -7.000, eps 0.02, beta 1.00, speed 143.08 f/s
305264: done 157 games, mean reward -6.710, eps 0.02, beta 1.00, speed 139.92 f/s
307984: done 158 games, mean reward -6.490, eps 0.02, beta 1.00, speed 136.42 f/s
310206: done 159 games, mean reward -6.220, eps 0.02, beta 1.00, speed 139.40 f/s
312181: done 160 games, mean reward -5.920, eps 0.02, beta 1.00, speed 139.03 f/s
314514: done 161 games, mean reward -5.630, eps 0.02, beta 1.00, speed 134.35 f/s
316618: done 162 games, mean reward -5.380, eps 0.02, beta 1.00, speed 139.34 f/s
319341: done 163 games, mean reward -5.250, eps 0.02, beta 1.00, speed 142.09 f/s
320934: done 164 games, mean reward -4.880, eps 0.02, beta 1.00, speed 136.19 f/s
323784: done 165 games, mean reward -4.710, eps 0.02, beta 1.00, speed 136.38 f/s
326626: done 166 games, mean reward -4.610, eps 0.02, beta 1.00, speed 140.76 f/s
328759: done 167 games, mean reward -4.280, eps 0.02, beta 1.00, speed 135.88 f/s
331464: done 168 games, mean reward -4.100, eps 0.02, beta 1.00, speed 140.77 f/s
333268: done 169 games, mean reward -3.740, eps 0.02, beta 1.00, speed 136.02 f/s

335064: done 170 games, mean reward -3.430, eps 0.02, beta 1.00, speed 137.26 f/s
338003: done 171 games, mean reward -3.170, eps 0.02, beta 1.00, speed 134.66 f/s
339779: done 172 games, mean reward -2.810, eps 0.02, beta 1.00, speed 137.99 f/s
341997: done 173 games, mean reward -2.600, eps 0.02, beta 1.00, speed 139.69 f/s
344285: done 174 games, mean reward -2.330, eps 0.02, beta 1.00, speed 137.24 f/s
347242: done 175 games, mean reward -2.170, eps 0.02, beta 1.00, speed 138.42 f/s
349420: done 176 games, mean reward -1.940, eps 0.02, beta 1.00, speed 138.24 f/s
351557: done 177 games, mean reward -1.680, eps 0.02, beta 1.00, speed 134.38 f/s
353373: done 178 games, mean reward -1.370, eps 0.02, beta 1.00, speed 137.53 f/s
355082: done 179 games, mean reward -1.080, eps 0.02, beta 1.00, speed 136.75 f/s
357021: done 180 games, mean reward -0.840, eps 0.02, beta 1.00, speed 136.46 f/s
359030: done 181 games, mean reward -0.550, eps 0.02, beta 1.00, speed 137.39 f/s
361401: done 182 games, mean reward -0.310, eps 0.02, beta 1.00, speed 137.88 f/s
364361: done 183 games, mean reward -0.140, eps 0.02, beta 1.00, speed 135.96 f/s
366310: done 184 games, mean reward 0.150, eps 0.02, beta 1.00, speed 137.74 f/s
368080: done 185 games, mean reward 0.460, eps 0.02, beta 1.00, speed 140.24 f/s
370099: done 186 games, mean reward 0.720, eps 0.02, beta 1.00, speed 138.44 f/s
373038: done 187 games, mean reward 0.910, eps 0.02, beta 1.00, speed 136.23 f/s
375049: done 188 games, mean reward 1.170, eps 0.02, beta 1.00, speed 137.87 f/s
377702: done 189 games, mean reward 1.420, eps 0.02, beta 1.00, speed 138.83 f/s
380891: done 190 games, mean reward 1.590, eps 0.02, beta 1.00, speed 139.17 f/s
382692: done 191 games, mean reward 1.860, eps 0.02, beta 1.00, speed 138.25 f/s
384417: done 192 games, mean reward 2.140, eps 0.02, beta 1.00, speed 138.12 f/s
386519: done 193 games, mean reward 2.370, eps 0.02, beta 1.00, speed 137.37 f/s
388461: done 194 games, mean reward 2.640, eps 0.02, beta 1.00, speed 135.07 f/s
390047: done 195 games, mean reward 2.950, eps 0.02, beta 1.00, speed 142.94 f/s
391846: done 196 games, mean reward 3.240, eps 0.02, beta 1.00, speed 138.22 f/s
393780: done 197 games, mean reward 3.490, eps 0.02, beta 1.00, speed 136.19 f/s
395527: done 198 games, mean reward 3.790, eps 0.02, beta 1.00, speed 139.40 f/s
397464: done 199 games, mean reward 4.050, eps 0.02, beta 1.00, speed 138.34 f/s
399526: done 200 games, mean reward 4.330, eps 0.02, beta 1.00, speed 136.90 f/s
401681: done 201 games, mean reward 4.530, eps 0.02, beta 1.00, speed 140.46 f/s
403882: done 202 games, mean reward 4.510, eps 0.02, beta 1.00, speed 138.36 f/s
406178: done 203 games, mean reward 4.720, eps 0.02, beta 1.00, speed 133.27 f/s
407987: done 204 games, mean reward 4.970, eps 0.02, beta 1.00, speed 137.29 f/s
410104: done 205 games, mean reward 5.190, eps 0.02, beta 1.00, speed 139.08 f/s
411815: done 206 games, mean reward 5.540, eps 0.02, beta 1.00, speed 136.51 f/s
413566: done 207 games, mean reward 5.810, eps 0.02, beta 1.00, speed 137.26 f/s
415857: done 208 games, mean reward 5.990, eps 0.02, beta 1.00, speed 139.20 f/s
418697: done 209 games, mean reward 6.200, eps 0.02, beta 1.00, speed 136.29 f/s
421644: done 210 games, mean reward 6.360, eps 0.02, beta 1.00, speed 140.92 f/s
424398: done 211 games, mean reward 6.510, eps 0.02, beta 1.00, speed 135.35 f/s
426529: done 212 games, mean reward 6.790, eps 0.02, beta 1.00, speed 139.94 f/s
428663: done 213 games, mean reward 7.010, eps 0.02, beta 1.00, speed 141.26 f/s

431096: done 214 games, mean reward 7.230, eps 0.02, beta 1.00, speed 135.77 f/s
433946: done 215 games, mean reward 7.260, eps 0.02, beta 1.00, speed 142.85 f/s
436236: done 216 games, mean reward 7.410, eps 0.02, beta 1.00, speed 133.92 f/s
438451: done 217 games, mean reward 7.650, eps 0.02, beta 1.00, speed 140.12 f/s
440302: done 218 games, mean reward 7.850, eps 0.02, beta 1.00, speed 136.78 f/s
442466: done 219 games, mean reward 8.070, eps 0.02, beta 1.00, speed 140.75 f/s
444999: done 220 games, mean reward 8.090, eps 0.02, beta 1.00, speed 135.77 f/s
447294: done 221 games, mean reward 8.240, eps 0.02, beta 1.00, speed 139.80 f/s
449769: done 222 games, mean reward 8.480, eps 0.02, beta 1.00, speed 140.43 f/s
452200: done 223 games, mean reward 8.610, eps 0.02, beta 1.00, speed 134.45 f/s
454601: done 224 games, mean reward 8.790, eps 0.02, beta 1.00, speed 140.33 f/s
457240: done 225 games, mean reward 8.910, eps 0.02, beta 1.00, speed 137.74 f/s
459459: done 226 games, mean reward 9.180, eps 0.02, beta 1.00, speed 136.69 f/s
462809: done 227 games, mean reward 9.280, eps 0.02, beta 1.00, speed 137.91 f/s
464501: done 228 games, mean reward 9.570, eps 0.02, beta 1.00, speed 137.40 f/s
466998: done 229 games, mean reward 9.600, eps 0.02, beta 1.00, speed 139.97 f/s
469196: done 230 games, mean reward 9.760, eps 0.02, beta 1.00, speed 139.51 f/s
471314: done 231 games, mean reward 9.970, eps 0.02, beta 1.00, speed 138.53 f/s
473541: done 232 games, mean reward 10.090, eps 0.02, beta 1.00, speed 136.95 f/s
475378: done 233 games, mean reward 10.230, eps 0.02, beta 1.00, speed 139.09 f/s
477641: done 234 games, mean reward 10.280, eps 0.02, beta 1.00, speed 141.12 f/s
479782: done 235 games, mean reward 10.510, eps 0.02, beta 1.00, speed 139.05 f/s
481855: done 236 games, mean reward 10.690, eps 0.02, beta 1.00, speed 138.38 f/s
484699: done 237 games, mean reward 10.750, eps 0.02, beta 1.00, speed 138.64 f/s
487276: done 238 games, mean reward 10.820, eps 0.02, beta 1.00, speed 138.53 f/s
489810: done 239 games, mean reward 10.800, eps 0.02, beta 1.00, speed 138.50 f/s
492099: done 240 games, mean reward 10.970, eps 0.02, beta 1.00, speed 140.61 f/s
494232: done 241 games, mean reward 11.140, eps 0.02, beta 1.00, speed 139.95 f/s
496236: done 242 games, mean reward 11.420, eps 0.02, beta 1.00, speed 132.26 f/s
498098: done 243 games, mean reward 11.560, eps 0.02, beta 1.00, speed 137.06 f/s
499817: done 244 games, mean reward 11.710, eps 0.02, beta 1.00, speed 139.66 f/s
501689: done 245 games, mean reward 11.800, eps 0.02, beta 1.00, speed 137.28 f/s
503525: done 246 games, mean reward 11.910, eps 0.02, beta 1.00, speed 139.94 f/s
505300: done 247 games, mean reward 11.980, eps 0.02, beta 1.00, speed 139.54 f/s
507387: done 248 games, mean reward 11.920, eps 0.02, beta 1.00, speed 137.19 f/s
509986: done 249 games, mean reward 11.940, eps 0.02, beta 1.00, speed 139.96 f/s
512669: done 250 games, mean reward 11.810, eps 0.02, beta 1.00, speed 131.64 f/s
514384: done 251 games, mean reward 11.950, eps 0.02, beta 1.00, speed 139.27 f/s
516326: done 252 games, mean reward 12.040, eps 0.02, beta 1.00, speed 139.96 f/s
517920: done 253 games, mean reward 12.080, eps 0.02, beta 1.00, speed 137.20 f/s
519837: done 254 games, mean reward 12.130, eps 0.02, beta 1.00, speed 138.68 f/s
521423: done 255 games, mean reward 12.140, eps 0.02, beta 1.00, speed 134.34 f/s
523263: done 256 games, mean reward 12.290, eps 0.02, beta 1.00, speed 140.00 f/s
526156: done 257 games, mean reward 12.220, eps 0.02, beta 1.00, speed 140.22 f/s

528008: done 258 games, mean reward 12.360, eps 0.02, beta 1.00, speed 137.99 f/s
529819: done 259 games, mean reward 12.400, eps 0.02, beta 1.00, speed 137.85 f/s
532200: done 260 games, mean reward 12.300, eps 0.02, beta 1.00, speed 132.07 f/s
534427: done 261 games, mean reward 12.300, eps 0.02, beta 1.00, speed 136.06 f/s
536619: done 262 games, mean reward 12.280, eps 0.02, beta 1.00, speed 135.89 f/s
539352: done 263 games, mean reward 12.290, eps 0.02, beta 1.00, speed 140.42 f/s
541625: done 264 games, mean reward 12.030, eps 0.02, beta 1.00, speed 141.67 f/s

Traceback (most recent call last):

File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/05_per_dqn_pong.py", line 277, in <module>

```
loss_t, td_errors = calc_loss(batch, weights, net, tgt_net, device=device)
```

[illegible]

File "/content/Deep-Reinforcement-Learning-Hands-On/Chapter06/05_per_dqn_pong.py", line 195, in calc_loss

```
states_v = torch.tensor(states).to(device)
```

AA

KeyboardInterrupt

